

COURSE SCHEME AND SYLLABI

SEMESTER – I

S N	COURSE NO	COURSE TITLE	CODE	L	T	P	CR
1	UEC001	ELECTRONIC ENGINEERING	CF	3	1	2	4.5
2	UHU003	PROFESSIONAL COMMUNICATION	CF	2	0	2	3.0
3	UMA003	MATHEMATICS-I	CF	3	1	0	3.5
4	UPH004	APPLIED PHYSICS	CF	3	1	2	4.5
5	UTA015	ENGINEERING DRAWING	CF	2	4	0	4.0
6	UTA017	COMPUTER PROGRAMMING - I	CF	3	0	2	4.0
		TOTAL		16	7	8	23.5

SEMESTER – II

S N	COURSE NO	COURSE TITLE	CODE	L	T	P	CR
1	UCB008	APPLIED CHEMISTRY	CF	3	1	2	4.5
2	UEE001	ELECTRICAL ENGINEERING	CF	3	1	2	4.5
3	UEN002	ENERGY AND ENVIRONMENT	CF	3	0	0	3.0
4	UES009	MECHANICS *	CF	2	1	2*	2.5
5	UMA004	MATHEMATICS-II	CF	3	1	0	3.5
6	UTA013	ENGINEERING DESIGN PROJECT-I (Mangonel) (6 Self-Effort Hours)	PR	1	0	2	5.0
7	UTA018	OBJECT ORIENTED PROGRAMMING	CF	3	0	2	4.0
		TOTAL		18	4	8	27.0

* Only one Lab session per semester

SEMESTER – III

S N	COURSE NO	COURSE TITLE	CODE	L	T	P	CR
1	UEE304	TRANSMISSION AND DISTRIBUTION OF ELECTRIC POWER	CP	3	1	0	3.5
2	UEE305	ELECTROMAGNETIC FIELDS AND WAVES	CP	3	1	0	3.5
3	UES010	SOLIDS AND STRUCTURES *	CF	3	1	2	4.5
4	UES011	THERMO-FLUIDS *	CF	3	1	2	4.5
5	UMA031	OPTIMIZATION TECHNIQUES	CF	3	1	0	3.5
6	UTA002	MANUFACTURING PROCESSES	CF	2	0	3	3.5
7	UTA014	ENGINEERING DESIGN PROJECT-II (Buggy) (6 Self-Effort Hours)	PR	1	0	4	6.0
		TOTAL		18	5	11	29.0

* The lab sessions will be on every alternate week

SEMESTER – IV

S N	COURSE NO	COURSE TITLE	CODE	L	T	P	CR
1	UEE301	DIRECT CURRENT MACHINES AND TRANSFORMERS	CP	3	1	2	4.5
2	UEE406	POWER SYSTEM PRACTICES	CP	3	0	0	3.0
3	UEE407	NETWORK THEORY AND DESIGN	CP	3	1	2	4.5
4	UEE505	ANALOG AND DIGITAL SYSTEMS	CP	3	1	2	4.5
5	UES012	ENGINEERING MATERIALS	CF	3	1	2	4.5
6	UMA007	NUMERICAL ANALYSIS	CF	3	1	2	4.5
		TOTAL		18	5	10	25.5

SEMESTER – V

S N	COURSE NO	COURSE TITLE	CODE	L	T	P	CR
1	UEE401	ALTERNATING CURRENT MACHINES	CP	3	1	2	4.5
2	UEE403	MEASUREMENT AND TRANSDUCERS	CP	3	0	2	4.0
3	UEE504	POWER ELECTRONICS	CP	3	1	2	4.5
4	UEI501	CONTROL SYSTEMS	CP	3	1	2	4.5
5	UEI610	FUNDAMENTALS OF MICROPROCESSORS AND MICROCONTROLLERS	CP	3	0	2	4.0
6	UTA012	INNOVATION AND ENTREPRENEURSHIP (5 self-effort hour)	CF	1	0	2	4.5
		TOTAL		16	3	12	26.0

SEMESTER – VI

S N	COURSE NO	COURSE TITLE	CODE	L	T	P	CR
1	UEE603	SWITCHGEAR AND PROTECTION	CP	3	0	2	4.0
2	UEE605	POWER SYSTEM ANALYSIS AND STABILITY	CP	3	1	2	4.5
3	UEE801	ELECTRIC DRIVES	CP	3	1	2	4.5
4	UHU005	HUMANITIES FOR ENGINEERS	CF	2	0	2	3.0
5		ELECTIVE-I	PE	3	0	0	3.0
6		GENERIC ELECTIVE	GE	3	0	0	3.0
		TOTAL		17	2	10	22.0

SEMESTER – VII

S N	COURSE NO	COURSE TITLE	CODE	L	T	P	CR
1	UEE502	HIGH VOLTAGE ENGINEERING	CP	3	0	2	4.0
2	UEE604	FLEXIBLE AC TRANSMISSION SYSTEMS	CP	3	1	0	3.5
3	UEE608	SOFT COMPUTING IN ELECTRICAL ENGINEERING	CP	3	0	2	4.0
4	UEE795	CAPSTONE PROJECT	PR	0	0	2	8.0
5	UEE804	OPERATION AND CONTROL OF POWER SYSTEMS	CP	3	1	2	4.5
6		ELECTIVE-II	PE	3	1	0	3.5
		TOTAL		15	3	8	27.5

SEMESTER – VIII

S N	COURSE NO	COURSE TITLE	CODE	L	T	P	CR
1	UEE891	PROJECT	PR				20.0

OR

1	UEE806	ALTERNATE SOURCES OF ENERGY		3	0	2	4.0
2	UEE892	DESIGN PROJECT	PR				13.0
3	UEI805	ENVIRONMENTAL INSTRUMENTATION	CP	3	0	0	3.0
		TOTAL		6	0	2	20.0

OR

1	UEE893	START- UP SEMESTER	PR				20.0
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ELECTIVE-I

S N	COURSE NO	COURSE TITLE	CODE	L	T	P	CR
1	UCS049	DATA STRUCTURES AND ALGORITHMS	PE	2	0	2	3.0
2	UEE631	HVDC TRANSMISSION SYSTEMS	PE	3	0	0	3.0
3	UEE632	POWER GENERATION AND ECONOMICS	PE	3	0	0	3.0
4	UEE633	GENERALIZED THEORY OF ELECTRICAL MACHINES	PE	3	0	0	3.0
5	UEE634	REAL TIME POWER SYSTEMS	PE	2	0	2	3.0

ELECTIVE-II

S N	COURSE NO	COURSE TITLE	CODE	L	T	P	CR
1	UEC408	PROBABILITY AND INFORMATION THEORY	PE	3	1	0	3.5
2	UEE521	ELECTRIC MACHINE DESIGN	PE	3	1	0	3.5
3	UEE524	POWER QUALITY MONITORING AND CONDITIONING	PE	3	1	0	3.5
4	UEE714	DIGITAL SIGNAL PROCESSING FUNDAMENTALS	PE	2	1	2	3.5
	UEE803	LOAD DISPATCH AND COMMUNICATION	PE	3	1	0	3.5
5	UEE841	INDUSTRIAL ELECTRONICS	PE	3	1	0	3.5
6	UEE850	SMART GRID	PE	3	1	0	3.5
7	UEI841	ADVANCED CONTROL SYSTEMS	PE	3	1	0	3.5

GENERIC ELECTIVES

S N	COURSE NO	TITLE	CODE	L	T	P	Cr.
1	UBT509	BIOLOGY FOR ENGINEERS	GE	3	0	0	3.0
2	UCS001	INTRODUCTION TO CYBER SECURITY	GE	3	0	0	3.0
3	UEN004	TECHNOLOGIES FOR SUSTAINABLE DEVELOPMENT	GE	3	0	0	3.0
4	UHU006	INTRODUCTORY COURSE IN FRENCH	GE	2	2	0	3.0
5	UHU007	EMPLOYABILITY DEVELOPMENT SKILLS	GE	0	3	3	3.0
6	UHU008	INTRODUCTION TO CORPORATE FINANCE	GE	3	0	0	3.0
7	UHU009	INTRODUCTION TO COGNITIVE SCIENCE	GE	3	0	0	3.0

8	UMA061	ADVANCED NUMERICAL METHODS	GE	3	0	0	3.0
9	UMA066	GRAPH THEORY AND APPLICATIONS	GE	3	0	0	3.0
10	UPH063	NANOSCIENCE AND NANOMATERIALS	GE	3	0	0	3.0

Course Syllabi (2017 Scheme)
SEMESTER – I

1. **Course number and name:** UEC001; Electronic Engineering
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electronics and Communication Engineering
4. **Text book, title, author, and year****Text Books / Reference Books**
 - *M. M. Mano and M.D. Ciletti, Digital Design, Pearson, Prentice Hall, 2013.*
 - *Milliman, J. and Halkias, C.C., Electronic Devices and Circuits, Tata McGraw Hill, 2007.*
 - *Donald D Givone, Digital Principles and Design, McGraw-Hill, 2003.*
 - *John F Wakerly, Digital Design: Principles and Practices, Pearson, (2000).*
 - *N Storey, Electronics: a Systems Approach, Pearson, Prentice Hall, (2009).*
 - *Boylestad, R.L. and Nashelsky, L., Electronic Devices & Circuit Theory, Perason (2009).*
 - *M. M. Mano and M.D. Ciletti, Digital Design, Pearson, Prentice Hall, 2013.*
 - *Milliman, J. and Halkias, C.C., Electronic Devices and Circuits, Tata McGraw Hill, 2007.*
 - *Donald D Givone, Digital Principles and Design, McGraw-Hill, 2003.*
 - *John F Wakerly, Digital Design: Principles and Practices, Pearson, (2000).*
 - *N Storey, Electronics: a Systems Approach, Pearson, Prentice Hall, (2009).*
 - *Boylestad, R.L. and Nashelsky, L., Electronic Devices & Circuit Theory, Perason (2009).*
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Semiconductor Devices: P-N junction diode: Ideal diode, V-I characteristics of diode, Diode small signal model, Diode switching characteristics, Zener diode

Electronics Devices and Circuits: P-N Diode as a rectifier, Clipper and clamper, Operation of Bipolar Junction Transistor and Transistor Biasing, CB, CE, CC (Relationship between α , β , γ) circuit configuration Input-output characteristics, Equivalent circuit of ideal and real amplifiers, Low frequency response of amplifiers, Introduction to Field Effect Transistor and its characteristics

Operational Amplifier Circuits: The ideal operational amplifier, the inverting, non-inverting amplifiers, Op-Amp Characteristics, Frequency response of op-amp, Application of op-amp

Digital Systems and Binary Numbers: Introduction to Digital signals and systems, Number systems, Positive and negative representation of numbers, Binary arithmetic, Definitions and basic theorems of Boolean Algebra, Algebraic simplification, Sum of products and product of sums formulations (SOP and POS), Gate primitives, AND, OR, NOT and Universal Gate, Minimization of logic functions, Karnaugh maps.

Combinational and Sequential Logic: Code converters, multiplexors, decoders, Addition circuits and priority encoder, Master-slave and edge-triggered flip-flops, Synchronous and Asynchronous counters, Registers.

Logic families: N and P channel MOS transistors, CMOS inverter, NAND and NOR gates,

General CMOS Logic, TTL and CMOS logic families, and their interfacing.

Laboratory Work:

Familiarization of CRO and Electronic Components, Diodes characteristics Input-Output and Switching characteristics, BJT and MOSFET Characteristics, Zener diode as voltage regulator, Transistorized Series voltage regulator. Half and Full wave Rectifiers with and without filter circuit, Half and full adder circuit implementation, Decoder, DMUX and MUX, Binary/BCD up/down counters.

- b. Prerequisites
 - None
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. Specific goals for the course

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to

- Demonstrate the use of semiconductor diodes in various applications.
 - Discuss and explain the working of transistors and Operational Amplifiers, their configurations and applications.
 - Recognize and apply the number systems and Boolean algebra.
 - Reduce Boolean Expressions and implement them with Logic Gates.
 - Analyze, design and implement combinational and sequential circuits.
 - Analyze and differentiate logic families, TTL and CMOS.
- b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.
E3	Use analytical, computational and/or experimental methods to obtain solutions.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Semiconductor devices
- Transistor
- FET
- Op-amp
- Digital circuits and Sequential circuits

1. **Course number and name:** UHU003; Professional Communication
2. **Credits and contact hours:** 3.0 and 4
3. **Instructor's or course coordinator's name:** Faculty from the School of Humanities and Social Sciences
4. **Text book, title, author, and year****Text Books / Reference Books**
 - *Lesikar R.V and Flatley M.E., Basic Business Communication Skills for the Empowering the Internet Generation. Tata McGraw Hill. New Delhi (2006).*
 - *Raman, M & Sharma, S., Technical Communication Principles and Practice, Oxford University Press New Delhi. (2011).*
 - *Mukherjee H.S., Business Communication-Connecting at Work, Oxford University Press New Delhi, (2013).*
 - *Butterfield, Jeff., Soft Skills for everyone, Cengage Learning New Delhi, (2013).*
 - *Robbins, S.P., &Hunsaker, P.L., Training in Interpersonal Skills, Prentice Hall of India New Delhi, (2008).*
 - *Di Sianza, J. J & Legge, N.J., Business and Professional Communication, Pearson Education India New Delhi, (2009).*
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Effective Communication: Meaning, Barriers, Types of Communication and Essentials Interpersonal Communication Skills.

Effective Spoken Communication: Understanding essentials of spoken communication, Public speaking, Discussion techniques, Presentation strategies.

Effective Professional and Technical writing: Paragraph development, Forms of writing, Abstraction and Summarization of a text; Technicalities of letter writing, internal and external organizational communication. Technical reports, proposals and papers.

Effective Non-verbal communication: Knowledge and adoption of the right non-verbal cues of body language, interpretation of the body language in professional context. Understanding Proxemics and other forms of Non-verbal communication.

Communicating for Employment: Designing Effective Job application letter and resumes; Success strategies for Group discussions and Interviews.

Communication Networks in organizations: Types, barriers and overcoming the barriers.

Laboratory Work:

- Needs-assessment of spoken and written communication and feedback.
- Training for Group Discussions through simulations and role plays.
- Training for effective presentations.
- Project based team presentations.
- Proposals and papers-review and suggestions.

Minor Project (if any):

Team projects on technical report writing and presentations.

- b. Prerequisites
 - None
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. Specific goals for the course

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Understand and appreciate the need of communication training.
- Use different strategies of effective communication.
- Select the most appropriate mode of communication for a given situation.
- Speak assertively and effectively.
- Correspond effectively through different modes of written communication.
- Write effective reports, proposals and papers.
- Present himself/ herself professionally through effective resumes and interviews.

- b Specific outcomes addressed by the course.

F1	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
G2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
D1	Share responsibility and information schedule with others in team.

SO3.2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
SO3.3	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
SO5.1	Share responsibility and information schedule with others in team.

7. Brief list of topics to be covered

- Effective Communication
- Effective Spoken Communication
- Effective Professional and Technical Writing
- Effective Non-Verbal Communication
- Communicating for Employment
- Communication Networks in Organizations

1. **Course number and name:** UMA003; Mathematics – I
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from School of Mathematics
4. **Text book, title, author, and year****Text Books / Reference Books**
 - Thomas, G.B. and Finney, R.L., *Calculus and Analytic Geometry*, Pearson Education (2007), 9thed.
 - Stewart James, *Essential Calculus*; Thomson Publishers (2007), 6thed.
 - Wider David V, *Advanced Calculus: Early Transcendentals*, Cengage Learning (2007).
 - Apostol Tom M, *Calculus, Vol I and II*, John Wiley (2003).
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Applications of Derivatives: Mean value theorems and their geometrical interpretation, Cartesian graphing using first and second order derivatives, Asymptotes and dominant terms, graphing of polar curves, Applied minimum and maximum problems.

Sequences and Series: Introduction to sequences and Infinite series, Tests for convergence/divergence, Limit comparison test, Ratio test, Root test, Cauchy integral test, Alternating series, Absolute convergence and conditional convergence.

Series Expansions: Power series, Taylor series, Convergence of Taylor series, Error estimates, Term by term differentiation and integration.

Partial Differentiation: Functions of several variables, Limits and continuity, Chain rule, change of variables, Partial differentiation of implicit functions, Directional derivatives and its properties, Maxima and minima by using second order derivatives.

Multiple Integrals: Change of order of integration, Change of variables, Applications of multiple integrals.
 - b. Prerequisites
 - None
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required
6. **Specific goals for the course**
 - a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

 - Apply the knowledge of calculus to plot graphs of functions and solve the problem of maxima and minima.
 - Determine the convergence/divergence of infinite series, approximation of functions using power and Taylor's series expansion and error estimation.
 - Evaluate multiple integrals and their applications to engineering problems.

- Examine functions of several variables, define and compute partial derivatives, directional derivatives and their use in finding maxima and minima.
 - Analyze some mathematical problems encountered in engineering applications.
- b Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
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SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
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7. Brief list of topics to be covered

- Applications of Derivatives
- Sequences and Series
- Series Expansions
- Partial Differentiation
- Multiple Integrals

1. **Course number and name:** UPH004; Applied Physics
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the School of Physics & Material Science
4. **Text book, title, author, and year** **Text Books / Reference Books**
 - Beiser, A., *Concept of Modern Physics*, Tata McGraw Hill (2007) 6th ed.
 - Griffiths, D.J., *Introduction to Electrodynamics*, Prentice Hall of India (1999) 3rd ed.
 - Jenkins, F.A. and White, H.E., *Fundamentals of Optics*, McGraw Hill (2001) 4th ed.
 - Wehr, M.R, Richards, J.A., Adair, T.W., *Physics of The Atom*, Narosa Publishing House (1990) 4th ed.
 - Verma, N.K., *Physics for Engineers*, Prentice Hall of India (2014) 1st ed.
 - Pedrotti, Frank L., Pedrotti, Leno S., and Pedrotti, Leno M., *Introduction to Optics*, Pearson Prentice HallTM (2008) 3rd ed.
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

Brief description of the content of the course (catalog description)

Oscillations and Waves: Oscillatory motion and damping, Applications - Electromagnetic damping – eddy current; **Acoustics:** Reverberation time, absorption coefficient, Sabine's and Eyring's formulae (Qualitative idea), Applications - Designing of hall for speech, concert, and opera; **Ultrasonics:** Production and Detection of Ultrasonic waves, Applications - green energy, sound signaling, dispersion of fog, remote sensing, Car's airbag sensor.

Electromagnetic Waves: Scalar and vector fields; Gradient, divergence, and curl; Stokes' and Green's theorems; Concept of Displacement current; Maxwell's equations; Electromagnetic wave equations in free space and conducting media, Application - skin depth.

Optics: Interference: Parallel and wedge-shape thin films, Newton rings, Applications as Non-reflecting coatings, Measurement of wavelength and refractive index.

Diffraction: Single and Double slit diffraction, and Diffraction grating, Applications - Dispersive and Resolving Powers.

Polarization: Production, detection, Applications – Anti-glare automobile headlights, Adjustable tint windows.

Lasers: Basic concepts, Laser properties, Ruby, HeNe, and Semiconductor lasers, Applications – Optical communication and Optical alignment.

Quantum Mechanics: Wave function, Steady State Schrodinger wave equation, Expectation value, Infinite potential well, Tunneling effect (Qualitative idea), Application - Quantum computing.

Laboratory Work:

- Determination of damping effect on oscillatory motion due to various media.
- Determination of velocity of ultrasonic waves in liquids by stationary wave method.
- Determination of wavelength of sodium light using Newton's rings method.
- Determination of dispersive power of sodium-D lines using diffraction grating.

- Determination of specific rotation of cane sugar solution.
 - Study and proof of Malus' law in polarization.
 - Determination of beam divergence and beam intensity of a given laser.
 - Determination of displacement and conducting currents through a dielectric.
 - Determination of Planck's constant
- a. Prerequisites
- None
- b. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
- Required

6. Specific goals for the course

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Understand damped and simple harmonic motion, the role of reverberation in designing a hall and generation and detection of ultrasonic waves.
 - Use Maxwell's equations to describe propagation of EM waves in a medium.
 - Demonstrate interference, diffraction and polarization of light.
 - Explain the working principle of Lasers.
 - Use the concept of wave function to find probability of a particle confined in a box.
- b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Oscillations and Waves
- Ultrasonics
- Electromagnetic Waves
- Optics, Polarization, Lasers
- Quantum Mechanics

1. **Course number and name:** UTA015; Engineering Design-I
2. **Credits and contact hours:** 4.0 and 6
3. **Instructor's or course coordinator's name:** Faculty from the Department of Mechanical Engineering
4. **Text book, title, author, and year****Text Books / Reference Books**
 - Jolhe, D.A., *Engineering Drawing*, Tata McGraw Hill, 2008
 - Davies, B. L., Yarwood, A., *Engineering Drawing and Computer Graphics*, Van Nostrand Reinhold (UK), 1986
 - Gill, P.S., *Geometrical Drawings*, S.K. Kataria & Sons, Delhi (2008).
 - Gill, P.S., *Machine Drawings*, S.K. Kataria & Sons, Delhi (2013).
 - Mohan, K.R., *Engineering Graphics*, Dhanpat Rai Publishing Company (P) Ltd, Delhi(2002).
 - French, T. E., Vierck, C. J. and Foster, R. J., *Fundamental of Engineering Drawing & Graphics Technology*, McGraw Hill Book Company, New Delhi (1986).
 - Rowan, J. and Sidwell, E. H., *Graphics for Engineers*, Edward Arnold, London (1968).
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Engineering Drawing

- 1 Introduction
- 2 Orthographic Projection: First angle and third angle projection system
- 3 Isometric Projections
- 4 Auxiliary Projections
- 5 Perspective Projections
- 6 Introduction to Mechanical Drawing
- 7 Sketching Engineering Objects
- 8 Sections, Dimensions and Tolerances

AutoCAD

- 1 Management of screen menus commands
- 2 Introduction to drawing entities
- 3 Co-ordinate systems: Cartesian, polar and relative coordinates
- 4 Drawing limits, units of measurement and scale
- 5 Layering: organizing and maintaining the integrity of drawings
- 6 Design of prototype drawings as templates.
- 7 Editing/modifying drawing entities: selection of objects, object snap modes, Editing commands.
- 8 Dimensioning: use of annotations, dimension types, properties and placement, adding text to drawing

Micro Projects /Assignments:

1. Completing the views - Identification and drawing of missing lines in the projection of objects
2. Missing Views – using two views to draw the projection of the object in the third view,

primarily restricting to Elevation, Plan and Profile views

3. Projects related to orthographic and isometric projections
 - a. Using wax blocks or soap bars to develop three-dimensional object from given orthographic projections
 - b. Using wax blocks or soap bars to develop three-dimensional object, section it and color the section
 - c. Use of AUTOCAD as a complementary tool for drawing the projections of the objects created in (1) and (2).
4. Develop the lateral surface of different objects involving individual or a combination of solids like Prism, Cone, Pyramid, Cylinder, Sphere etc.
5. To draw the detailed and assembly drawings of simple engineering objects/systems with due sectioning (where ever required) along with bill of materials.
e.g. Rivet joints, simple bearing, wooden joints, Two plates connected with nut and bolt etc.
 - b. Prerequisites
 - None
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. Specific goals for the course

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Creatively comprehend geometrical details of common engineering objects.
- Draw dimensioned orthographic and isometric projections of simple engineering objects.
- Interpret the meaning and intent of toleranced dimensions and geometric tolerance symbolism.
- Create the engineering drawings for simple engineering objects using AutoCAD.
- Manage screen menus and commands using AutoCAD.
- Operate data entry modes and define drawings geometrically in terms of cartesian, polar and relative coordinates in AutoCAD.
- Create and edit drawings making selections of objects, discriminating by layering and using entities, object snap modes, editing commands, angles and displacements using AutoCAD.

- b. Specific outcomes addressed by the course.

D1	Share responsibility and information schedule with others in team.
D2	Participate in the development and selection of ideas.
I3	Recognize the need to embrace personal responsibility for lifelong learning
F1	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
F2	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.

SO2.3	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.
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SO3.3	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
SO5.1	Share responsibility and information schedule with others in team.
SO5.2	Participate in the development and selection of ideas on a team whose members together provide leadership.
SO7.3	Recognize the need to embrace personal responsibility for lifelong learning

7. Brief list of topics to be covered

- Orthographic Projection: First angle and third angle projection system
- Isometric Projections
- Auxiliary Projections

1. **Course number and name:** UTA007; Computer Programming-I
2. **Credits and contact hours:** 4.0 and 5
3. **Instructor's or course coordinator's name:** Faculty from the Department of Computer Science and Engineering
4. **Text book, title, author, and year****Text Books / Reference Books**
 - “Brain W. Kernighan, Dennis M. Ritchie”, *The C Programming Language* (2nd Edition), Prentice Hall, 1988.
 - “Ajay Mittal”, *Programming in C-A Practical Approach*, Pearson, 2010.
 - “Reema Thareja”, *Computer Fundamentals and Programming in C*, Oxford University Press, 2012.
 - “Zed. A. Shaw”, *Learn C the Hard Way*, Pearson, 2015.
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Computers Fundamentals: Classification of Computers, Application of Computers, Basic organization of computer, Input and Output Devices, Binary Number System, Computer memory, Computer Software.

Algorithms and Programming Languages: Algorithm, Flowcharts, Pseudocode, Generation of Programming Languages.

C Language: Structure of C Program, Life Cycle of Program from Source code to Executable, Compiling and Executing C Code, Keywords, Identifiers, Primitive Data types in C, variables, constants, input/output statements in C, Operators, type conversion and type casting. Conditional branching statements, Iterative statements, Nested loops, break and continue statements.

Functions: Declaration, Definition, Call and return, Call by value, Call by reference, showcase stack usage with help of debugger, Scope of variables, Storage classes, Recursive functions, Recursion vs Iteration.

Arrays, Strings and Pointers: One-dimensional, Two-dimensional and Multi-dimensional arrays, operations on array: traversal, insertion, deletion, merging and searching, Inter-function communication via arrays: passing a row, passing the entire array, matrices. Reading, writing and manipulating Strings, understanding computer memory, accessing via pointers, pointers to arrays, dynamic allocation, drawback of pointers.

Linear and Non-Linear Data Structures: Linked lists, Stacks and Queues.
 - b. Prerequisites
 - None
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required
6. **Specific goals for the course**
 - a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

 - Comprehend concepts related to computer hardware and software, draw flowcharts and

write algorithm/pseudocode.

- Write, compile and debug programs in C language, use different data types, operators and console I/O function in a computer program.
- Design programs involving decision control statements, loop control statements, case control structures, arrays, strings, pointers, functions and implement the dynamics of memory by the use of pointers.
- Comprehend the concepts of linear and non-Linear data structures by implementing linked lists, stacks and queues.

b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
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SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
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7. Brief list of topics to be covered

- Functions
- Arrays
- Strings
- Pointers

SEMESTER – II

1. **Course number and name:** UCB008; Applied Chemistry
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the School of Chemistry and Biochemistry

4. **Text book, title, author, and year** **Text Books / Reference Books**

- Ramesh, S. and Vairam S. *Engineering Chemistry*, Wiley India (2012) 1sted.
- Jain, P.C. and Jain, M. *Engineering Chemistry*, Dhanpat Rai Publishing Co. (2005) 15thed.
- Puri, B.R., Sharma and L.R., Pathania, M.S. *Principles of Physical Chemistry*, Vishal Publishing Co. (2008).
- Brown, Holme, *Chemistry for engineering students*, Thompson, 1sted.
- Shulz, M.J. *Engineering Chemistry*, Cengage Learnings, (2007) 1sted.
 - a. Other supplemental materials
- Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Atomic Structure and Bonding: Chemical change; elements, compounds and mixtures, Atomic structure, dual nature of electron, concept of atomic orbitals, Pauli's Exclusion principle, Concept of chemical bonding: covalent, ionic, metallic, hydrogen bond, Vander Waal's, Hybridization and shapes of molecule, electronic structure and periodic table.

Chemical Equilibrium: Law of mass action, Factors that influence the position of equilibrium. Ionic equilibria: ionic equilibria in aqueous solutions; strong and weak acids and bases; buffer solution and indicators.

Electrochemistry: Migration of ions, Transference number, Specific, Equivalent and Molar Conductivity of electrolytic solutions, Conductometric titrations, Electrode potential and types of electrodes, Introduction to galvanic and concentration cells, Liquid junction potential.

Colligative Properties of Dilute Solutions: Depression of freezing point and elevation of boiling point.

Phase Rule: States of matter, Phase, Component and Degree of freedom, Gibbs phase rule, One component and two component systems.

Water Treatment and Analysis: Hardness and alkalinity of water: Units and determination, External and internal method of Softening of water: Lime-soda Process, Ion exchange process, Desalination of brackish water.

Fuels: Classification of fuels, Calorific value, Cetane and Octane number, fuel quality, Comparison of solid liquid and gaseous fuel, properties of fuel, alternative fuels: Biofuels, Power alcohol, Synthetic petrol.

Application of Atomic and Molecular Spectroscopic Methods: Structure determination of certain model compounds of industrial importance.

Assignments based on working and applications of advanced instruments will be given in the tutorial class.

Laboratory Work

Electrochemical measurements: Experiments involving use of pH meter, conductivity meter, potentiometer.

Acid and Bases: Determination of mixture of bases

Spectroscopic techniques: Colorimeter, UV-Vis spectrophotometer.

Kinetics: Kinetics of oxidation of iodine ion by peroxydisulphate ion.

Thermochemistry: Cloud point and pour point determination

Water and its treatment: Determination of hardness, alkalinity, chloride, chromium, iron and copper in aqueous medium.

b. Prerequisites

- None

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Analyse trends in periodic table with electronic and atomic structure.
- Interpret phase diagrams of pure and binary substances.
- Demonstrate the working of electrodes and their applications.
- Calculate various parameters defining water and fuel quality.
- Identify the various functional groups through IR spectra.
- Carry out basic experimental procedure and to emphasize need for safety and safety procedure in laboratory.

b Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Atomic Structure and Bonding
- Chemical Equilibrium
- Electrochemistry
- Colligative Properties of Dilute Solutions
- Phase Rule
- Water Treatment and Analysis and Fuels

1. **Course number and name:** UEE001; Electrical Engineering
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year****Text Books / Reference Books**
 - *Hughes, E., Smith, I.M., Hiley, J. and Brown, K., Electrical and Electronic Technology, PHI (2008).*
 - *Nagrath, I.J. and Kothari, D.P., Basic Electrical Engineering, Tata McGraw Hill (2002).*
 - *Naidu, M.S. and Kamashaiah, S., Introduction to Electrical Engineering, Tata McGraw Hill (2007).*
 - *Chakraborti, A., Basic Electrical Engineering, Tata McGraw-Hill (2008).*
 - *Del Toro, V., Electrical Engineering Fundamentals, Prentice-Hall of India Private Limited (2004)*
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

DC Circuits: Kirchhoff's Voltage and Current Laws; Power Dissipation; Voltage Source and Current Source; Mesh and Nodal Analysis; Star-Delta Transformation; Superposition Theorem; Thevenin's Theorem; Norton's Theorem; Maximum Power Transfer Theorem; Millman's Theorem and Reciprocity Theorem.

Steady state analysis of DC Circuits: The Ideal Capacitor, Permittivity; The Multi-Plate Capacitor, Variable Capacitor; Capacitor Charging and Discharging, Current-Voltage Relationship, Time-Constant, Rise-Time, Fall-Time; Inductor Energization and De-Energization, Inductance Current-Voltage Relationship, Time-Constant; Transient Response of RL, RC and RLC Circuits.

AC Circuits: Sinusoidal sources, RC, RL and RLC circuits, Concept of Phasor, Phasor representation of circuit elements, Complex notation representation, Single phase AC Series and parallel circuits, Power dissipation in AC circuits, Power factor correction, Resonance in Series and Parallel circuits, Balanced and unbalanced 3-phase circuit - voltage, current and power relations, 3-phase power measurement, Comparison of single phase and three phase supply systems.

Electromagnetism: Electromagnetic induction, Dot convention, Equivalent inductance, Analysis of Magnetic circuits, AC excitation of magnetic circuit, Iron losses, Fringing and stacking applications: solenoids and relays.

Single Phase Transformers: Constructional features of transformer, operating principle and applications, equivalent circuit, phasor analysis and calculation of performance indices.

Motors and Generators: DC motor operating principle, construction, energy transfer, speed-torque relationship, conversion efficiency, applications, DC generator operating principle, reversal of energy transfer, emf and speed relationship, applications.

Laboratory Work: Network laws and theorems, Measurement of R,L,C parameters, A.C. series and parallel circuits, Measurement of power in 3 phase circuits, Reactance calculation of variable reactance choke coil, Open circuit and Short circuit tests on single phase transformer, Starting of rotating machines.

- b. Prerequisites
 - UEC001
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. Specific goals for the course

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Apply networks laws and theorems to solve electric circuits.
 - Analyze transient and steady state response of DC circuits.
 - Signify AC quantities through phasor and compute AC system behavior during steady state.
 - Explain and analyze the behavior of transformer.
 - Elucidate the principle and characteristics of DC motor and DC generator.
- b. Specific outcomes addressed by the course.

D1	Share responsibility and information schedule with others in team.
D2	Participate in the development and selection of ideas.
E1	Classify information to identify engineering problems.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.
E3	Use analytical, computational and/or experimental methods to obtain solutions.

SO1.1	Ability to identify and formulate problems for electrical systems
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO5.1	Share responsibility and information schedule with others in team.
SO5.2	Participate in the development and selection of ideas on a team whose members together provide leadership.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- DC Network
- DC Transients
- AC Circuits
- Electromagnetism
- Single Phase Transformers
- DC Motors and Generators

1. **Course number and name:** UEN002; Energy and Environment
2. **Credits and contact hours:** 3.0 and 3
3. **Instructor's or course coordinator's name:** Faculty from the School of Energy and Environment
4. **Text book, title, author, and year****Text Books / Reference Books**
 - *Bharucha, E., Textbook of Environmental Studies, Universities Press (2005).*
 - *Chapman, J.L. and Reiss, M.J., Ecology-Principles and Application, Cambridge University Press (LPE) (1999).*
 - *Joseph, B., Environmental Studies, Tata McGraw-Hill (2006).*
 - *Eastop, T.P. and Croft, D.R. Energy Efficiency for Engineers and Technologists, Longman and Harrow (2006).*
 - *Miller, G.T., Environmental Science- Working with Earth, Thomson (2006).*
 - *Wright, R.T., Environmental Science-Towards a sustainable Future, Prentice Hall (2008) 9thed.*
 - *O'Callagan, P.W., Energy Management, McGraw Hill Book Co. Ltd. (1993).*
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Environment pollution, global warming and climate change: Air pollution (local, regional and global); Water pollution problems; Land pollution and food chain contaminations; Carbon cycle, greenhouse gases and global warming; Climate change – causes and consequences; Carbon footprint; Management of greenhouse gases at the source and at the sinks.

Ecology, Structure and functioning of natural ecosystems: Ecology, ecosystems and their structure, functioning and dynamics; Energy flow in ecosystems; Biogeochemical cycles and climate; Population and communities.

Natural resources: Human settlements and resource consumption; Biological, mineral and energy resources; Land, water and air; Natural resources vis-à-vis human resources and technological resources; Concept of sustainability; Sustainable use of natural resources.

Agricultural, industrial systems and environment: Agricultural and industrial systems vis-à-vis natural ecosystems; Agricultural systems, and environment and natural resources; Industrial systems and environment.

Energy technologies and environment: Electrical energy and steam energy; Fossil fuels, hydropower and nuclear energy; Solar energy, wind energy and biofuels; Wave, ocean thermal, tidal energy and ocean currents; Geothermal energy; Future energy sources; Hydrogen fuels; Sustainable energy.

Group assignments: Assignments related to Sanitary landfill systems; e-waste management; Municipal solid waste management; Biodiversity and biopiracy; Air pollution control systems; Water treatment systems; Wastewater treatment plants; Solar heating systems; Solar power plants; Thermal power plants; Hydroelectric power plants; Biofuels; Environmental status assessments; Energy status assessments, etc.
 - b. Prerequisites

- UPH004
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required
- 6. Specific goals for the course**
 - a. Specific outcomes of instructions.
After the completion of the course, the students will be able to:
 - Correlate major local and regional environmental issues with changes in ecology and human health.
 - Monitor and document the development and dynamics of ecosystems in experimental or natural microcosms.
 - Define and document local resource consumption patterns and conservation strategies.
 - Define opportunities available for energy conservation and for use of renewable energy resources in local and regional entities.
 - b. Specific outcomes addressed by the course.

D1	Share responsibility and information schedule with others in team.
F2	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and well-being of society.
H3	Evaluate engineering solutions considering environmental constraints.
J2	Recognize the impact of engineering decisions on energy resources and environment.

SO4	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
SO4.1	Recognize the impact of engineering decisions on energy resources and environment.
SO4.2	Evaluate engineering solutions considering environmental constraints.
SO5.1	Share responsibility and information schedule with others in team.

- 7. Brief list of topics to be covered**
 - Environment pollution, global warming and climate change
 - Ecology, Structure and functioning of natural ecosystems
 - Natural resources
 - Agricultural, industrial systems and environment
 - Energy technologies and environment
 - Group assignments

1. **Course number and name:** UES009; Mechanics
2. **Credits and contact hours:** 2.5 and 3
3. **Instructor's or course coordinator's name:** Faculty from the Department of Civil Engineering
4. **Text book, title, author, and year** **Text Books / Reference Books**

- Shames, I. H. *Engineering Mechanics: Dynamics*, Pearson Education India (2002).
- Beer, Johnston, Clausen and Staab, *Vector Mechanics for Engineers, Dynamics*, McGraw-Hill Higher Education (2003).
- Hibler, T.A., *Engineering Mechanics: Statics and Dynamics*, Prentice Hall (2012).
- Timoshenko and Young, *Engineering Mechanics*, Tata McGraw Hill Education Private Limited (2000).
 - a. Other supplemental materials
- Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Review of Newton's law of motion and vector algebra

Equilibrium of bodies: Free-body diagrams, conditions of equilibrium, torque due to a force, statical determinacy.

Plane trusses: Forces in members of a truss by method of joints and method of sections.

Friction: Sliding, belt, screw and rolling.

Properties of plane surfaces: First moment of area, centroid, second moment of area etc.

Virtual work: Principle of virtual work, calculation of virtual displacement and virtual work.

Work and energy: Work and energy, work-energy theorem, principle of conservation of energy, collisions, principles of momentum etc.

Dynamics of Rigid Bodies: Newton's Laws, D'Alembert's Principle, Energy Principles.

Experimental project assignment/ Micro project: Students in groups of 4/5 will do project on

Model Bridge Experiment: This will involve construction of a model bridge using steel wire and wood.

- b. Prerequisites

- UTA015

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. **Specific goals for the course**

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Determine resultants in plane force systems.
- Identify and quantify all forces associated with a static framework.
- Solve problems in kinematic and dynamic systems.

b. Specific outcomes addressed by the course.

C2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Plane trusses
- Friction
- Virtual work
- Work and energy
- Dynamics of rigid bodies

1. **Course number and name:** UMA004; Mathematics - II
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from the School of Mathematics
4. **Text book, title, author, and year****Text Books / Reference Books**
 - Simmons, G.F., *Differential Equations (With Applications and Historical Notes)*, Tata McGraw Hill (2009).
 - Krishnamurthy, V.K., Mainra, V.P. and Arora, J.L., *An introduction to Linear Algebra*, Affiliated East West Press (1976).
 - Kreyszig Erwin, *Advanced Engineering Mathematics*, John Wiley (2006), 8th ed.
 - Jain, R.K. and Iyenger, S.R.K., *Advanced Engineering Mathematics*, Narosa Publishing House(2011), 11th ed.
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Linear Algebra: Row reduced echelon form, Solution of system of linear equations, Matrix inversion, Linear spaces, Subspaces, Basis and dimension, Linear transformation and its matrix representation, Eigen-values, Eigen-vectors and Diagonalisation, Inner product spaces and Gram-Schmidt orthogonalisation process.

Ordinary Differential Equations: Review of first order differential equations, Exact differential equations, Second and higher order differential equations, Solution techniques using one known solution, Cauchy - Euler equation, Method of undetermined coefficients, Variation of parameters method, Engineering applications of differential equations.

Laplace Transform: Definition and existence of Laplace transforms and its inverse, Properties of the Laplace transforms, Unit step function, Impulse function, Applications to solve initial and boundary value problems.

Fourier Series: Introduction, Fourier series on arbitrary intervals, Half range expansions, Applications of Fourier series to solve wave equation and heat equation

- b. Prerequisites
 - UMA003
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. **Specific goals for the course**

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Solve the differential equations of first and second order and basic application problems described by these equations.
- Find the Laplace transformations and inverse Laplace transformations for various functions. Using the concept of Laplace transform students will be able to solve the initial value and boundary value problems.

- Find the Fourier series expansions of periodic functions and subsequently will be able to solve heat and wave equations.
 - Solve systems of linear equations by using elementary row operations.
 - Identify the vector spaces/subspaces and to compute their bases/orthonormal bases. Further, students will be able to express linear transformation in terms of matrix and find the eigen values and eigen vectors.
- b Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
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SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
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7. Brief list of topics to be covered

- Linear Algebra
- Ordinary Differential Equations
- Laplace Transform
- Fourier Series

Course Syllabi: UTA013 Engineering Design Project-I (L : T : P :: 1 : 0 : 2)

1. **Course number and name:** UTA0113: Engineering Design Project-I
2. **Credits and contact hours:** 3 and 5.0
3. **Instructor's or course coordinator's name:** Faculty from the Department of Mechanical Engineering and Electronics & Communication Engineering
4. **Text book, title, author, and year****Text Books / Reference Books**
 - *Michael McRoberts, Beginning Arduino, Technology in action publications.*
 - *Alan G. Smith, Introduction to Arduino: a piece of cake, CreateSpace Independent Publishing Platform (2011)*
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Breakup of lecture details to be taken up by MED:

Lec No.	Topic	Contents
Lec 1	Introduction	The Mangonel Project. History. Spreadsheet.
Lec 2	PROJECTILE MOTION	no DRAG, Design spread sheet simulator for it.
Lec 3	PROJECTILE MOTION	with DRAG, Design spread sheet simulator for it.
Lec 4	STRUCTURES FAILURE	STATIC LOADS
Lec 5	STRUCTURES FAILURE	DYNAMIC LOADS
Lec 6	REDESIGNING THE MANGONEL	Design constraints and limitations of materials for redesigning the Mangonel for competition as a group.
Lec 7	MANUFACTURING	Manufacturing and assembling the Mangonel.
Lec 8	SIMULATION IN ENGINEERING DESIGN	Simulation as an Analysis Tool in Engineering Design.
Lec 9	ROLE OF MODELLING & PROTOTYPING	The Role of Modelling in Engineering Design.

Breakup of lecture details to be taken up by ECED:

Lec No.	Topic	Contents
Lec 1-5	Digital Electronics	Prototype, Architecture, Using the Integrated Development Environment (IDE) to Prepare an Arduino Sketch, structuring an Arduino Program, Using Simple Primitive Types (Variables), Simple programming examples. Definition of a sensor and actuator.

Tutorial Assignment / Laboratory Work:

Associated Laboratory/Project Program: T- Mechanical Tutorial, L- Electronics Laboratory, W- Mechanical Workshop of "Mangonel" assembly, redesign, operation and reflection.

Title for the weekly work in 15 weeks	Code
Using a spread sheet to develop a simulator	T1
Dynamics of projectile launched by a Mangonel - No Drag	T2
Dynamics of projectile launched by a Mangonel - With Drag	T3
Design against failure under static actions	T4
Design against failure under dynamic actions	T5
Electronics hardware and Arduino controller	L1
Electronics hardware and Arduino controller	L2
Programming the Arduino Controller	L3
Programming the Arduino Controller	L4
Final project of sensors, electronics hardware and programmed Arduino, controller based measurement of angular velocity of the “Mangonel” throwing arm.	L5
Assembly of the Mangonel by group	W1
Assembly of the Mangonel by group	W2
Innovative redesign of the Mangonel and its testing by group	W3
Innovative redesign of the Mangonel and its testing by group	W4
Final inter group competition to assess best redesign and understanding of the “Mangonel”.	W5

Project: The Project will facilitate the design, construction and analysis of a “Mangonel”. In addition to some introductory lectures, the content of the students’ work during the semester will consist of:

1. the assembly of a Mangonel from a Bill of Materials (BOM), detailed engineering drawings of parts, assembly instructions, and few prefabricated parts.
2. the development of a software tool to allow the trajectory of a “missile” to be studied as a function of various operating parameters in conditions of no-drag and drag due to air;
3. a structural analysis of certain key components of the Mangonel for static and dynamic stresses using values of material properties which will be experimentally determined;
4. the development of a micro-electronic system to allow the angular velocity of the throwing arm to be determined;
5. testing the Mangonel;
6. redesigning the throwing arm of the Mangonel to optimise for distance without compromising its structural integrity;
7. an inter-group competition at the end of the semester with evaluation of the group redesign strategies.

b. Prerequisites

- UTA015

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- simulate trajectories of a mass with and without aerodynamic drag using a spreadsheet-based

- software tool to allow trajectories be optimized;
 - perform a test to acquire an engineering material property of strength in bending and analyze the throwing arm of the “Mangonel” under conditions of static and dynamic loading;
 - develop and test software code to process sensor data;
 - design, construct and test an electronic hardware solution to process sensor data;
 - construct and operate a Roman catapult “Mangonel” using tools, materials and assembly instructions, in a group, for a competition;
 - operate and evaluate the innovative redesign of elements of the “Mangonel” for functional and structural performance.
- b Specific outcomes addressed by the course.

F2	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and well-being of society.
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SO2.3	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.
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Course Syllabi: UTA018 Object Oriented Programming (L : T : P :: 3 : 0 : 2)

1. **Course number and name:** UTA018; Object Oriented Programming
2. **Credits and contact hours:** 4.0 and 5
3. **Instructor's or course coordinator's name:** Faculty from the Department of Computer Science and Engineering
4. **Text book, title, author, and year** **Text Books / Reference Books**

- “Balaguruswamy”, *Object Oriented Programming with C++ 6th Edition*, Tata McgrawHill, 2013.
- “Bruce Eckel”, *Thinking in C++*, Prentice-Hall of India Pvt. Ltd, 2000.
- “Joyce Farrell”, *Object Oriented Programming Using C++ 4th Edition*, Cengage Learning, 2013.

- a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Object Oriented Programming with C++: Class declaration, creating objects, accessing objects members, nested member functions, memory allocation for class, objects, static data members and functions. Array of objects, dynamic memory allocation, this pointer, nested classes, friend functions, constructors and destructors, constructor overloading, copy constructors, operator overloading and type conversions.

Inheritance and Polymorphism: Single inheritance, multi-level inheritance, multiple inheritance, runtime polymorphism, virtual constructors and destructors.

File handling: Stream in C++, Files modes, File pointer and manipulators, type of files, accepting command line arguments.

Templates and Exception Handling: Use of templates, function templates, class templates, handling exceptions.

Introduction to Windows Programming in C++: Writing program for Windows, using COM in Windows Program, Windows Graphics, User Input

Laboratory work: to implement Programs for various kinds of programming constructs in C++ Language.

- b. Prerequisites
 - UTA017
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. Specific goals for the course

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Write, compile and debug programs in C++, use different data types, operators and I/O function in a computer program.
- Comprehend the concepts of classes, objects and apply basics of object-oriented programming, polymorphism and inheritance.
- Demonstrate use of file handling.

- Demonstrate use of templates and exception handling.
- Demonstrate use of windows programming concepts using C++.

b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
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SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
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7. Brief list of topics to be covered

- Class
- Object
- Inheritance
- Polymorphism
- File handling
- Exception handling

SEMESTER – III

1. **Course number and name:** UEE304; Transmission and Distribution of Electric Power
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Chakrabarti, A., Soni, M.L., Gupta, P.V. and Bhatnagar, U.S., *a Text Book on Power System Engineering*, Dhanpat Rai (2008).
- Wadhwa, C.L., *Electrical Power Systems*, New Age International (P) Limited, Publishers (2008).
- Gupta, B.R., *Power System Analysis and Design*, S. Chand (2009).
- Nagrath, I.J. and Kothari, D.P., *Power System Engineering*, Tata McGraw-Hill (2007).
- Pabla, A.S., *Electric Power Distribution*, McGraw Hill (2008).
- Stevenson, W.D., *Power System Analysis*, McGraw-Hill (2007).

a. Other supplemental materials

- Nil

5. **Specific course information**

a. Brief description of the content of the course (catalog description)

Introduction: Structure of power systems, Growth of power systems-Indian overview, Interconnections and their advantages.

Transmission Line Parameters: Choice of voltage and frequency, Types of conductor, Size of conductor, Resistance, Inductance and capacitance of single phase and three phase transmission lines.

Mechanical design of overhead transmission lines: Tension and sag calculations, Factors affecting Sag, Sag template, Stringing charts, Vibrations and vibration damper.

Insulators: Insulator types, String efficiency, Improvement of String Efficiency Grading rings, Insulator arc Failure, Arcing horns, Armored rods and Bushing.

Transmission Line Performance: Characteristics and performance of power transmission lines: short, medium, long lines, generalized constants, power flow, regulation, Power circle diagrams, Series and shunt compensation, Corona visual and disruptive, Critical voltage, Phenomenon of Corona, Corona loss, Factors affecting Corona, Ferranti Effect, Electrostatic and Electromagnetic interference with communication lines.

Insulated Cables: Constructional features, Parameters, Cable laying procedures, Fault location Methods, High voltage cables, Thermal characteristics, Ratings of Cables, Introduction to XLPE cables.

Distribution Systems: Classification of distribution system, Primary and secondary distribution, Ring main and radial systems, Systematic design of distribution systems.

EHV transmission and HVDC transmission: Need of EHV transmission system, types of DC links, advantages of DC transmission, HVDC systems in India.

b. Prerequisites

- UEE001

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Analyse the transmission line models and evaluate its performance parameters.
- Design the transmission lines under various working conditions.
- Describe and select the configurations of different line insulators and evaluate their performance.
- Supervise the laying of cables and fault detection in cables.
- Design the distribution system network.

- b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
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SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
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7. Brief list of topics to be covered

- Transmission Line Parameters
- Mechanical design of overhead transmission lines
- Insulators
- Transmission Line Performance
- Insulated Cables
- EHV transmission and HVDC transmission

1. **Course number and name:** UEE305; Electromagnetics Field and Waves
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Hayt, W.H., *Engineering Electromagnetics*, Tata McGraw-Hill (2008).
- Kraus, J.D., *Electromagnetics*, McGraw-Hill (2006).
- Sadiku, M.N.O, *Elements of Electromagnetics*, Oxford University Press (2009).
- Jordan, E.C. and Balmain K.G., *Electromagnetic Waves and Radiating Systems*, Prentice Hall of India (2008).
- Paramanik, A, *Electromagnetism: Theory and Applications*, Prentice-Hall of India (2006).
 - a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Vector Analysis: Review of vector algebra, Review of cartesian, Cylindrical and spherical coordinate systems, Introduction to ∇ (operator, Use of ∇ operator as gradient, divergence, curl).

Electrostatic fields: Introduction to coulomb's law, Gaussian law and its applications in determination of field of spherical and cylindrical geometries, Laplace's and poisson's equation in various coordinate systems. Effect of dielectric on capacitance, Boundary conditions at electric interfaces, Method of images and its applications.

Magnetostatics: Introduction to ampere's law, Magnetic vector potential, Magnetic forces, Boundary conditions at magnetic interfaces.

Time Varying Fields and Maxwell's Equations: Continuity of charge, Concept of displacement current, Maxwell's equation in integral and differential form: for static fields, for time varying fields, for free space, for good conductors, for harmonically varying fields, Poynting theorem: Energy stored and radiated power, Complex poynting vector, Properties of conductor and dielectrics, Wave equations for free space, Wave equations for conductors.

Uniform Plane Waves: Introduction, Uniform plane wave propagation: Wave equations, Transverse nature of uniform plane waves, Perpendicular relation between $\vec{E}$ and $\vec{H}$, EM waves in charge free, Current free dielectric, Reflection by ideal conductor: Normal incidence, reflection and transmission with normal incidence at another dielectric, Plane wave in lossy dielectric, Wave impedance and propagation constant, Depth of penetration, Surface impedance and surface resistance, Application of EM propagation through Transmission Lines and Rectangular Waveguides

- b. Prerequisites

- UPH004

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Appraise need analysis for different coordinate systems in electromagnetics and their interrelations.
 - Apply vector calculus to solve field theory problems.
 - Calculate electric and magnetic fields in different coordinates for various charge and current configurations.
 - Exhibit the concept of time varying fields.
 - Demonstrate different aspects of plane wave in dielectric and conducting media.
 - Realize the analogy of wave with transmission line and determine the transmission line performance.
- b. Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
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SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
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7. Brief list of topics to be covered

- Vector Analysis
- Electrostatic fields
- Magnetostatics
- Time Varying Fields and Maxwell's Equations
- Uniform Plane Waves

1. **Course number and name:** UES010; Solids and Structures
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from Department of Civil Engineering
4. **Text book, title, author, and year****Text Books / Reference Books**
 - Popov, E.P. and Balan, T.A., *Engineering Mechanics of Solids*, Prentice Hall of India (2012).
 - Singh, D.K., *Mechanics of Solids*, Pearson Education (2008).
 - Shames, I. H. and Pitarresi, J. M., *Solid Mechanics*, Prentice Hall of India (1996).
 - Crandall, S.H., Dahl, N.C. and Lardner, T.J., *An Introduction to Mechanics of Solids*, McGraw Hill International, Tokyo (1969).
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Elastic Plastic Behavior

Axial Stress and Strain: Concept of stress, strain, elasticity and plasticity; one-dimensional stress-strain relationships; Young's modulus of elasticity, shear modulus and Poisson's ratio; two-dimensional elasticity; isotropic and homogeneous materials; ductile and brittle materials; statically determinate and indeterminate problems, compound and composite bars; thermal stresses. Torsion of shafts; buckling of struts, concept of factor of safety.

Shear Force and Bending Moment Diagrams: Types of load on beams, classification of beams; axial, shear force and bending moment diagrams: simply supported, overhung and cantilever beams subjected to any combination of point loads, uniformly distributed and varying load and moment, equation of condition, load function equation, qualitative analysis for two-dimensional frames.

Bending & Shear Stresses in beams: Derivation of flexural formula for straight beams, concept of second moment of area, bending stress calculation for beams of simple and built up sections, Flitched beams. Shear stress formula for beams, shear stress distribution in beams

Transformation of Stress and Strain: Transformation equations for plane stress and plane strain, Mohr's stress circle, relation between elastic constants, strain measurements, strain rosettes.

Deformations: Governing differential equation for deflection of straight beams having constant flexural rigidity, double integration and Macaulay's methods for slopes and deflection, unit load method for deflection of trusses

Laboratory Work

Experimental Project Assignment: Students in groups of 4/5 will do projects:

1. Calculation of tensile strength using UTM.
2. Buckling of struts.
3. Experimental verification of Theory of bending (calculation of bending stress and deflections at various points in the beam theoretically and verifying the same experimentally) and indirect evaluation of the modulus of elasticity.
4. Torsion: Study the behavior of circular shafts under torsion and analysis of failure and indirect evaluation of the modulus of rigidity.

Micro Project:

Model Bridge Experiment: This will involve construction of a model bridge using steel wire and wood.

b. Prerequisites

- UES009

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

a. Specific outcomes of instruction

After completion of this course, the students will be able to:

- Evaluate axial stresses and strains in various determinate and indeterminate structural systems.
- Draw Shear Force Diagram and Bending Moment Diagram in various kinds of beams subjected to different kinds of loads.
- Calculate load carrying capacity of columns and struts and their buckling strength.
- Evaluate various kinds of stresses (axial, bending, torsional and shearing) in various structural elements due to different type of external loads.
- Determine deformations and deflections in various kinds of beams and trusses.

b. Specific outcomes addressed by the course.

B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.

SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Axial stress and strain
- Shear force
- Bending moment
- Bending and shear stress in beams

1. **Course number and name:** UES011; Thermo Fluids
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from Department of Civil Engineering
4. **Text book, title, author, and year**

Text Books / Reference Books

- Kumar, D. S, Fluid Mechanics and Fluid Power Engineering, S. K. Kataria (2009)
 - Cengel and Boles, Thermodynamics: an Engineering Approach, McGraw-Hill (2011)
 - Jain, A. K. , Fluid Mechanics: including Hydraulic Machines, Khanna Publishers (2003)
 - Rao, Y.V. C, An Introduction to Thermodynamics, Universities Press (2004)
- b. Other supplemental materials
- Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Fluid Mechanics

Introduction: Definition of a fluid and its properties

Hydrostatics: Measurement of pressure, thrust on submerged surfaces

Principles of Fluid Motion: Description of fluid flow; continuity equation; Euler and Bernoulli equations; Pitot total head and static tubes, venturi-meter, orifice-meter, rotameter; Momentum equation and its applications

Pipe Flow: Fully developed flow; laminar pipe flow; turbulent pipe flow, major and minor losses; Hydraulic gradient line (HGL) and total energy line (TEL)

Boundary Layer: Boundary layer profile; displacement, momentum and energy thickness

Thermodynamics

Introduction: Properties of matter, the state postulate, energy, processes and thermodynamic systems;

Properties of Pure Substances: property tables, property diagrams, phase change, equations of state (ideal gas);

Energy: Energy transfer by heat, work and mass;

First Law of Thermodynamics: Closed system, open system, steady-flow engineering devices;

Second Law of Thermodynamics: Statements of the Second Law, heat engines, refrigeration devices, reversible versus irreversible processes, the Carnot cycle.

Laboratory/Project programme

List of Experiments

1. Verification of Bernoulli's theorem
2. Determination of hydrostatic force and its location on a vertically immersed surface
3. Determination of friction factor for pipes of different materials
4. Determination of loss coefficients for various pipe fittings

5. Verification of momentum equation
6. Visualization of laminar and turbulent flow, and rotameter
7. Calibration of a venturi-meter
8. Boundary layer over a flat plate

Sample List of Micro-Projects

Students in a group of 4/5 members will be assigned a micro project.

1. Design a physical system to demonstrate the applicability of Bernoulli's equation.
2. Determine the pressure distribution around the airfoil body with the help of wind tunnel.
3. Demonstrate the first law of thermodynamics for an open system, for example: a ordinary hair dryer.
4. Develop a computer program for solving pipe flow network.

b. Prerequisites

- UPH004
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
- Required

6. Specific goals for the course

a. Specific outcomes of instructions.

After the completion of this course, the students will be able to:

- Analyze and solve problems of simple fluid-based engineering systems including pressures and forces on submerged surfaces.
- Analyze fluid flow problems with the application of the mass, momentum and energy equations.
- Evaluate practical problems associated with pipe flow systems.
- Conceptualize and describe practical flow systems such as boundary layers and their importance in engineering analysis.
- Estimate fluid properties and solve basic problems using property tables, property diagrams and equations of state.
- Analyze and solve problems related to closed systems and steady-flow devices by applying the conservation of energy principle.
- Analyze the second law of thermodynamics for various systems and to evaluate the performance of heat engines, refrigerators and heat pumps.

b. Specific outcomes addressed by the course.

B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.

SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Hydrostatics
- Pipe flow
- Fluid mechanics
- Thermostat

1. **Course number and name:** UMA031; Optimization Techniques
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from the School of Mathematics
4. **Text book, title, author, and year****Text Books / Reference Books**
 - Chandra, S., Jayadeva, Mehra, A., *Numerical Optimization and Applications*, Narosa Publishing House, (2013).
 - Taha H.A., *Operations Research-An Introduction*, PHI (2007).
 - Pant J. C., *Introduction to optimization: Operations Research*, Jain Brothers (2004)
 - Bazaarra Mokhtar S., Jarvis John J. and Shirali Hanif D., *Linear Programming and Network flows*, John Wiley and Sons (1990)
 - Swarup, K., Gupta, P. K., Mammohan, *Operations Research*, Sultan Chand & Sons, (2010).
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Scope of Operations Research: Introduction to linear and non-linear programming formulation of different models.

Linear Programming: Geometry of linear programming, Graphical method, Linear programming (LP) in standard form, Solution of LP by simplex method, Exceptional cases in LP, Duality theory, Dual simplex method, Sensitivity analysis.

Integer Programming: Branch and bound technique.

Transportation and Assignment Problem: Initial basic feasible solutions of balanced and unbalanced transportation/assignment problems, Optimal solutions.

Project Management: Construction of networks, Network computations, Floats (free floats and total floats), Critical path method (CPM), Crashing.

Game Theory: Two-person zero-sum game, Game with mixed strategies, Graphical method and solution by linear programming.
 - b. Prerequisites
 - UMA003
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required
6. **Specific goals for the course**
 - a. Specific outcomes of instruction

After completion of this course, the students will be able to:

 - Formulate and solve linear programming problems.
 - Solve the transportation and assignment problems
 - Solve the project management problems using cpm.
 - Solve two-person zero-sum games.

b Specific outcomes addressed by the course.

A3	Apply numerical and statistical methods in analyzing data
D2	Participate in the development and selection of ideas.
I2	Ability to identify directions for continuing education opportunities.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO5.2	Participate in the development and selection of ideas on a team whose members together provide leadership.
SO7.2	Ability to identify directions for continuing education opportunities.

7. Brief list of topics to be covered

- Scope of Operations Research
- Linear Programming
- Integer Programming
- Transportation and Assignment Problem
- Project Management
- Game Theory

1. **Course number and name:** UTA002: Manufacturing Processes
2. **Credits and contact hours:** 3.5 and 5
3. **Instructor's or course coordinator's name:** Faculty from the Department of Mechanical Engineering
4. **Text book, title, author, and year****Text Books / Reference Books**
 - Chandra, S., Jayadeva, Mehra, A., *Numerical Optimization and Applications*, Narosa Publishing House, (2013).
 - Taha H.A., *Operations Research-An Introduction*, PHI (2007).
 - Pant J. C., *Introduction to optimization: Operations Research*, Jain Brothers (2004)
 - Bazaarra Mokhtar S., Jarvis John J. and Shirali Hanif D., *Linear Programming and Network flows*, John Wiley and Sons (1990)
 - Swarup, K., Gupta, P. K., Mammohan, *Operations Research*, Sultan Chand & Sons, (2010).
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Machining Processes: Principles of metal cutting, Cutting tools, Cutting tool materials and applications, Geometry of single point cutting tool, Introduction to multi-point machining processes – milling, drilling and grinding, Tool Life, Introduction to computerized numerical control (CNC) machines, G and M code programming for simple turning and milling operations, introduction of canned cycles.

Metal Casting: Principles of metal casting, Introduction to sand casting, Requisites of a sound casting, Permanent mold casting processes.

Metal Forming: Forging, Rolling, Drawing, Extrusion, Sheet Metal operations.

Joining Processes: Electric arc, Resistance welding, Soldering, Brazing.

Laboratory Work:

Relevant shop floor exercises involving practices in Sand casting, Machining, Welding, Sheet metal fabrication techniques, CNC turning and milling exercises, Experiments on basic engineering metrology and measurements to include measurements for circularity, ovality, linear dimensions, profiles, radius, angular measurements, measurement of threads, surface roughness.

Basic knowledge and derivations related to above measurements, uncertainties, statistical approaches to estimate uncertainties, Line fitting, static and dynamic characteristics of instruments will be discussed in laboratory classes.

- b. Prerequisites
 - UES009
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. Specific goals for the course

a. Specific outcomes of instruction

After the completion of this module, students will be able to:

- Develop simple CNC code, and use it to produce components while working in groups.
- Analyse various machining processes and calculate relevant quantities such as velocities, forces.
- Recognize cutting tool wear and identify possible causes and solutions.
- Understand the basic principle of bulk and sheet metal forming operations for analysis of forces.
- Analyse various shearing operations for tooling design.
- Apply the knowledge of metal casting for different requirements.
- Analyse and understand the requirements to achieve sound welded joint while welding different similar and dissimilar engineering materials.

b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
F2	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and well-being of society.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.3	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.

7. Brief list of topics to be covered

- Machining process
- Metal casting
- Metal forming

1. **Course number and name:** UTA014; Engineering Design-II
2. **Credits and contact hours:** 6 and 5
3. **Instructor's or course coordinator's name:** Faculty from the Department of Mechanical Engineering and Electronics & Communication Engineering
4. **Text book, title, author, and year****Text Books / Reference Books**
 - Michael McRoberts, *Beginning Arduino, Technology in action publications.*
 - Alan G. Smith, *Introduction to Arduino: a piece of cake, CreateSpace Independent Publishing Platform (2011)*
 - John Boxall, *Arduino Workshop - a Hands-On Introduction with 65 Projects, No Starch Press; 1 edition (2013).*
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Arduino Microcontroller: Features of Arduino Microcontroller, Architecture of Arduino, Different boards of Arduino, Arduino Interfacing and Applications, Anatomy of an Interactive Device like Sensors and Actuators, a to D converters and their comparison, Blinking an LED, LCD Display, Driving a DC and stepper motor, Temperature sensors, Serial Communications, Sending Debug Information from Arduino to Your Computer, Sending Formatted Text and Numeric Data from Arduino, Receiving Serial Data in Arduino, Sending Multiple Text Fields from Arduino in a Single Message, Receiving Multiple Text Fields in a Single Message in Arduino. Light controlling with PWM.

Introduction to ARM processor: Features of ARM processor, ARM Architecture, Instruction set, ARM Programming

Programming of Arduino: The Code designing step by step. Taking a Variety of Actions Based on a Single Variable, Comparing Character and Numeric Values, Comparing Strings, Performing Logical Comparisons, Performing Bitwise Operations, Combining Operations and Assignment, Using Embedded techniques to program Arduino microcontroller, Understanding the libraries of Arduino programming language and applying for circuit design

Laboratory work: Introduction to Arduino board. Programming examples of Arduino board. Interfacing of LED, seven segment display, ADC and DAC with Arduino board. Introduction to ARM processor kit.

Projects: Arduino and ARM based projects to be allocated by concerned faculty.

- b. Prerequisites
 - None
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. **Specific goals for the course**

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Understand of features of Arduino board.
 - Analyze of internal Architecture of Arduino board.
 - Apply Arduino board programming concepts.
 - Design and implement Buggy project based on different goals and challenges defined.
- b Specific outcomes addressed by the course.

F1	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
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SO3.3	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
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7. Brief list of topics to be covered

- ARM Processor
- Arduino Microcontroller
- Programming of Arduino

SEMESTER – IV

1. **Course number and name:** UEE301; Direct Current Machines and Transformers
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Bimbhra, P.S., *Electrical Machinery*, Khanna Publishers (2008).
- Mukherjee, P.K. and Chakravorty, S., *Electrical Machines*, Dhanpat Rai (2004).
- Nagrath, I.J. and Kothari, D.P., *Electric Machines*, Tata McGraw Hill (2004).
- Bimbhra, P.S., *Generalized Theory of Electrical Machines*, Khanna Publishers (2007).
- Toro, Vincert, *Electromechanical Devices for Energy Conversion*, Prentice Hall of India (2004).
- Fitzgerald, A.E., Kingsley, C. Jr. and Umans, Stephen, *Electric Machinery*, McGraw Hill (2002).
- a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

General Concepts of Rotating Electrical Machines: Electromagnetic torque, Reluctance torque, Constructional features of rotating electrical machines, Classifications of rotating electrical machines, Construction of DC machines.

DC Generators: Classification of DC generator, Armature reaction, compensating windings, Commutation, Methods of improving commutation, Characteristic of DC generators, Voltage buildup of shunt generators, Voltage regulation, Parallel operation of DC generators, Condition for maximum efficiency, Applications of DC generators.

DC Motors: Characteristic of DC motors, Speed control of DC motors, Ward-Leonard control (Voltage control), Three-point starter, four-point starter, DC shunt motor starter design, Electric breakings of DC shunt and series motors, Condition for maximum mechanical power, Testing of DC machines: Brake test, Swinburne's test, Hopkinson's test or back-to-back test, Retardation test or Running test, Field's test, Applications of DC motors.

Single Phase Transformers: Introduction, Basic principle, Types of transformers, Construction, Equivalent circuit, Open circuit and short circuit, Separation of core losses, Per unit representation, Voltage regulation of a transformer, Losses in a transformer, Efficiency of a transformer, Condition for maximum efficiency, All day efficiency, Polarity test of a single-phase transformer, Sumpner's test, Parallel operation, Auto transformer.

Three-Phase Transformer: Advantages of three phase transformer, Principle of operation, Construction, Three-phase transformer connections, Open delta or V-V connection, Scott connection or T-T connection, Three-phase to two-phase conversion, Three-phase to six-phase conversion, Three-winding transformer, Parallel operation of transformers.

Special Purpose Transformers: Instrument transformers (CT and PT), Earthing transformer, Pulse transformer, High frequency transformer, Converter transformer.

Laboratory Work: DC Machines: Characteristics of generators and motors, Speed control, Efficiency, DC generators in parallel. Transformers: Open and short circuit tests, Parallel

operation, Harmonics in no-load current, Three-phase connections, 3-phase to 2-phase and 6-phase conversions.

b. Prerequisites

- UEE001

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Test the transformer and calculate its efficiency and performance in distribution system.
- Compare the performance of auto-transformer with that of two winding transformers.
- Use special purpose transformer for measurement and protection.
- Compute the performance of DC motors and generators in various modes.
- Explain the advantages of increasing load with parallel operation.
- Explain the speed control and starting methods of DC motors for specific purpose(s).

b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
B1	Identify the constraints, assumptions and models for the experiments.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.
D1	Share responsibility and information schedule with others in team.
E1	Classify information to identify engineering problems.
K1	Able to operate modern engineering equipment.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
E2	Develop appropriate models to formulate solutions.
E3	Use analytical, computational and/or experimental methods to obtain solutions.

SO1.1	Ability to identify and formulate problems for electrical systems
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO5.1	Share responsibility and information schedule with others in team.
SO6.1	Identify the constraints, assumptions and models for the experiments.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- General Concepts of Rotating Electrical Machines
- DC Generators
- DC Motors
- Single Phase Transformers
- Three-Phase Transformer
- Special purpose transformers

Course Syllabi: UEE406: Power System Practices (L : T : P :: 3 : 0 : 0)

1. **Course number and name:** UEE406; Power System Practices
2. **Credits and contact hours:** 3 and 3
1. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
3. **Text book, title, author, and year**
 - Kothari D. P., Nagrath I. J., *Modern Power System Analysis*, Tata McGraw Hill Education Private Limited (2009).
 - Shahedepour M., Yamin H., Zuyi Li., *Market operations in power systems: Forecasting, Scheduling, and Risk Management*, John Wiley & Sons, New York.
 - a. Other supplemental materials
 - Nil
4. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Energy Scenario: Energy needs of growing economy, Long term energy scenario, Energy pricing, Energy sector reforms, Energy and environment: Air pollution, Climate change, Energy security, Energy conservation and its importance, Energy strategy for the future, Energy conservation Act- 2001 and its features.

Energy Management and Audit: Definition, Energy audit- need, Types of energy audit, Energy management (audit) approach-understanding energy costs, Bench marking, Energy performance, Matching energy use to requirement, maximizing system efficiencies, Optimizing the input energy requirements, Fuel and energy substitution, Energy audit instruments

Financial Management: Investment-need, Appraisal and criteria, Financial analysis techniques- Simple payback period, Return on investment, Net present value, Internal rate of return, Cash flows, Risk and sensitivity analysis, Financing options, Energy performance contracts and role of ESCOs.

Introduction to Deregulation: Introduction, Reasons for restructuring / deregulation of power industry, Understanding the restructuring process: Entities involved, the levels of competition, the market place mechanisms, Sector-wise major changes required, Reasons and objectives of deregulation of various power systems across the world: The US, The UK and India. Market models based on contractual arrangements: Monopoly model, Single buyer model, Wholesale competition model, Retail competition model.

Electricity vis-à-vis Other Commodities: Distinguishing features of electricity as a commodity, Four pillars of market design: Imbalance, Scheduling and Dispatch, Congestion Management, Ancillary Services. Framework of Indian power sector and introduction to the availability-based tariff (ABT), Familiarization with applicable IEEE standards.
 - b. Prerequisites
 - UEE304
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Selected Elective
5. **Specific goals for the course**
 - a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Analyze about energy scenario nationwide and worldwide
- Decide about energy management in more effective way.
- Carry out financial management.
- Analyze about deregulation of power industry.
- Explain about various pillars of electricity market design.

a. Specific outcomes addressed by the course

A1	Apply mathematics (partial differentiation, vector calculus, linear algebra, complex variables, Laplace transform, probability, statistics, discrete mathematics etc.) to obtain analytical, numerical and statistical solutions.
A2	Demonstrate and apply knowledge of fundamentals, scientific and/or engineering principles towards solving engineering problems.
B3	Analyze and validate experimental results using appropriate techniques.
C1	Ability to formulate problems for electrical systems.
C4	Build prototypes, wherever needed, that meet design specifications.
K2	Able to develop systems for engineering applications.
K3	Able to analyze engineering problems using software tools.

SO1.1	Ability to identify and formulate problems for electrical systems
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

6. Brief list of topics to be covered

- a. Energy Scenario
- b. Energy Management and Audit
- c. Financial Management
- d. Nonlinear data structures
- e. Deregulation

2. **Course number and name:** UEE407; Network Theory and Design
3. **Credits and contact hours:** 4.5 and 6
4. **Instructor's or course coordinator's name:** Faculty from Department of Electrical and Instrumentation Engineering.
5. **Text book, title, author, and year**

Text Books / Reference Books

- Hayt, W., *Engineering Circuit Analysis*, Tata McGraw-Hill (2006).
- Hussain, A., *Networks and Systems*, CBS Publications (2004).
- Valkenberg, Van, *Network Analysis*, Prentice-Hall of India Private Limited (2007).
- Gayakwad, A. *Op-Amps and Linear Integrated Circuits*, Prentice-Hall of India (2006).
- Chakrabarti, A., *Circuit Theory*, Dhanpat Rai and Co. (P) Ltd. (2006).
- Roy Chowdhury, D., *Networks and Systems*, New Age International (P) Limited, Publishers (2007).
- Sudhakar, A., *Circuits and Networks*, Tata McGraw-Hill (2006).
- Suresh Kumar, K.S. *Electrical circuits and Networks*, Pearson Education, (2009).
- b. Other supplemental materials
 - Nil

6. Specific course information

- a. Brief description of the content of the course (catalog description)

Graph Theory: Graph, Tree and link branches, Network matrices and their relations, Choice of linearly independent network variables, Topological equations for loop current and topological equation for nodal voltage, Duality

Network Theorems: Source transformation, Superposition Theorem, Thevenin's theorem, Norton's theorem, Millman's theorem, Reciprocity theorem and Maximum power transfer theorem as applied to A.C. circuits, Compensation theorem, Tellegen's theorem and their applications.

Two Port Networks: Two port network description in terms of open circuits impedance, Short circuit admittance, Hybrid and inverse hybrid, ABCD and inverse ABCD parameters, Inter-connection of two port network, Indefinite admittance matrix and its applications.

Network Functions: Concepts of complex frequency, Transform impedance, Networks function of one port and two port network, concepts of poles and zeros, property of driving point and transfer function.

Passive Network Synthesis: Introduction, Positive Real Functions: Definition, Necessary and sufficient conditions for a function to be positive real, Synthesis vs. analysis, Elements of circuit synthesis, Foster and cauer forms of LC Networks, Synthesis of RC and RL networks.

Filters and Attenuators: Classification of filters, Analysis of a prototype low pass filter, High pass filter, Band pass filter, Band stop filter, M-derived filter, Attenuation, Types of attenuators: symmetrical and asymmetrical.

Active Filters and Oscillators: Introduction to Active filters, first and second order low pass Butterworth filter, First and second order high pass Butterworth filter, Band pass filter, Oscillators: Principles, types, Phase shift Oscillators, Wein Bridge Oscillators, Square wave generators.

Laboratory Work:

Verification of Network Theorems, Determination of Z, Y, hybrid and ABCD parameters of two port network, Inter-connection of two port networks, Analysis of T and Π -Attenuator Networks.

b. Prerequisites

- UEE001, UMA004.

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

7. Specific goals for the course

a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Apply the various laws and theorems related to electric networks.
- Explain the concept of two port networks.
- Familiarization with network synthesis.
- Theory and designing of passive filters and attenuators.
- Design of active filters.

b. Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.
C1	Ability to formulate problems for electrical systems.
E2	Develop appropriate models to formulate solutions.
E3	Use analytical, computational and/or experimental methods to obtain solutions.

SO1.1	Ability to identify and formulate problems for electrical systems
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

8. Brief list of topics to be covered

- Graph Theory
- Network Theorems
- Two Port Networks
- Network Functions
- Passive Network Synthesis
- Filters and Attenuators

1. **Course number and name:** UEE505; Analog and Digital Systems
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Boylestad R. L., *Electronic Devices and Circuit Theory*, Pearson Education (2007).
- Millman, J. and Halkias, C.C., *Integrated Electronics*, Tata McGraw Hill (2006).
- Floyd, T.L. and Jain, R. P., *Digital Fundamentals*, Pearson Education (2008).
- Tocci, R. and Widmer, N., *Digital Systems: Principles and Applications*, Pearson Education (2007).
- Neamen, Donald A., *Electronic Circuit Analysis and Design*, McGraw Hill (2006).
- Sedra A. S. and Smith K. C., *Microelectronic Circuits*, Oxford University Press (2006).
- Mano, M. M. and Ciletti, M., *Digital Design*, Pearson Education (2008).
- Kumar, A., *Fundamentals of Digital Circuits*, Prentice Hall (2007).

a. Other supplemental materials

- Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Bipolar Junction Transistor and Field Effect Transistor: Different configurations and their static characteristics; CE configuration as two port network: h-parameters, h-parameter equivalent circuit; Biasing and load line analysis; High frequency operation of BJT; Structure and working of JFET and MOSFET; output and transfer characteristics, Applications of JFET and MOSFET

Oscillators and Wave Shaping Circuits: Condition for sustained oscillation, R-C phase shift, Hartley, Colpitts, Crystal and Wien Bridge Oscillators, Negative Resistance oscillator; Switching characteristics of diodes and transistors including square wave response, High pass and low pass filters using R-C Circuits; R-L, R-L-C circuits, Attenuators; Clipping and clamping circuits; Clamping circuit theorem; Comparators; Multivibrators.

Simplification of Boolean Expressions: Quine-McClusky method in SOP and POS forms, determination of prime implications, simplification using Map-entered variables.

Combinational and Sequential Circuits: Introduction, Adders: Parallel Binary adder, Serial adder, BCD adder, Subtractors, Binary multiplier, Dividers, ALU, Code converters, Magnitude comparators, Parity Generators/checkers, Encoders, Decoders, Multiplexers, Demultiplexer; Introduction of sequential circuits, Flip-flops, Registers: Serial/Parallel in/out, Bi-directional, Universal shift register, Counters: Synchronous, Asynchronous, Decade, Binary, Modulo-n, Shift register counters; Design of Synchronous sequential circuits, FSM, Concept of Moore and Mealy machines, Synchronous detector.

Memories: Introduction and classification of ROM, ROM organization, Static and Dynamic RAM, DRAM Refreshing, Representative circuits for cells using BJT and FET's, Timing diagrams of memories, Memory expansion using IC's, Flash memory, CCD, Magnetic Memories.

Converters: Digital to Analog conversion, R2R ladder DAC, Weighted Resistor DAC, Analog-Digital conversion, Flash type, Counter type ADC, Dual-slope ADC, Successive

approximation type ADC.

Laboratory Work: Series voltage regulator, RC coupled amplifier in CE mode, Use of Bistable, Astable and monostable multivibrator, Hartley and Colpitts Oscillator, shift register and binary counting using JK flip flop, asynchronous/synchronous up/down counters, Variable modulus counters, Usage of IC tester, Computer simulation using EDA tools.

Minor Project: Design of LED lighting system for household application; street lighting system; soft starting of DC machine.

- b. Prerequisites
 - UEC001
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Design different type of circuits such as rectifiers, clippers, clampers, filters etc.
- Design power supplies and solve problems related to amplifiers and oscillators.
- Design combinational and sequential circuits.
- Differentiate various type of memories and their use in different applications.
- Demonstrate the concept of logic circuits and converters.
- b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.
E1	Classify information to identify engineering problems.
E2	Develop appropriate models to formulate solutions.
E3	Use analytical, computational and/or experimental methods to obtain solutions.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.

SO1.1	Ability to identify and formulate problems for electrical systems
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Bipolar Junction Transistor and Field Effect Transistor
- Oscillators and Wave Shaping Circuits
- Combinational and Sequential Circuits

1. **Course number and name:** UES012; Engineering Materials
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from School of Physics and Materials Science
4. **Text book, title, author, and year**

Text Books / Reference Books

- W.D. Callister, *Materials Science and Engineering*; John Wiley & Sons, Singapore, 2002.
- W.F. Smith, *Principles of Materials Science and Engineering: An Introduction*; Tata Mc-Graw Hill, 2008.
- V. Raghavan, *Introduction to Materials Science and Engineering*; PHI, Delhi, 2005.
- S. O. Kasap, *Principles of Electronic Engineering Materials*; Tata Mc-Graw Hill, 2007.
- L. H. Van Vlack, *Elements of Material Science and Engineering*; Thomas Press, India, 1998.
- K. G. Budinski, *Engineering Materials – Properties and selection*, Prentice Hall India, 1996.
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Structure of solids: Classification of engineering materials, Structure-property relationship in engineering materials, Crystalline and non-crystalline materials, Miller Indices, Crystal planes and directions, Determination of crystal structure using X-rays, Inorganic solids, Silicate structures and their applications. Defects; Point, line and surface defects.

Mechanical properties of materials: Elastic, Inelastic and Viscoelastic behaviour, Engineering stress and engineering strain relationship, True stress - true strain relationship, review of mechanical properties, Plastic deformation by twinning and slip, Movement of dislocations, Critical shear stress, Strengthening mechanism and Creep.

Equilibrium diagram: Solids solutions and alloys, Gibbs phase rule, Unary and binary eutectic phase diagram, Examples and applications of phase diagrams like Iron - Iron carbide phase diagram.

Electrical and magnetic materials: Conducting and resistor materials, and their engineering application; Semiconducting materials, their properties and applications; Magnetic materials, Soft and hard magnetic materials and applications; Superconductors; Dielectric materials, their properties and applications. Smart materials: Sensors and actuators, piezoelectric, magnetostrictive and electrostrictive materials.

Corrosion process: Corrosion, Cause of corrosion, Types of corrosion, Protection against corrosion.

Materials selection: Overview of properties of engineering materials, Selection of materials for different engineering applications.

Laboratory Work and Micro-Project:

Note: The micro-project will be assigned to the group(s) of students at the beginning of the semester. Based on the topic of the project the student will perform any of the six experiments from the following list:

- To determine Curie temperature of a ferrite sample and to study temperature dependence of permeability in the vicinity of Curie temperature.

- To study cooling curve of a binary alloy.
 - Determination of the elastic modulus and ultimate strength of a given fiber strand.
 - To determine the dielectric constant of a PCB laminate.
 - Detection of flaws using ultrasonic flaw detector (UFD).
 - To determine fiber and void fraction of a glass fiber reinforced composite specimen.
 - To investigate creep of a given wire at room temperature.
 - To estimate the Hall coefficient, carrier concentration and mobility in a semiconductor crystal.
 - To estimate the band-gap energy of a semiconductor using four probe techniques.
 - To measure grain size and study the effect of grain size on hardness of the given metallic specimens.
- b. Prerequisites
- UES009, UES010.
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
- Required

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Classify engineering materials based on its structure.
- Draw crystallographic planes and directions.
- Distinguish between elastic and plastic behavior of materials.
- Distinguish between Isomorphous and eutectic phase diagram.
- Classify materials based on their electrical and magnetic properties.
- Propose a solution to prevent corrosion.

- b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Structure of solids
- Mechanical properties of materials
- Equilibrium diagram
- Electrical and magnetic materials
- Corrosion process
- Materials selection

1. **Course number and name:** UMA007; Numerical Analysis
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the School of Mathematics
4. **Text book, title, author, and year**

Text Books / Reference Books

- *Curtis F. Gerald and Patrick O. Wheatley, Applied Numerical Analysis, Pearson, (2003) 7th Edition,*
- *M. K. Jain, S .R. K. Iyengar and R. K. Jain, Numerical Methods for Scientific and Engineering Computation, New Age International Publishers (2012), 6th edition.*
- *Steven C. Chappra, Numerical Methods for Engineers, McGraw-Hill Higher Education; 7th edition (1 March 2014)*
- *J. H. Mathew, Numerical Methods for Mathematics, Science and Engineering, Prentice Hall, (1992) 2nd edition,*
- *Richard L. Burden and J. Douglas Faires, Numerical Analysis, Brooks Cole (2004), 8th edition.*
- *K. Atkinson and W. Han, Elementary Numerical Analysis, John Willey & Sons (2004), 3rd Edition.*
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Floating-Point Numbers: Floating-point representation, rounding, chopping, error analysis, conditioning and stability.

Non-Linear Equations: Bisection, secant, fixed-point iteration, Newton method for simple and multiple roots, their convergence analysis and order of convergence.

Linear Systems and Eigen-Values: Gauss elimination method using pivoting strategies, LU decomposition, Gauss-Seidel and successive-over-relaxation (SOR) iteration methods and their convergence, ill and well-conditioned systems, Rayleigh's power method for eigen-values and eigen-vectors.

Interpolation and Approximations: Finite differences, Newton's forward and backward interpolation, Lagrange and Newton's divided difference interpolation formulas with error analysis, least square approximations.

Numerical Integration: Newton-Cotes quadrature formulae (Trapezoidal and Simpson's rules) and their error analysis, Gauss-Legendre quadrature formulae.

Differential Equations: Solution of initial value problems using Picard, Taylor series, Euler's and Runge-Kutta methods (up to fourth-order), system of first-order differential equations.

Probability Distributions: Binomial, Poisson, Geometric, Uniform, Normal, Exponential and Log-Normal distribution.

Hypothesis Testing: General concepts, Testing a Statistical Hypothesis, one and two tailed tests, Critical region, Confidence interval estimation. Single and two sample tests on proportion, mean and variance.

Linear Regression and Correlation: Linear Regression, Least Square principle and the Fitted models, Karl Pearson's Correlation Coefficient, Rank Correlation, Lines of

Regression (two variables only).

Laboratory Work: Lab experiments will be set in consonance with materials covered in the theory. Implementation of numerical techniques using MATLAB.

b. Prerequisites

- UMA003, UMA004

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Understand the errors, source of error and its effect on any numerical computations and also analysis the efficiency of any numerical algorithms.
- Learn how to obtain numerical solution of nonlinear equations using bisection, secant, newton, and fixed-point iteration methods.
- Solve system of linear equations numerically using direct and iterative methods.
- Understand how to approximate the functions using interpolating polynomials.
- Learn how to solve definite integrals and initial value problems numerically.

b. Specific outcomes addressed by the course.

A3	Apply numerical and statistical methods in analyzing data
I2	Ability to identify directions for continuing education opportunities.
K3	Able to analyze engineering problems using software tools.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.
SO7.2	Ability to identify directions for continuing education opportunities.

7. Brief list of topics to be covered

- Floating-Point Numbers
- Non-Linear Equations
- Linear Systems and Eigen-Values
- Interpolation and Approximations
- Numerical Integration
- Differential Equations

SEMESTER – V

1. **Course number and name:** UEE401; Alternating Current Machines
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Bimbhra, P.S., *Electrical Machinery*, Khanna Publishers (2008).
- Mukherjee, P.K. and Chakravorty, S., *Electrical Machines*, Dhanpat Rai and Co. (P) Ltd.(2004).
- Nagrath, I.J. and Kothari, D.P., *Electric Machines*, Tata McGraw Hill (2004).
- Bimbhra, P.S., *Generalized Theory of Electrical Machines*, Khanna Publishers (2007).
- Toro, Vincert, *Electromechanical Devices for Energy Conversion*, Prentice Hall of India (2004).
- Fitzgerald, A.E., Kingsley, C. Jr., and Umans, Stephen, *Electric Machinery*, McGraw-Hill (2002).
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Three-Phase Induction Motors: Construction, working principle, Slip and its effect on rotor parameters: rotor frequency, Torque-slip characteristics, Power flow diagram, Efficiency, Synchronous watt, Measurement of slip, Equivalent circuit, No-load test, Blocked rotor test, Circle diagram, Starting methods, Speed control methods, Crawling, Cogging, Deep cage and Double cage rotors, Applications, self-excited and grid connected Induction generator.

Fractional kW Motors and Special Machines: Classification, Production of rotating field, Double revolving field theory, Equivalent circuit, Determination of equivalent circuit parameters, Split phase induction motor, Capacitor motor, Permanent split capacitor motor; Shaded pole motor, Universal motor, Stepper motor.

Synchronous Generators/Alternators: Introduction, Comparison with DC generator, Advantages of rotating field over rotating armature, Constructional features, Excitation systems, Armature windings, EMF equation, Winding factor, Harmonics, Armature resistance, Armature reaction: Unity power factor, Zero lagging and Zero leading power factor, Armature reaction reactance, Equivalent circuit of an alternator, Voltage equation, Phasor diagram of a loaded alternator for various types of loads, Voltage regulation and methods of estimation of voltage regulation, Load characteristic of alternators, power equation, Two reaction theory and Torque-angle characteristic of a salient-pole alternator, Maximum reactive power for a salient-pole alternator, Losses and efficiency, Determination of X_d and X_q , Parallel operation of alternators, Synchronising procedures, Synchronising power and Torque co-efficient, Damper Windings, Hunting.

Synchronous Motors: Voltage equation, Phasor diagram, Operation at constant load with variable excitation, Power equations, salient pole Synchronous motor, Starting of synchronous motors, Applications, Synchronous condensers, Familiarization with relevant IEEE standards.

Laboratory work: Voltage regulation, Direct and quadrature axis reactances, Operating characteristics, Synchronizing, Parallel operation and load division, Sudden short circuit

analysis and determination of sub transient, Transient and steady state reactances and various time constants, Determination of positive, negative and zero sequence reactances, Synchronous motor starting, Efficiency. Three phase induction motors: starting methods, Equivalent circuit parameters, Load test, Polarity test, Single phasing, Efficiency, Schrage motor, Single-phase induction motors: Equivalent circuit parameters and performance indices.

b. Prerequisites

- UEE001, UEE301.

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Simulate the steady-state and transient state performance of induction and synchronous machines.
- Validate and identify the machine parameters.
- Select the appropriate AC motor for different large power application.
- Analyse the stability of single machine – infinite bus system and form the grid to supply large load.
- Choose the appropriate fractional horse power motor as per the usage in daily life.

b. Specific outcomes addressed by the course.

B1	Identify the constraints, assumptions and models for the experiments.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.
D1	Share responsibility and information schedule with others in team.
E1	Classify information to identify engineering problems.
E2	Develop appropriate models to formulate solutions.
E3	Use analytical, computational and/or experimental methods to obtain solutions.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
K1	Able to operate modern engineering equipment.
K2	Able to develop systems for engineering applications.

SO1.1	Ability to identify and formulate problems for electrical systems.
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO5.1	Share responsibility and information schedule with others in team.
SO6.1	Identify the constraints, assumptions and models for the experiments.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Three-Phase Induction Motors
- Fractional kW Motors and Special Machines
- Synchronous Generators/Alternators

1. **Course number and name:** UEE403; Measurement and Transducers
2. **Credits and contact hours:** 4.0 and 5
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**
Text Books / Reference Books
 - *Golding, E.W., and Widdis, F.C., Electrical Measurements and Measuring Instruments, Pitman (2003).*
 - *Sawhney, A.K., a Course in Electrical and Electronic Measurements and Instrumentation, DhanpatRai and Co. (P) Ltd. (2007).*
 - *Nakra, B. C. and Chaudhry, K. K., Instrumentation Measurement and Analysis, Tata McGraw-Hill (2003).*
 - *Murthy, D.V.S., Transducers and Instrumentation, Prentice-Hall of India Private Limited (2003).*
 - *Doebelin, E.O., Measurement systems, Applications and Design, McGraw-Hill (1982).*
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)
Units, Systems and Standards: SI units, Classification of standards, Time and frequency standards, Electrical standards.
Electromechanical Indicating Instruments: PMMC galvanometer, Ohmmeter, Electrodynamometer, Moving iron meter, Rectifier and thermo-instruments, Comparison of various types of indicating instruments, error analysis, uncertainties in measurement.
Power and Energy Measurement: Electrodynamometer type of wattmeter and power factor meter, Single-phase induction and Electronic energy meters.
Bridges for Measurement: Kelvin double bridge, AC bridges: Maxwell's bridge, Hay's bridge, Schering bridge, Wien's bridge, Low and High resistance measurement.
Electronic Instruments: Electronic multi-meter, Quantization error, Digital frequency meter, Q meter, Spectrum Analyzer, Digital Storage Oscilloscopes.
Sensors and Transducers: Basic principle and applications of Resistive, Inductive, Capacitive and, Piezoelectric sensors, Synchros and Resolvers, Fiber optic sensors, Hall-Effect, Photo transducer, Photovoltaic, Digital transducers, Tacho-generators, shaft parameters measurement in rotating shafts.
 - b. Prerequisites
 - UEC001, UEE001.
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required
6. **Specific goals for the course**
 - a. Specific outcomes of instructions.
After the completion of the course, the students will be able to:
 - Select various types of instruments for measurement of variables.
 - Select and use various types of sensors in different conditions.
 - Select and use various types of bridge circuits with different sensors.

- Explain the working of electronic instruments.
 - Explain the working of sensors and transducers.
- b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.
K2	Able to develop systems for engineering applications.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Units, Systems and Standards
- Electromechanical Indicating Instruments
- Power and Energy Measurement
- Bridges for Measurement
- Electronic Instruments
- Sensors and Transducers

1. **Course number and name:** UEE504; Power Electronics
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Dubey, G.K., Doradla, S.R., Joshi, A. and Sinha, R.M.K., *Thyristorised Power Controllers*, New Age International (P) Limited, Publishers (2004).
- Rashid, M., *Power Electronics*, Prentice-Hall of India (2006).
- Bimbhra, P.S., *Power Electronics*, Khanna Publishers (2012).
- Mohan, N., Underland, T. and Robbins, W. P., *Power Electronics: Converter Applications and Design*, John Wiley (2007) 3rd ed.
- Bose, B.K., *Handbook of Power Electronics*, IEEE Publications.
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Introduction: Introduction to Thyristors and its family, static and dynamic characteristics, turn-on and turn-off methods and firing circuits, Ratings and protection of SCRs, series and parallel operation.

Phase Controlled Converters: Principle of phase control, Single phase and three phase converter circuits with different types of loads, continuous and discontinuous conduction, effect of source inductance, Dual converters and their operation.

DC Choppers: Principle of chopper operation, control strategies, types of choppers, step up and step-down choppers, steady state time domain analysis with R, L, and C type loads, voltage, current and load commutated choppers.

Inverters: Single phase voltage source bridge inverters and their steady state analysis, modified McMurray half bridge inverter, series inverters, three phase bridge inverters with 180° and 120° modes. single-phase PWM inverters, current source inverters, CSI with R load (qualitative approach).

AC Voltage Controllers: Types of single-phase voltage controllers, single-phase voltage controller with R and RL type of loads.

Cycloconverters: Principles of operation, single phase to single phase step up and step down cycloconverters, three phase to single phase cycloconverters, output voltage equation for a cycloconverter.

Laboratory Work: SCR V-I characteristics, Gate firing circuit, DC -DC chopper, Semi converter and Full converter with R, RL and RLC type of loads, DC shunt motor speed control, Single phase AC voltage controller with R load, Inverters, Simulation of power electronics converters.

Minor Project: Design and development of power converters

- b. Prerequisites
 - UEC001.
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

a. Specific outcomes of instructions

After the completion of the course, the students will be able to:

- Select the power devices as per the usage for energy conversion and control.
- Exhibit the designing of firing and commutation circuits for different converter configurations.
- Analyse various converter configuration / topology with different types of load.
- Identify converter configurations for various power applications.
- Exhibit the usage of power converters for harmonic mitigation, voltage and frequency control.

b. Specific outcomes addressed by the course.

B1	Identify the constraints, assumptions and models for the experiments.
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.
C1	Ability to formulate problems for electrical systems.
C2	Design process to satisfy project objective for electrical systems.
C4	Build prototypes where needed, that meet design specifications.
D1	Share responsibility and information schedule with others in team.
E2	Develop appropriate models to formulate solutions.
E3	Use analytical, computational and/or experimental methods to obtain solutions.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
J1	Comprehend the relevance of contemporary issues.
K1	Able to operate modern engineering equipment.
K3	Able to analyze engineering problems using software tools.

SO1.1	Ability to identify and formulate problems for electrical systems
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO5.1	Share responsibility and information schedule with others in team.
SO6.1	Identify the constraints, assumptions and models for the experiments.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Thyristors
- Phase Controlled Converters
- DC Choppers
- Inverters
- AC Voltage Controllers
- Cycloconverters

1. **Course number and name:** UEI501; Control Systems
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Gopal, M., *Digital Control System*, Wiley Eastern (1986).
- Nagrath, I.J. and Gopal, M., *Control System Engineering*, New Age International (P)Limited, Publishers (2003).
- Ogata, K., *Modern Control Engineering*, Prentice-Hall of India Private Limited (2001).
- Kuo, B.C., *Automatic Control System*, Prentice-Hall of India Private Limited (2002).
- Sinha, N.K., *Control System*, New Age International (P) Limited, Publishers (2002).
- a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Basic Concepts: Historical review, Definitions, Classification, Relative merits and demerits of open and closed loop systems, Linear and non-linear systems, Transfer function, , Block diagrams and signal flow graphs.

Components: D.C. and A.C. Servomotors, D.C. and A.C. Tachogenerators, Potentiometers and optical encoders, Synchros and stepper motors

Analysis: Steady-state errors and error constants, Concepts and applications of P, PD, PI and PID types of control.

Stability: Definition, Routh-Hurwitz criterion, Root locus techniques, Nyquist criterion, Bode plots, Relative stability, Gain margin and phase margins.

Compensation: Lead, Lag and lag-lead compensators, Design of compensating networks for specified control system performance.

State Space Analysis: Concepts of state, State variables and state models, State space equations, Transfer function, Transfer model, State space representation of dynamic systems, State transition matrix, Decomposition of transfer function, Controllability and observability.

Laboratory: Linear system simulator, Compensation design, D.C. position control and speed control, Synchro characteristics, Servo demonstration, Stepper motor, Potentiometer error detector, Rate control system, Series control system, Temperature control system.

- b. Prerequisites

- UMA004.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Develop the mathematical model of the physical systems.
- Analyze the response of the closed and open loop systems.
- Analyze the stability of the closed and open loop systems.

- Design the various kinds of compensator.
 - Develop and analyze state space models.
- b. Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.
E1	Classify information to identify engineering problems.
E2	Develop appropriate models to formulate solutions.
E3	Use analytical, computational and/or experimental methods to obtain solutions.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
K1	Able to operate modern engineering equipment.
K2	Able to develop systems for engineering applications.

SO1.1	Ability to identify and formulate problems for electrical systems.
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Open loop and closed loop system
- Steady state analysis
- Stability
- Compensation
- State space analysis

1. **Course number and name:** UEI610; Fundamentals of Microprocessors and Microcontrollers
2. **Credits and contact hours:** 4 and 5
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Hall, D.V., *Microprocessor- Interfacing Programming and Hardware*, Tata McGraw-Hill (1997).
- Ayala, K.J., *The 8051 Microcontroller Architecture, Programming and applications*, Penram International Publishing (India) Pvt. Ltd. (2007).
- Mazidi, M.A., *The 8051 Microcontroller and Embedded System*, Pearson Education (2008).
- Brey, B.B., *The INTEL Microprocessors*, Prentice-Hall of India Private Limited (2002).
- Liu, Y. C. and Gibson, G.A., *Microcomputer Systems: The 8086/8088 Family- Architecture, Programming and Design*, Prentice-Hall of India Private Limited (2007).
- Uffenbeck, J., *The 8086/ 8088 Family*, Prentice-Hall of India Private Limited (1994).
- Predko, M., *Customizing The 8051 Microcontroller*, Tata McGraw-Hill (2002).
- a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

INTEL 8086 Microprocessor: Pin Functions, Architecture, Characteristics and Basic Features of Family, Segmented Memory, Addressing Modes, Instruction Set, Data Transfer Instructions, Arithmetic, Logical, Shift and Rotate Instructions, String Instructions, Flag Control Instructions, Transfer of Control Instructions, Processor Control Instructions, Programming Examples, Interrupt Structures, Multitasking and Multiprogramming, MIN/MAX Modes of 8086, Co-processors 8087 and 8089.

Introduction to 8051 Microcontroller: 8051 architecture and pin diagram, Registers, Timers, Counters, Flags, Special Function Registers, Addressing Modes, Data types, instructions and programming, Single-bit operations, Timer and Counter programming, Interrupts programming, Serial communication, Memory accessing and their simple programming applications.

Hardware interfacing: I/O Port programming, Bit manipulation, Interfacing to a LED, LCD, Keyboard, ADC, DAC, Stepper Motors and sensors, Familiarization with relevant IEEE/IEC standards.

Laboratory work: Introduction to INTEL kit, Programming examples of 8086, Interfacing using 8086 kits, ADC, DAC, 8253, Microprocessor based project, Programming and Application development around 8051, Interfacing to LED, LCD, Keyboard, ADC, DAC, Stepper Motors and sensors etc.

- b. Prerequisites

- UEE505.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Demonstrate the concept of microprocessor and to be able to design a microprocessorbased system to get desired results.
 - Use 8086 microprocessor in advanced applications, which will give them a good platform to work further.
 - Graduates will be able to update with current trends through self-study and show genuine need to learn on continuous basis.
 - Students will be able to use hardware interfacing of 8051 to develop solutions of realworld electrical problems.
- b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
C1	Ability to formulate problems for electrical systems.
C3	Work with real time systems within realistic constraints.
E3	Use analytical, computational and/or experimental methods to obtain solutions.
K3	Able to analyze engineering problems using software tools.

SO1.1	Ability to identify and formulate problems for electrical systems
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.2	Work with real time systems within realistic constraints
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- 8086 Microprocessor
- 8051 microcontrollers
- Hardware interfacing

1. **Course number and name:** UTA012; Innovation and Entrepreneurship
2. **Credits and contact hours:** 4.5 and 3
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering
4. **Text book, title, author, and year**

Text Books / Reference Books

- Ries, Eric(2011), *The lean Start-up: How constant innovation creates radically successful businesses*, Penguin Books Limited.
- Blank, Steve (2013), *The Startup Owner's Manual: The Step by Step Guide for Building a Great Company*, K&S Ranch.
- S. Carter and D. Jones-Evans, *Enterprise and small business- Principal Practice and Policy*, Pearson Education (2006)
- T. H. Byers, R. C. Dorf, A. Nelson, *Technology Ventures: From Idea to Enterprise*, McGraw Hill (2013)
- Osterwalder, Alex and Pigneur, Yves (2010) *Business Model Generation*.
- Kachru, Upendra, *India Land of a Billion Entrepreneurs*, Pearson
- Bagchi, Subroto, (2008), *Go Kiss the World: Life Lessons for the Young Professional*, Portfolio Penguin
- Bagchi, Subroto, (2012). *MBA At 16: a Teenager's Guide to Business*, Penguin Books
- Bansal, Rashmi, *Stay Hungry Stay Foolish*, CIIE, IIM Ahmedabad
- Bansal, Rashmi, (2013). *Follow Every Rainbow*, Westland.
- Mitra, Sramana (2008), *Entrepreneur Journeys (Volume 1)*, Booksurge Publishing
- Abrams, R. (2006). *Six-week Start-up*, Prentice-Hall of India.
- Verstraete, T. and Laffitte, E.J. (2011). *a Business Model of Entrepreneurship*, Edward Elgar Publishing.
- Johnson, Steven (2011). *Where Good Ideas comes from*, Penguin Books Limited.
- Gabor, Michael E. (2013), *Awakening the Entrepreneur Within*, Primento.
- Guillebeau, Chris (2012), *The \$100 startup: Fire your Boss, Do what you love and work better to live more*, Pan Macmillan
- Kelley, Tom (2011), *The ten faces of innovation*, Currency Doubleday
- Prasad, Rohit (2013), *Start-up sutra: what the angels won't tell you about business and life*, Hachette India.
- Ries, Eric(2011), *The lean Start-up: How constant innovation creates radically successful businesses*, Penguin Books Limited.
- Blank, Steve (2013), *The Startup Owner's Manual: The Step by Step Guide for Building a Great Company*, K&S Ranch.
- S. Carter and D. Jones-Evans, *Enterprise and small business- Principal Practice and Policy*, Pearson Education (2006)
- T. H. Byers, R. C. Dorf, A. Nelson, *Technology Ventures: From Idea to Enterprise*, McGraw Hill (2013)
- Osterwalder, Alex and Pigneur, Yves (2010) *Business Model Generation*.
- Kachru, Upendra, *India Land of a Billion Entrepreneurs*, Pearson
- Bagchi, Subroto, (2008), *Go Kiss the World: Life Lessons for the Young Professional*, Portfolio Penguin
- Bagchi, Subroto, (2012). *MBA At 16: a Teenager's Guide to Business*, Penguin Books

- Bansal, Rashmi, *Stay Hungry Stay Foolish*, CIIE, IIM Ahmedabad
- Bansal, Rashmi, (2013). *Follow Every Rainbow*, Westland.
- Mitra, Sramana (2008), *Entrepreneur Journeys (Volume 1)*, Booksurge Publishing
- Abrams, R. (2006). *Six-week Start-up*, Prentice-Hall of India.
- Verstraete, T. and Laffitte, E.J. (2011). *a Business Model of Entrepreneurship*, Edward Elgar Publishing.
- Johnson, Steven (2011). *Where Good Ideas comes from*, Penguin Books Limited.
- Gabor, Michael E. (2013), *Awakening the Entrepreneur Within*, Primento.
- Guillebeau, Chris (2012), *The \$100 startup: Fire your Boss, Do what you love and work better to live more*, Pan Macmillan
- Kelley, Tom (2011), *The ten faces of innovation*, Currency Doubleday
- Prasad, Rohit (2013), *Start-up sutra: what the angels won't tell you about business and life*, Hachette India.
- a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Introduction to Entrepreneurship: Entrepreneurs; entrepreneurial personality and intentions - characteristics, traits and behavioral; entrepreneurial challenges.

Entrepreneurial Opportunities: Opportunities. discovery/ creation, Pattern identification and recognition for venture creation: prototype and exemplar model, reverse engineering.

Entrepreneurial Process and Decision Making: Entrepreneurial ecosystem, Ideation, development and exploitation of opportunities; Negotiation, decision making process and approaches, Effectuation and Causation.

Crafting business models and Lean Start-ups: Introduction to business models; Creating value propositions-conventional industry logic, value innovation logic; customer focused innovation; building and analyzing business models; Business model canvas, Introduction to lean startups, Business Pitching.

Organizing Business and Entrepreneurial Finance: Forms of business organizations; organizational structures; Evolution of Organisation, sources and selection of venture finance options and its managerial implications. Policy Initiatives and focus; role of institutions in promoting entrepreneurship.

- b. Prerequisites
- None
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
- Required

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Comprehend the role of bounded rationality, framing, causation and effectuation in entrepreneurial decision making.
- Demonstrate an ability to design a business model canvas.
- Evaluate the various sources of raising finance for startup ventures.
- Understand the fundamentals of developing and presenting business pitching to potential investors.

- b. Specific outcomes addressed by the course.

F2	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.
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SO4.2	Evaluate engineering solutions considering environmental constraints.
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7. Brief list of topics to be covered

- Entrepreneurial Opportunities
- Entrepreneurial Process and Decision Making
- Crafting business models and Lean Start-ups
- Organizing Business and Entrepreneurial Finance

SEMESTER – VI

1. **Course number and name:** UEE603; Switchgear and Protection
2. **Credits and contact hours:** 4.0 and 5
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Chakraborti, A., Soni, M.L., Gupta, P.V. and Bhatnagar, U.S., *a Text Book on Power System Engineering*, DhanpatRai and Co. (P) Ltd. (2008).
- Pathinkar, Y.G. and Bhide, S.R., *Fundamentals of Power System Protection*, PHI Learning Pvt. Limited (2008).
- Rao, S.S., *Switchgear and Protection*, Khanna Publishers (2007).
- Deshpande, M.V., *Switchgear and Protection*, Tata McGraw-Hill (2005).
- Elmore, W.A., *Protective Relaying Theory and Applications*, ABB Power T and D Company Inc. (2003).
 - a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Introduction: a protection system and its attributes, System transducers, duties of switchgear, various power system elements that needs protection.

Fuses: Types, ratings and characteristics, construction and application of HRC fuses, limitations and application of fuses, Introduction to MCBs.

Circuit Breakers: Theory of arc formation and its extinction (AC and DC), re-striking and recovery voltage, Current chopping, circuit breakers: specifications of circuit breakers, different types of circuit breakers like oil, Air, Vacuum and SF₆, comparative merits and demerits, HVDC circuit breaker system.

Earthing: Earthing requirements, Earthing practices, Earth resistivity and earth gradient, Neutral shift.

Protective Relays: Functions, Constructional and operating principles of electromagnetic type like over-current, Directional, Differential and distance relays, Characteristics, General equation. Basic principles of static relaying, Phase and amplitude comparator, Microprocessor based relays.

Protection Schemes: Overcurrent and Overvoltage protection of transmission lines, differential protection, transformer protection, Bus bar protection, distance protection of transmission line, carrier aided protection of transmission lines, generator protection, induction motor protection.

Laboratory work: Sequence impedance and their calculations, Symmetrical fault level measurement on a D.C. network analyzer, Unsymmetrical fault level measurement on a D.C. network analyzer for various types of faults, Measurement of ground resistivity and resistance of a ground electrode, Plotting of characteristics of different types of relays, Performance of different types of protection schemes, ABCD constants of an artificial transmission line, String efficiency of insulator string, use of standard software package for short circuit studies and relay co-ordination.

- b. Prerequisites
 - UEE304, UEE406.
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Explain various protection strategies applied for power system protection.
- Select the protection elements namely fuse, circuit breakers and relays for a given configuration.
- Design the basic Earthing requirement for residential and other purposes.
- Select required protection measures against overcurrent, overvoltage in transmission lines.
- Select suitable protection scheme for different power system equipment.

- b. Specific outcomes addressed by the course.

C2	Design process to satisfy project objective for electrical systems.
E1	Classify information to identify engineering problems.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
J1	Comprehend the relevance of contemporary issues.
K1	Able to operate modern engineering equipment.
K2	Able to develop systems for engineering applications.

SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO1.1	Ability to identify and formulate problems for electrical systems.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Introduction of switchgear protection.
- Fuses
- Circuit Breakers
- Earthing
- Protective Relays
- Protection Schemes

1. **Course number and name:** UEE605; Power System Analysis and Stability
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Chakraborti, A., Soni, M.L., Gupta, P.V. and Bhatnagar, U.S., *a Text Book on Power System Engineering*, Dhanpat Rai and Co. (P) Ltd. (2008).
- Nagrath, I.J. and Kothari, D.P., *Power System Engineering*, Tata McGraw-Hill (2007).
- Stevenson, W.D., *Power System Analysis*, McGraw-Hill (2007).
- Gupta, B.R., *Power System Analysis and Design*, S.Chand and Company Limited (2009).
- Pabla, A.S., *Electric Power Distribution*, Tata McGraw-Hill (2008).
- Wadhwa, C.L., *Electrical Power Systems*, New Age International (P) Limited, Publishers (2008).
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Representation of Power System: Representation of power system components, regulating transformers generators, transmission line and loads, phase shift in star-delta transformer, sequence impedance of transmission line, transformer and generators, sequence networks of power system, Y-Bus and Z-Bus building algorithm.

Load Flow Study: Load flow problem, power flow equations, load flow solution using Gauss Seidal and Newton Raphson methods, decoupling between real and reactive power control, decoupled and fast decoupled methods, comparison of load flow methods.

Fault Analysis: Symmetrical fault, algorithm for symmetrical fault analysis, unbalanced faults (Single line to ground fault, Line to line and double line to ground, Open conductor), Bus Impedance matrix method for the analysis of unsymmetrical shunt faults.

Power System Stability: Concepts of types of stability limits, steady state stability analysis, transient stability analysis, Swing equation and its solution by point-by-point method, Equal area criterion, critical clearing angle and improvement of transient stability.

Laboratory work: Develop software for various matrix inversion techniques, load flow problems with all methods, Fault analysis and stability studies; Use of standard software for simulation and steady state analysis of power system.

- b. Prerequisites
 - UEE304, UEE406, UMA031, UMA007.
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. **Specific goals for the course**

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Develop an appropriate mathematical model of power system
 - Carry out power flow analysis of practical power system for balanced system.
 - Conduct studies during balanced faults to decide the fault levels and circuit breaker ratings.
 - Analyze the stability of single machine-infinite bus system and can decide the critical clearing time of circuit breakers.
- b. Specific outcomes addressed by the course.

A3	Apply numerical and statistical methods in analyzing data
E1	Classify information to identify engineering problems.
E2	Develop appropriate models to formulate solutions.
K3	Able to analyze engineering problems using software tools.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.1	Ability to identify and formulate problems for electrical systems.
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Representation of Power System
- Load Flow Study
- Fault Analysis
- Power System Stability

1. **Course number and name:** UEE801; Electric Drives
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Dubey, G.K., *Power Semiconductor Controlled Drives*, Prentice Hall Inc. (1989).
- Pillai, S.K., *a Course in Electric Drives*, New Age International (P) Limited, Publishers (1989).
- Bose, B.K., *Modern Power Electronics and AC Drives*, Prentice-Hall of India Private Limited (2006).
- Dubey, G.K., *Fundamentals of Electric Drives*, Narosa Publications (2001).
- Sen, P.C., *Thyristor DC Drives*, John Wiley and Sons (1981).

- a. Other supplemental materials

- Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Definitions and Dynamics of Electric Drives: Concept of electric drive and its classifications, Types of loads, Four-quadrant drive, Dependence of load torque on various factors, Dynamics of motor-load combination, Steady state stability of an electric drive system, Load Equalization.

Drive Features of Importance: Multi-quadrant operations of DC and AC motors, Energy relations during starting and braking.

Static Control of Motors: Contactors and relays for electric drives, Control circuits for automatic starters of DC and AC motors.

Estimation of Motors Rating: Thermal modeling of motors, Types of duty cycles, Calculation of motor rating for duty cycles, Overload factor calculation for short and intermittent duty cycle, Use of load diagrams.

Solid State Controlled Drives: Control of DC drives fed through single-phase and three-phase semi-converter and full-converter phase-controlled configurations, their analysis, Regeneration and braking through static power converters, control of three phase induction motors by stator voltage and frequency control for speeds below and above synchronous speed, Static rotor resistance control, Static Kramer and Scherbius drives, V/f and Vector control, Energy efficient drives, losses in electrical drive system, Energy conservation in electric drives.

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Laboratory work: Starting and running characteristics of converter fed AC and DC motor control, Harmonic analysis of AC and DC Drives, V/f-based drive, Microprocessor based Drive, PLC based drive, Project on drives using standard software.

- b. Prerequisites or co-requisites

- UEE301, UEE401, UEI501, UEE504.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Conceptualize the basic drive system and analyse it for different types of loads.
- Analyse the motor situation during starting and braking.
- Develop control circuitry and devices for control of motor.
- Estimate the motor rating for different condition of load.
- Design the converter circuit for control purpose along with its different configuration.
- Use PLC and converter control to drive on the basis of energy efficiency.

b. Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
B2	Utilize suitable hardware equipment for data collection.
B3	Analyze and validate experimental results using appropriate techniques.
C1	Ability to formulate problems for electrical systems.
D1	Share responsibility and information schedule with others in team.
E1	Classify information to identify engineering problems.
E3	Use analytical, computational and/or experimental methods to obtain solutions.
F1	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
F2	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
K1	Able to operate modern engineering equipment.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO1.1	Ability to identify and formulate problems for electrical systems
SO5.1	Share responsibility and information schedule with others in team.
SO1.1	Ability to identify and formulate problems for electrical systems.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO4.2	Evaluate engineering solutions considering environmental constraints.
SO4.2	Evaluate engineering solutions considering environmental constraints.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Definitions and Dynamics of Electric Drives
- Drive Features of Importance
- Static Control of Motors
- Estimation of Motors Rating
- Solid State Controlled Drives

1. **Course number and name:** UHU005; Humanities for Engineers
2. **Credits and contact hours:** 3.0 and 4
3. **Instructor's or course coordinator's name:** Faculty from the School of Humanities and Social Sciences
4. **Text book, title, author, and year**

Text Books / Reference Books

- Morgan, C.T., King, R.A., Weisz, J.R., & Schopler, J. *Introduction to Psychology*, McGraw Hill Book Co. (International Student) (1986).
- N. Tripathi, *Human Values*, New Age International (P) Ltd (2009).
- Krugman, Paul and Wells Robin, *Economics*, W.H. Freeman & Co Ltd. Fourth Edition (2015).
- Rubinfeld Pindyck. *Microeconomic Theory and application*, Pearson Education New Delhi (2012).
- Samuelson, Paul, A. and Nordhaus, William, D. *Economics*, McGraw Hill, (2009).
- Mankiw, Gregory N. *Principles of Macroeconomics*, South-Western College Pub., (2014).
- Gregory, Paul R. and Stuart, Robert C. *The Global Economy and Its Economic Systems*, 2013 South-Western College Pub (2013).
- Atkinson, R.L., Atkinson, R.C., Smith, E.E., Bem, D.J. and Nolen-Hoeksema, S. (2000). *Hilgard's Introduction to Psychology*, New York: Harcourt College Publishers.
- Berne, Eric (1964). *Games People Play – The Basic Hand Book of Transactional Analysis*. New York: Ballantine Books.
- Ferrell, O. C and Ferrell, John Fraedrich *Business Ethics: Ethical Decision Making & Cases*, Cengage Learning (2014).
- Duane P. Schultz and Sydney Ellen Schultz, *Theories of Personality*, Cengage Learning, (2008).
- Saleem Shaikh. *Business Environment*, Pearson (2007).
- Chernilam, Francis *International Business-Text and Cases*, Prentice Hall (2013).
- Salvatore, Dominick, Srivastav, Rakesh., *Managerial Economics: Principles with Worldwide Applications*, Oxford, 2012.
- Peterson H. Craig. and. Lewis, W. Cris. *Managerial Economics*, Macmillan Pub Co; (1990).
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Introduction to Psychology: Historical Background, Psychology as a science. Different perspectives in Psychology.

Perception and Learning: Determinants of perception, Learning theories, Behavior Modification.

Motivational and Affective basis of Behavior: Basic Motives and their applications at work. Components of emotions, Cognition and Emotion. Emotional Intelligence Group Dynamics and Interpersonal relationships.

Development of self and personality.

Transactional Analysis.

Culture and Mind.

Practicals:

1. Experiments on learning and behavior modification.
2. Application of Motivation Theories: Need based assessment.
3. Experiments on understanding Emotions and their expressions.
4. Personality Assessment.
5. Exercises on Transactional analysis.
6. Role plays, case studies, simulation tests on human behavior

HUMAN VALUES and ETHICAL PERSPECTIVE

Values: Introduction to Values, Allport-Vernon Study of Values, Rokeach Value Survey, Instrumental and Terminal Values.

Value Spectrum for a Good Life: Role of Different Types of Values such as Individual, Societal, Material, Spiritual, Moral, and Psychological in living a good life.

Moral and Ethical Values: Types of Morality, Kant's Principles of Morality, Factors for taking ethical decisions, Kohlberg's Theory of Moral Development.

Analyzing Individual human values such as Creativity, Freedom, Wisdom, Love and Trust. Professional Ethics and Professional Ethos, Codes of Conduct, Whistle-blowing, Corporate Social Responsibility

Laboratory Work:

Practical application of these concepts by means of Discussions, Role-plays and Presentations, Analysis of Case studies on ethics in business and CSR.

ECONOMIC PERSPECTIVE

Basics of Demand and Supply

Production and cost analysis

Market Structure: Perfect and Imperfect Markets.

Investment Decisions: capital Budgeting, Methods of Project Appraisal

Macroeconomic Issues: Gross domestic product (GDP), Inflation and Financial Markets. **Globalisation:** Meaning, General Agreement on Trade and tariffs (GATT), World Trade Organisation (WTO). Global Liberalisation, and its impact on Indian Economy.

Laboratory Work:

The practicals will cover numerical on demand, supply, market structures and capital budgeting, Trading games on financial markets, Group discussions and presentations on macroeconomic issues. The practicals will also cover case study analysis on openness and globalization and the impact of these changes on world and Indian economy.

Micro Project: Global Shifts and the impact of these changes on world and Indian economy.

b. Prerequisites

- UHU003

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Improve the understanding of human behavior with the help of interplay of professional, psychological and economic activities.
- Able to apply the knowledge of basic principles of psychology, economics and ethics for the solution of engineering problems.

- Explain the impact of contemporary issues in psychology, economics and ethical principles on engineering.
- b. Specific outcomes addressed by the course.

G2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
H1	Aware of societal and global changes due to engineering innovations.

SO3.2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
SO4.4	Aware of societal and global changes due to engineering innovations.

7. Brief list of topics to be covered

- Perception and Learning
- Motivational and Affective basis of Behavior
- Value Spectrum for a Good Life
- Moral and Ethical Values
- Investment Decisions
- Globalisation, Macroeconomic Issues

SEMESTER – VII

1. **Course number and name:** UEE502; High Voltage Engineering
2. **Credits and contact hours:** 4.0 and 5
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Chakrabarti, A., Soni, M.L., Gupta, P.V. and Bhatnagar, U.S., *a Text Book on Power System Engineering*, Dhanpat Rai (2008).
- Wadhwa, C.L., *Electrical Power Systems*, New Age International (P) Limited, Publishers (2008).
- Gupta, B.R., *Power System Analysis and Design*, S. Chand (2009).
- Nagrath, I.J. and Kothari, D.P., *Power System Engineering*, Tata McGraw-Hill (2007).
- Pabla, A.S., *Electric Power Distribution*, McGraw Hill (2008).
- Stevenson, W.D., *Power System Analysis*, McGraw-Hill (2007).
- a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Introduction: Introduction to AC and DC impulse voltages and their use, Problems in dealing with high voltages.

Breakdown in Gases: Elementary ideas on ionization by electron collision, Townsend mechanism, Townsend first and second ionization coefficients, Paschen law, breakdown in non-uniform fields and corona discharges, vacuum breakdown mechanisms, breakdown in liquids, fundamentals of insulating oils, conduction and breakdown in pure and commercial liquids.

Breakdown in Solids: Fundamentals of solid insulating materials intrinsic, electromechanical and thermal breakdown, breakdown in simple and composite dielectrics, types of insulating materials, temperature classification, factor affecting dielectric strength, insulation design of rotating machines, transformers, transmission lines, Switch gear, etc.

Generation of High Voltages: Generation of high voltages, testing transformers in cascade, series resonant circuits and their advantages, half and full wave rectifier circuits, voltage doubler and cascade circuits, electrostatic generator, characteristics parameters of impulse voltages, single stage impulse generator circuits, multistage impulse generation circuits.

Measurement of High Voltages: Measurement of direct, alternating and impulse voltages by electrostatic voltmeters, sphere gap, uniform field gap, ammeter in series with high voltage resistors and voltage divider

Non-Destructive High Voltage Tests: Loss in a dielectric and its measurement, dielectric loss measurement by Schering bridge, partial discharges at alternating voltages, external and internal partial discharges and discharge measurements.

Laboratory work: Voltage measurement by sphere gap and Chubb and Fortesque methods, Insulation resistance measurement using Meggar, Experimental setup for standard lightning wave, Efficiency and peak voltage measurement by sphere gap impulse voltage time curves,

Breakdown voltage, Conductivity and dissipation factor measurement with Schering bridge, partial discharge measurements.

b. Prerequisites

- UPH004, UES012.

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Conceptualize the idea of high voltage and safety measures involved.
- Analyse the breakdown mechanism of solids, liquids and gases.
- Analyse and calculate the circuit parameters involved in generation of high voltages.
- Measure direct, alternating and impulse high voltage signals.
- Measure the dielectric loss and partial discharge involved in non-destructive high voltage tests.

b. Specific outcomes addressed by the course.

B1	Identify the constraints, assumptions and models for the experiments.
B3	Analyze and validate experimental results using appropriate techniques.
C2	Design process to satisfy project objective for electrical systems.
D1	Share responsibility and information schedule with others in team.
E1	Classify information to identify engineering problems.
E2	Develop appropriate models to formulate solutions.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
J1	Comprehend the relevance of contemporary issues.
K1	Able to operate modern engineering equipment.

SO6.1	Identify the constraints, assumptions and models for the experiments.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO5.1	Share responsibility and information schedule with others in team.
SO1.1	Ability to identify and formulate problems for electrical systems
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Breakdown in Gases
- Breakdown in Solids
- Generation of High Voltages
- Measurement of High Voltages

1. **Course number and name:** UEE604; Flexible AC Transmission Systems
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**
Text Books / Reference Books
 - Hingorani, N.G. and Gyragyi, L., *Understanding FACTS : Concepts and Technology of Flexible AC Transmission System*, Standard Publishers and Distributors (2005).
 - Sang, Y.H. and John, A.T., *Flexible AC Transmission Systems*, IEEE Press (2006).
 - Ghosh, A. and Ledwich, G., *Power Quality Enhancement Using Custom Power Devices*, Kluwer Academic Publishers (2005).
 - Mathur, R.M. and Verma, R.K., *Thyristor Based FACTS Controllers for Electrical Transmission Systems*, IEEE Press (2002).
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Power Transmission control: Fundamentals of AC power transmission, Transmission problems and needs, Overview of stability, the emergence of FACTS, FACTS controller and consideration.

Static power convertor: Review of Power Electronics fundamentals: Static power convertor structures, AC controller-based structure, DC link convertor topologies, Convertor output and harmonic control.

Shunt Compensation: Shunt SVC principles, Configuration and control, STATCOM, Configuration applications.

Series Compensation: Fundamental of series compensation, Principle of operation, Application of TCSC for different problems of power system, TCSC lay out, SSSC principle of operation.

Phase Shifter: Principle of operation, Steady state model of static phase shifter, Operating characteristics of SPS, Power current configuration of SPS application.

Unified Power Flow Controllers: Basic operating principles and characteristics, Control UPFC installation applications, UPFC model for power flow studies.
 - b. Prerequisites
 - UEE504, UEE605.
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required
6. **Specific goals for the course**
 - a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

 - Describe the converter configuration for different power systems applications such as HVDC, FACTS etc.
 - Evaluate the converters, harmonics on AC and DC side and filtering.

- Classify various compensators suited for various power system purposes.
 - Analyze power system behavior with different shunt compensators.
 - Appraise series compensated power system behavior with different series compensators.
 - Analyze system behavior with hybrid shunt-series compensators.
- b. Specific outcomes addressed by the course.

C3	Work with real time systems within realistic constraints.
E1	Classify information to identify engineering problems.
E3	Use analytical, computational and/or experimental methods to obtain solutions.

SO2.2	Work with real time systems within realistic constraints
SO1.1	Ability to identify and formulate problems for electrical systems
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.

7. Brief list of topics to be covered

- Power Transmission control
- Static power convertor
- Shunt Compensation
- Series Compensation
- Phase Shifter& Unified power flow controlled

1. **Course number and name:** UEE608; Soft Computing in Electrical Engineering
2. **Credits and contact hours:** 4.0 and 5
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Lin, C., Lee, G., *Neural Fuzzy Systems*, Prentice Hall International Inc. (2000).
- Rajashekran, S. and Vijaylaksmi Pai, G.A., *Neural Networks, Fuzzy Logic and Genetic Algorithm Synthesis and Applications*, PrenticeHall of India Private Limited (2004).
- Zurda, J.M., *C++ Neural Networks and Fuzzy Logics*, BPS Publication (2001).
- Kosko, B., *Neural Networks and Fuzzy Systems: a Dynamical systems Approach to Machine Intelligence*, Prentice Hall of India Private Limited (1992).
- a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Introduction: Concept of artificial intelligence, Introduction to classical problem-solving methods and heuristic search techniques.

Fuzzy Systems: Fuzzy sets, Operation on fuzzy sets, Fuzzy relations, measures, Fuzzy logic, Fuzzy logic controller (FLC).

Artificial Neural Networks: Fundamental concepts, Basic models, Learning rules, Single layer and multi-layer feed-forward and feedback networks, Supervised and unsupervised methods of training, Recurrent networks, Modular network.

Genetic Algorithm: Basic principle, Evolution of genetic algorithm, Hybrid genetic algorithm, trends in stochastic search.

Hybrid Systems: Integrated hybrid systems such as neuro-fuzzy, fuzzy-neuro.

Applications: Short term and long-term load forecasting, Identification, Classification, Faultlocation and fault diagnosis, Economic load dispatch, DC/AC four quadrant drive control.

Laboratory work: Training algorithms of neural networks and fuzzy logic, Implementation of fuzzy logic, Neural networks and genetic algorithms on various applications, Use of MATLAB tools of fuzzy logic and NN.

- b. Prerequisites

- UMA031, UEE605, UEE801.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Examine the fuzzy system and implement fuzzy controllers for control and classification.
- Explain neural networks behaviour and use them for classification, control system and optimization problem.
- Obtain the optimum solution of well formulated optimisation problem using

evolutionary approach.

- Formulate hybrid intelligent algorithms for typical electrical application.

b Specific outcomes addressed by the course.

B1	Identify the constraints, assumptions and models for the experiments.
D2	Participate in the development and selection of ideas.
E1	Classify information to identify engineering problems.
E2	Develop appropriate models to formulate solutions.
E3	Use analytical, computational and/or experimental methods to obtain solutions.
K2	Able to develop systems for engineering applications.
K3	Able to analyze engineering problems using software tools.

SO6.1	Identify the constraints, assumptions and models for the experiments.
SO5.2	Participate in the development and selection of ideas on a team whose members together provide leadership.
SO1.1	Ability to identify and formulate problems for electrical systems
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Fuzzy Systems
- Artificial Neural Networks
- Artificial Neural Networks
- Genetic Algorithm
- Hybrid Systems
- Applications

1. **Course number and name:** UEE795; Capstone Project
2. **Credits and contact hours:** 8 and 2
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering
4. **Text book, title, author, and year**
 - May be required as per the selected project title
 - a. Other supplemental materials
 - Electrical equipments, sensors and transducers etc.

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Course Description: Capstone Project is increasingly interdisciplinary, and requires students to function on multidisciplinary teams. It is the process of devising a system, component or process to meet desired needs. It is a decision-making process (often iterative), in which the basic sciences, mathematics, and the engineering sciences are applied to convert resources optimally to meet these stated needs. It typically includes both analysis and synthesis performed in an iterative cycle. Thus, students should experience some iterative design in the curriculum. As part of their design experience, students have an opportunity to define a problem, determine the problem scope and to list design objectives. The project must also demonstrate that students have adequate exposure to design, as defined, in engineering contexts. Engineering standards and realistic constraints are critical in engineering design. The students get familiarized with different IEEE/IEC standards on electric power and energy, electric machines, power electronics and drives, electronic components and circuits, robotics and control, sensors and signal processing for executing their project works. The program must clearly demonstrate where standards and constraints are taught and how they are integrated into the design component of the project. Each group will have 4-5 students. Each group should select their team leader and maintain daily diary. Each Group will work under mentorship of a Faculty supervisor. Each group must meet the assigned supervisor (2hrs slot/week) till the end of the semester (record of attendance will be maintained), as per the time slot which will be provided to them by the respective supervisor. This is mandatory requirement for the fulfillment of the attendance as well as the successful completion of the project. The faculty supervisor of the project will continuously assess the progress of the works of the assigned groups. Some part of the analysis and design of the system will be done in the first section of project in semester VI. The second section would comprise of completion of the project in semester VII in which each team will have to submit a detailed report of the project along with a poster.

- b. Prerequisites

- UTA013, UTA014, UEE401, UEE504, UEE505, UEI501, UEE605, UEI610, UEE801.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. **Specific goals for the course**

- a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- To identify design goals and analyse possible approaches to meet given specifications with realistic engineering constraints.
- To design an electrical engineering project implementing an integrated design approach applying knowledge accrued in various professional courses.
- To perform simulations and incorporate appropriate adaptations using iterative synthesis.
- To use modern engineering hardware and software tools.
- To work amicably as a member of an engineering design team.
- To improve technical documentation and presentation skills.

b. Specific outcomes addressed by the course.

A2	Demonstrate and apply knowledge of fundamentals, scientific and/or engineering principles towards solving engineering problems.
B1	Identify the constraints, assumptions and models for the experiments.
B3	Analyze and validate experimental results using appropriate techniques.
C1	Ability to formulate problems for electrical systems.
C4	Build prototypes, wherever needed, that meet design specifications.
D1	Share responsibility and information schedule with others in team.
D2	Participate in the development and selection of ideas.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
G2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
H1	Aware of societal and global changes due to engineering innovations.
I1	Able to use resources to adopt new technologies not included in curriculum.
J2	Recognize the impact of engineering decisions on energy resources and environment.

SO1.1	Ability to identify and formulate problems for electrical systems
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO3.2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
SO4.1	Recognize the impact of engineering decisions on energy resources and environment.
SO4.4	Aware of societal and global changes due to engineering innovations.
SO5.1	Share responsibility and information schedule with others in team.
SO5.2	Participate in the development and selection of ideas on a team whose members together provide leadership.
SO6.1	Identify the constraints, assumptions and models for the experiments.
SO6.2	Analyze and validate experimental results using appropriate techniques.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Capstone Project

1. **Course number and name:** UEE804; Operation and Control of Power Systems
2. **Credits and contact hours:** 4.5 and 6
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Chakraborti A., Soni, M.L., Gupta, P.V. and Bhatnagar, U.S., *a Text Book on Power System Engineering*, Dhanpat Rai and Co. (P) Ltd. (2008).
- Nagrath, I.J. and Kothari, D.P., *Power System Engineering*, Tata McGraw Hill (2007).
- Stevenson, W.D., *Power System Analysis*, McGraw Hill (2007).
- Kothari, D.P., Dhillon, J.S., *Power System Optimization*, PHI Learning (2010).
- Allen J. Wood, Bruce F. Wollenberg and Gerald B. Sheble, *Power Generation, Operation and Control*, Wiley-Interscience (2013).
- Kimbark, E. W., *Power System Stability, Volumes-I*, IEEE Press (1995).
- Jizhong Z., *Optimization of power system operation*, Edition Wiley (1996).
- Elgerd, O. *Electric Energy Systems Theory*, McGraw Hill Education Private Limited (2001).

a. Other supplemental materials

- Nil

5. Specific course information

a. Brief description of the content of the course (catalog description)

Economic Operation of Power Systems: Fuel consumption, Characteristics of thermal unit, Incremental fuel rate and their approximation, Minimum and maximum power generation limits.

Economic Dispatch: Economic dispatch problem with and without transmission line losses, Unit Commitment, methods for their solutions.

Hydrothermal Co-ordination: Hydro-scheduling, Plant models, Scheduling problems, Hydrothermal scheduling problems and its approach.

Power System Control: Ideas of load frequency and voltage control, Reactive power control, Block diagrams of P-f and Q-V controllers, ALFC control, Static and dynamic performance characteristics of ALFC and AVR controllers, Excitation systems model, concept of area and Tie-line operations.

Power System Security: Factors affecting security, Contingency analysis, Network sensitivity, correcting the generation dispatch by using sensitivity method and linear programming.

Small Scale Stability Analysis: d-q model of generator, State space representation, Eigen value and participation factor analysis.

Voltage Stability: Basic concepts, Voltage collapse, P-V and Q-V curves, Impact of load, Static and dynamic analysis of voltage stability, Prevention of voltage collapse.

Laboratory Work: Simulation of thermal scheduling with and without losses, Unit commitment by dynamic programming, simulation of hydro-thermal scheduling by gradient method, Stability analysis of single area frequency control, Bias control of two area system and AVR.

- b. Prerequisites
 - UEE304, UEE605.
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Develop small scale model of alternator, excitation and governing systems.
- Decide the scheduling of thermal units and hydro-thermal units for overall economy.
- Design and apply control for frequency and voltage of power system represented by multi area.
- Comprehend power system security and contingency.
- Computation of small scale and voltage stability.

- b. Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
A3	Apply numerical and statistical methods in analyzing data
B1	Identify the constraints, assumptions and models for the experiments.
C1	Ability to formulate problems for electrical systems.
C3	Work with real time systems within realistic constraints.
C4	Build prototypes where needed, that meet design specifications.
E1	Classify information to identify engineering problems.
E3	Use analytical, computational and/or experimental methods to obtain solutions
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
H3	Evaluate engineering solutions considering environmental constraints.
J2	Recognize the impact of engineering decisions on energy resources and environment.
K3	Able to analyze engineering problems using software tools.
K2	Able to develop systems for engineering applications.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO6.1	Identify the constraints, assumptions and models for the experiments.
SO1.1	Ability to identify and formulate problems for electrical systems
SO2.2	Work with real time systems within realistic constraints
SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO1.1	Ability to identify and formulate problems for electrical systems
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO3.1	Prepare and present variety of documents such as project or laboratory

	reports and inspection reports with discipline specific standards.
SO4.2	Evaluate engineering solutions considering environmental constraints.
SO4.1	Recognize the impact of engineering decisions on energy resources and environment.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Economic Operation of Power Systems
- Economic Dispatch
- Hydrothermal Co-ordination
- Power System Control
- Power System Security
- Small Scale Stability Analysis
- Voltage Stability

SEMESTER – VIII

1. **Course number and name:** UEE891; Project
2. **Credits and contact hours:** 20 and not applicable
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical & Instrumentation Engineering
4. **Text book, title, author, and year**
 - May be required as per the selected project title
 - a. Other supplemental materials
 - Electrical equipments, sensors and transducers, components etc.

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Course Objectives: The project semester is aimed at developing the undergraduate education programme in Electrical Engineering to include a practical training in a professional engineering set up (a company, top educational institution, research institute etc.) hereafter referred to as host "organization" as deemed appropriate. The participating organizations are selected that are either already visiting Thapar Institute of Engineering and Technology for placement or are forming new relationships of mutual benefit. The project semester gives the student the opportunity to translate engineering theory into practice in a professional engineering environment. The technical activity in the project semester should be related to both the student's engineering studies and to the host organization's activities and it should constitute a significant body of engineering work at the appropriate level. It should involve tasks and methods that are more appropriately completed in a professional engineering environment and should, where possible, make use of human and technology resources provided by the organization. It consolidates the student's prior learning and provides a context for later research studies. The student remains a full-time registered student at Thapar University during the project semester and this activity is therefore wholly distinct from any industrial interactions which may occur over vacation periods.

Assessment Details: Each student is assigned a faculty supervisor who is responsible for managing and assessment of the project semester. The faculty supervisor monitors the student's progress in a semester and interacts with the industry mentor during his/her visit to the host organization twice. This includes a Reflective Diary which is updated throughout the project semester, an Interim Project Report, a Final Report with Learning Agreement/Outcomes and a Final Presentation & Viva which involves the faculty Supervisor and some other members from the department. The mentor from the host organization is asked to provide his assessment on the designated form. The faculty supervisor is responsible for managing and performing the assessment of the project semester experience.

- b. Prerequisites
 - UEE401, UEE504, UEE505, UEE501, UEE605, UEE610, UEE801.
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required
6. **Specific goals for the course**
- a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Acquire knowledge and experience of software and hardware practices in the area of project.

- Carry out design calculations and implementations in the area of project.
 - Associate with the implementation of the project requiring individual and team work skills.
 - Communicate their work effectively through writing and presentation.
 - Demonstrate the knowledge of professional responsibilities and respect for ethics.
- b. Specific outcomes addressed by the course.

A2	Demonstrate and apply knowledge of fundamentals, scientific and/or engineering principles towards solving engineering problems.
B1	Identify the constraints, assumptions and models for the experiments.
C1	Ability to formulate problems for electrical systems.
C3	Work with real time systems within realistic constraints.
C4	Build prototypes, wherever needed, that meet design specifications.
D1	Share responsibility and information schedule with others in team.
D2	Participate in the development and selection of ideas.
F2	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and well-being of society
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
G2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
H1	Aware of societal and global changes due to engineering innovations.
H2	Analyze economic tradeoffs in engineering systems.
I1	Able to use resources to adopt new technologies not included in curriculum.
K2	Able to develop systems for engineering applications.
K3	Able to analyze engineering problems using software tools.

SO1.1	Ability to identify and formulate problems for electrical systems
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO2.2	Work with real time systems within realistic constraints
SO2.3	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO3.2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
SO3.3	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
SO4.3	Analyze economic trade-offs in engineering systems
SO4.4	Aware of societal and global changes due to engineering innovations.
SO5.1	Share responsibility and information schedule with others in team.
SO5.2	Participate in the development and selection of ideas on a team whose members together provide leadership.
SO6.1	Identify the constraints, assumptions and models for the experiments.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

SO7.1	Able to use resources to adopt new technologies not included in curriculum.
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7. Brief list of topics to be covered

- Project

1. **Course number and name:** UEE806; Alternate Sources of Energy
2. **Credits and contact hours:** 4.0 and 5
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical & Instrumentation Engineering.

4. **Text book, title, author, and year**

Text Books / Reference Books

- Rai, G.D., *Non-Conventional Energy Sources*, Khanna Publishers (2005).
- Rao, S. and Parulekar, B.B., *Energy Technology: Non-Conventional, Renewable and Conventional*, Khanna Publishers (2005).
- Wadhwa, C.L., *Generation, Distribution and Utilization of Electric Energy*, New Age International (P) Limited, Publishers (2007).
- Simon, Christopher A., *Alternate Source of Energy*, Rowman and Littlefield Publishers Inc. (2007).
- Venikov, V.A. and Putyain, E.V., *Introduction to Energy Technology*, Mir Publishers (1990).
- a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Introduction: Global and national energy scenarios, concept of energy services, patterns of energy supply, energy resource availability, cultural, economic and national security aspects of energy consumption, forms and characteristics of renewable energy sources, energy classification, source and utilization, thermodynamic power cycles and binary cycles.

Solar Energy: Solar radiation, flat plate collectors, solar concentration, thermal applications of solar energy, photovoltaic technology and applications, energy storage.

Biomass Energy: Energy from biomass, thermo chemical, biochemical conversion to fuels, biogas and its applications.

Wind Energy: Wind characteristics, resource assessment, horizontal and vertical axis wind turbines, electricity generation and water pumping, Micro/Mini hydropower system, water pumping and conversion to electricity, hydraulic pump.

Other Alternate Sources: Ocean thermal energy conversion, Geothermal, Tidal, Wave energy, MHD, Fuel cells, environmental issues of energy services.

Standalone generating units: Synchronous generator and induction generator, operation and characteristics, voltage regulation, lateral aspects of renewable energy technologies and systems.

- b. Prerequisites

- UEC001, UEN002.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. **Specific goals for the course**

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Explain the basic renewable energy sources like solar, wind, biomass etc.
- Explain various advantages and disadvantages of renewable energy sources.

- Familiarization with different standalone, off grid energy sources.
 - Explain different technology associate with solar, wind, biomass and other renewable energy sources.
 - Describe the working of micro/mini hydropower system.
- b Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
H3	Evaluate engineering solutions considering environmental constraints.
K1	Able to operate modern engineering equipment.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO4.2	Evaluate engineering solutions considering environmental constraints.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Solar Energy
- Biomass Energy
- Wind Energy
- Other Alternate Sources
- Standalone generating units

1. **Course number and name:** UEE892; Design Project
2. **Credits and contact hours:** 13 and not applicable
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical and Instrumentation Engineering
4. **Text book, title, author, and year**
 - May be required as per the selected project title
 - a. Other supplemental materials
 - Electrical equipment, sensors and transducers, components etc.
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Course Objectives: The design project is introduced in Electrical Engineering undergraduate program to include a practical training in the university itself for six months. The project offers the student the opportunity to demonstrate engineering theory into practice under the supervision of a faculty supervisor in electrical engineering department. The students are also offered with two courses. The technical activity in the project semester should be related to both the student's engineering studies and the faculty supervisor's guide lines to make working model in the area of application of electrical engineering. It involves tasks and methods that are more appropriately completed in an academic practical environment and should, where possible, make use of human and technology resources provided by the university. It consolidates the student's prior learning and provides a context for later research studies. The student remains a full-time registered student at Thapar Institute of Engineering and Technology during the project semester and this activity is, therefore, wholly distinct from any industrial interactions which may occur over vacation periods.

Assessment Details: Each student is assigned a faculty supervisor who is responsible for managing and assessment of the alternate project semester. The faculty supervisor guides the students till the end of semester and monitors the student's progress throughout the same. This includes a Reflective Diary which is updated throughout the alternate project semester, an Interim Project Report, a Final Report with Learning Agreement/Outcomes and a Final Presentation & Viva which involves the faculty Supervisor and some other faculty members from the department.

- b. Prerequisites
 - UEE401, UEE504, UEE505, UEI501, UEE605, UEI610, UEE801.
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required
6. **Specific goals for the course**
 - a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Acquire knowledge and experience of software and hardware practices in the area of project.
- Carry out design calculations and implementations in the area of project.
- Associate with the implementation of the project requiring individual and teamwork skills.

- Communicate their work effectively through writing and presentation.
- Demonstrate the professional responsibilities and respect for ethics in university ambiance.

b. Specific outcomes addressed by the course.

A2	Demonstrate and apply knowledge of fundamentals, scientific and/or engineering principles towards solving engineering problems.
B1	Identify the constraints, assumptions and models for the experiments.
C1	Ability to formulate problems for electrical systems.
C3	Work with real time systems within realistic constraints.
C4	Build prototypes, wherever needed, that meet design specifications.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
G2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
H1	Aware of societal and global changes due to engineering innovations.
H2	Analyze economic tradeoffs in engineering systems.
I1	Able to use resources to adopt new technologies not included in curriculum.
I3	Recognize the need to embrace personal responsibility for lifelong learning.
K2	Able to develop systems for engineering applications.
K3	Able to analyze engineering problems using software tools.

SO1.1	Ability to identify and formulate problems for electrical systems
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO2.2	Work with real time systems within realistic constraints
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO3.2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
SO4.3	Analyze economic trade-offs in engineering systems
SO4.4	Aware of societal and global changes due to engineering innovations.
SO6.1	Identify the constraints, assumptions and models for the experiments.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Design Project

1. **Course number and name:** UEI805; Environmental Instrumentation
2. **Credits and contact hours:** 3.0 and 3
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electronics and Communication Engineering.

4. **Text book, title, author, and year**

Text Books / Reference Books

- Bhatia, H.S., *a Text Book in Environmental Pollution and control*, Galgotia Publication(1998).
- Dhameja, S.K., *Environmental Engineering and Management*, S.K Kataria (2000).
- Rao, M.N. and Rao, H.V., *Air Pollution*, Tata McGraw Hill (2004).
- Rao. C.S., *Environmental Pollution Control*, New Age International (P) Limited, Publishers (2006) 2nd ed
- a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Air Pollution: Impact of man of the environment: An overview, Air pollution sources and effects, Metrological aspect of air pollutant dispersion, Air pollution sampling and measurement, Air pollution control methods and equipment, Air sampling techniques, soil pollution and its effects, Gas analyzer, Gas chromatography, Control of specific gaseous pollutants, Measurement of automobile pollution, Smoke level meter, CO/HC analyzer.

Water pollution: Sources and classification of water pollution, Waste water sampling and analysis, Waste water sampling techniques and analyzers: Gravimetric, Volumetric, Calometric, Potentiometric, Flame photometry, Atomic absorption spectroscopy, Ion chromatography, Instruments used in waste water treatment and control, Latest methods of waste water treatment plants.

Pollution Management: Management of radioactive pollutants, Noise level measurement techniques, Noise pollution and its effects, Solid waste management techniques, social and political involvement in the pollution management system

- b. Prerequisites
- UEN002, UEE403.
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
- Required

6. **Specific goals for the course**

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- explain sources and effects of air and water pollutants
- explain air pollution sampling and measurement techniques
- explain water sampling and analysis techniques
- explain solid waste management and noise level measurement techniques

b Specific outcomes addressed by the course.

H3	Evaluate engineering solutions considering environmental constraints.
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SO4.2	Evaluate engineering solutions considering environmental constraints.
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7. Brief list of topics to be covered

- Air pollution
- Water pollution
- Pollution management

1. **Course number and name:** UEE893; Startup Semester
2. **Credits and contact hours:** 20 and not applicable
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical and Instrumentation Engineering
4. **Text book, title, author, and year**
 - May be required as per the selected project title
 - a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)
- a. Activities during the Startup semester
 - Fundamentals of 'Entrepreneurship & Innovation'
 - Opportunity identification and evaluation, Customer validation
 - Developing a Business Model Canvas
 - Business Development Process related to the startup, relating theoretical framework with the business idea, Industry dynamics, opportunity canvas and regulatory aspects related to the business idea.
 - Design thinking
 - Technical development
 - Financial management
 - Entrepreneurial Marketing
 - Interaction with existing Startups and pitching of projects,
 - Presentation of Prototype/Working model/useful App or a working Software

Assessment Details: Each student is assigned a faculty supervisor and industry mentor. Faculty supervisor is responsible for managing and assessment of the Startup semester. The faculty supervisor monitors the student's progress in a semester and interacts with the industry mentor during his/her visit to the host organization twice.

The semester includes maintenance of a Reflective Diary, which is updated throughout the startup semester, an Interim Project Report, a Final Report with Learning Agreement/Outcomes and a Final Presentation & Viva, which involves the faculty Supervisor, and some other members from the department.

The mentor from the host organization is asked to provide the assessment on a designated form. The faculty supervisor is responsible for managing and performing the assessment of the startup semester experience.

- b. Prerequisites
 - None
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Required

6. Specific goals for the course

- a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Demonstrate an ability to develop a business plan.
- Carry out design calculations/simulations and implementations in the area of project.
- Develop a prototype/working model/software application.
- Comprehend the fundamentals of business pitching.
- Demonstrate the knowledge of professional responsibilities and respect for ethics.

b. Specific outcomes addressed by the course.

F1	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
F2	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and well-being of society.
G1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
G2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
H1	Aware of societal and global changes due to engineering innovations.
H2	Analyze economic tradeoffs in engineering systems.
I1	Able to use resources to adopt new technologies not included in curriculum.
I3	Recognize the need to embrace personal responsibility for lifelong learning.

SO2.3	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.
SO3.1	Prepare and present variety of documents such as project or laboratory reports and inspection reports with discipline specific standards.
SO3.2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
SO3.3	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
SO4.3	Analyze economic trade-offs in engineering systems
SO4.4	Aware of societal and global changes due to engineering innovations.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.
SO7.3	Recognize the need to embrace personal responsibility for lifelong learning

7. Brief list of topics to be covered

- Startup ideas and Innovation

ELECTIVE-I

1. **Course number and name:** UCS049; Data Structures and Algorithms
2. **Credits and contact hours:** 3 and 4
3. **Instructor's or course coordinator's name:** Faculty from the Department of Computer Science and Engineering
4. **Text book, title, author, and year**
 - *Seymour Lipschutz Data Structures, TATA McGraw Hill (2016).*
 - *Corman, Leiserson&Rivest, Introduction to Algorithms, MIT Press (2009).*
 - *Narasimha Karumanchi, Data Structures and Algorithms Made Easy (2014).*
 - *Sahni, Sartaj, Data Structures, Algorithms and Applications in C++, Universities Press (2005) 2nded.*
- a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Introduction and Overview: Basic Terminology, Elementary Data Organization, Data Structures, Control Structures, Asymptotic Notations for Algorithms, Big O notation: formal definition and use, Little o, big omega and big theta notation, Arithmetic Expressions, Polish

Notations, Arrays, Records, Pointers, Storing Strings, String Operations, Pattern Matching Algorithms, Stacks, Queues, Recursion, Towers of Hanoi.

Searching and Sorting: Linear Arrays, Traversing and Searching in Linear Arrays, Inserting and Deleting, Bubble Sort, Linear Search, Binary Search, Insertion Sort, and Selection Sort.

Non-Linear Data Structures: Trees, Binary Trees, Traversing Binary Trees, Binary Search Trees, Searching and Inserting in Binary Search Trees, Deleting in a Binary Search Tree, Preorder, Postorder and Inorder Traversal, Heaps, Graph, Graph Algorithms, Breadth First Search, Depth First Search.

Linked List: Introduction, Insertion into a linked list, Deletion into a linked list. Stack, Queues, trees using linked list, Hashing, Hash Functions.

- b. Prerequisites
 - UTA017, UTA018.
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Selected Elective
6. **Specific goals for the course**
 - Specific outcomes of instruction

After the completion of the course, the students will be able to:

- To implement and learn the basic data structures and learn the appropriate algorithmic approach to a problem and implement various search and sorting techniques.
- To implement the basic data structures and solve problems using fundamental algorithms.

- To implement various search and sorting techniques.
- To analyze the complexity of algorithms, to provide justification for that selection, and to implement the algorithm in a particular context.
- To analyze, evaluate and choose appropriate data structure and algorithmic technique to solve real-world problems.

b. Specific outcomes addressed by the course

A2	Demonstrate and apply knowledge of fundamentals, scientific and/or engineering principles towards solving engineering problems.
B1	Identify the constraints, assumptions and models for the experiments.
C3	Work with real time systems within realistic constraints.
K2	Able to develop systems for engineering applications.
K3	Able to analyze engineering problems using software tools.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.2	Work with real time systems within realistic constraints
SO6.1	Identify the constraints, assumptions and models for the experiments.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Stacks & Queues
- Linked lists
- Sorting
- Searching
- Nonlinear data structures

1. **Course number and name:** UEE631; HVDC Transmission Systems
2. **Credits and contact hours:** 3.0 and 3
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Arrillaga, J., *HVDC Transmission*, IEEE Press (2007).
- Edwart, K., *Direct Current Transmission (Vol. 1)*, John Wiley and Sons (2008).
- Padiyar, K.R., *HVDC Power Transmission System*, New Age International (P) Limited, Publishers (2008).
- Arrillaga, J. and Smith, B.C., *AC to DC Power System Analysis*, IEE Press (2008).
- a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

DC power transmission technology: Introduction, Comparison of HVAC and HVDC transmission system, Applications of DC transmission, Description of DC transmission system, Configurations, Modern trends in DC transmission.

Analysis of HVDC converters: Pulse number, Choice of converter configuration, Simplified analysis of Graetz circuit, Converter bridge characteristics, Characteristics of a twelve-pulse converter, Detailed analysis of converters with and without overlap.

Converter and HVDC system control: General, Principles of DC link control, Converter control characteristics, System control hierarchy, Firing angle control, Current and extinction angle control, Starting and stopping of DC link, Power control, Higher level controllers.

Converter faults and protection: Converter faults, Protection against over-currents, Over-voltages in a converter station, Surge arresters, Protection against over-voltages.

Smoothing reactor and DC line: Introduction, Smoothing reactors, DC line, Transient over voltages in DC line, Protection of DC line, DC breakers, Monopolar operation, Effects of proximity of AC and DC transmission lines.

Reactive power control: Reactive power requirements in steady state, Sources of reactive power, Static VAR systems, Reactive power control during transients, Harmonics and filters, Generation of harmonics, Design of AC filters, DC filters.

Component models for the analysis of ac/dc systems: General, Converter model, Converter control, Modelling of DC network, Modelling of AC networks.

Power flow analysis in AC/DC systems: General, Modelling of DC links, Solution of DC load flow, Discussion, Per unit system for DC quantities.

- b. Prerequisites

- UEE504.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Selected Elective

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Choose intelligently AC and DC transmission systems for the dedicated application(s).
 - Identify the suitable two-level/multilevel configuration for high power converters.
 - Select the suitable protection method for various converter faults.
 - Identify suitable reactive power compensation method.
 - Decide the configuration for harmonic mitigation on both AC and DC sides.
- b. Specific outcomes addressed by the course.

E1	Classify information to identify engineering problems.
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SO1.1	Ability to identify and formulate problems for electrical systems
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7. Brief list of topics to be covered

- DC power transmission technology
- Analysis of HVDC converters
- Converter and HVDC system control
- Converter faults and protection
- Smoothing reactor and DC line
- Reactive power control
- Component models for the analysis of ac/dc systems
- Power flow analysis in AC/DC systems

1. **Course number and name:** UEE632; Power Generation and Economics
2. **Credits and contact hours:** 3.0 and 3
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Chakrabarti, A., Soni, M.L., Gupta, P.V. and Bhatnagar, U.S., *a Text Book on Arora, S.C and Domkundawar, S., a course in Power Plant Engineering, Dhanpat Rai (2002).*
- Deshpande, M.V., *Power Plant Engineering, Tata McGraw Hill (2004).*
- Gupta, B.R., *Generation of Electrical Energy, S. Chand (1998).*
- Deshpande, M.V., *Electrical Power System Design, McGraw Hill (2004).*
- Wood, A.J. and Wollenberg, B.F., *Power Generation and Control, John Wiley (2004).*
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Introduction: Energy sources and their availability, Principle types of power plants, their special features and applications, Present status and future trends.

Hydro Electric Power Plants: Essentials, Classifications, Hydroelectric survey, Rainfall run-off, Hydrograph, Flow duration curve, Mass curve, Storage capacity, Site selection, Plant layout, various components, Types of turbines, Governor and speed regulation, Pumped storage, Small scale hydro-electric plants (mini and micro).

Thermal Power Plant: General developing trends, Essentials, Plant layout, Coal-its storage, Preparation, Handling, Feeding and burning, Cooling towers, Ash handling, Water treatment plant, High pressure boilers and steam turbines, Components of thermal power plant.

Gas Turbine Power Plants: Field of use, Components, Plant layout, Comparison with steam power plants, combined steam and gas power plants.

Nuclear Power Plant: Nuclear fuels, Nuclear energy, Main components of nuclear power plant, Nuclear reactors types and applications, Radiation shielding, Radioactive and waste disposal safety aspect.

Non-Conventional Power Generation: Geothermal power plants, Electricity from biomass, Direct energy conversion systems (Solar and Wind), Thermo-electric conversion system, Fuel cells, Magneto-Hydro dynamic system.

Cogeneration: Definition and scope, Cogeneration technologies, Allocation of costs, Sale of electricity and impact on cogeneration.

Power Plant Economics: Cost of electrical energy, Selection of type of generation and generation equipment, Performance and operating characteristics of power plants, Economic scheduling principle, Load curves, Effect of load on power plant design, Load forecasting, electric tariffs, Peak load pricing.

- b. Prerequisites

- UEE406.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Selected Elective

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Apply knowledge of India's power scenario, power system structure and related agencies.
- Explain about various types of power plants i.e., hydro, thermal, gas and nuclear.
- Harness power from conventional and renewable sources.
- Select the methods and size of plant generating power for overall economy.
- Decide the tariff structure for different type of users.

- b. Specific outcomes addressed by the course.

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
H1	Aware of societal and global changes due to engineering innovations.
H2	Analyze economic tradeoffs in engineering systems.
H3	Evaluate engineering solutions considering environmental constraints.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO4.4	Aware of societal and global changes due to engineering innovations.
SO4.3	Analyze economic trade-offs in engineering systems
SO4.2	Evaluate engineering solutions considering environmental constraints.

7. Brief list of topics to be covered

- Hydro Electric Power Plants
- Thermal Power Plant
- Gas Turbine Power Plants
- Nuclear Power Plant
- Non-Conventional Power Generation
- Cogeneration
- Power Plant Economics

1. **Course number and name:** UEE633; Generalized Theory of Electrical Machines
2. **Credits and contact hours:** 3.0 and 3
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- *Kraus, P.C., Analysis of Electric Machine, McGraw Hill (2000).*
- *Bimbhra, P.S., Generalized Theory of Electric Machines, Khanna Publishers (2006).*
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Introduction: Common essential constructional and operational features of electrical machines, basic two pole machine representation of different types of electrical machines, Kron's primitive machine, Voltage equations in matrix form for Kron's primitive machine, Impedance matrix.

Linear Transformations in Machines: Reference frame theory, 3-phase to 2-phase transformation, Transformation from rotating axes to stationary axes, Physical concept of park's transformation, Volt-ampere and torque equations, Space vector concept.

DC Machine: Transfer function for DC machine (Shunt, Series and compound), Linearization technique, Analysis under motoring and generating mode, Dynamic analysis.

Synchronous Machine: General machine equation in different frame, Dynamic analysis, Power angle characteristics, Phases diagram for cylindrical rotor and salient pole machine, Electromagnetic and reluctance torque, Electric braking of synchronous machine.

3-phase Induction Machine: Performance equations in different rotating frames, Equivalent circuit, Different inductance, Effect of voltage and frequency on the performance, Braking, Unbalance operations.

Advanced Machines: 1-phase synchronous motor, 2-phase servomotor, AC tachometers, Switched reluctance motor, Brushless DC machine.

- b. Prerequisites
 - UEE301, UEE401.
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Selected Elective

6. **Specific goals for the course**

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Express the revolving field and reference frame theory
- Develop mathematical model of three-phase AC machines and parameters in different reference frame
- Simulate the transient performance of three-phase ac machines in different reference frames.
- Investigate the transient performance of different DC machines.
- Select special purpose small machines for different applications.

Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
E1	Classify information to identify engineering problems.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.1	Ability to identify and formulate problems for electrical systems

7. Brief list of topics to be covered

- Linear Transformations in Machines
- DC Machine
- Synchronous Machine
- 3-phase Induction Machine
- Advanced Machines

1. **Course number and name:** UEE634; Real Time Power Systems
2. **Credits and contact hours:** 3 and 4
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering
4. **Text book, title, author, and year**
 - N. Hatziargyriou "Microgrids: Architectures and Control", Wiley-IEEE Press, January (2014).
 - HIL System catalogues; Opal-RT, RTDS and Typhoon.
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Introduction: Hardware-in-loop simulation systems, distributed control architecture, reliability enhancement by redundancy, Real time operating systems: Features, primary components, Structured design of real time systems.

Developing a mathematical model for Power system and control, Mathematical model of the real environment, Design of hardware device meant to be used in HIL.

Testing and parameter adjustment for real time implementation of real-time simulator, Design of desired control schemes for AC and DC electrical machine drives and other applications: Micro-grid and renewable and its testing in HIL.

Real time control strategy based on FPGA, dSpace, Understanding four-quadrant amplifier for HIL system.

Lab work: Off-line simulations for the various experiments related to hardware in-the-loop simulation system to predict ahead of conducting the lab experiment the operating characteristics and compare results; Microgrid operation and control using HIL; Implement hardware such as PV and Wind system on the simulated grid to test hardware device in the real environment.

- b. Prerequisites

- UEE406, UEE504, UEI501.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. **Specific goals for the course**

- a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Demonstrate about Hardware-in-loop simulation systems.
- Explain about mathematical model for power system and control in real environment.
- Design control schemes for AC and DC electrical machine drives.
- Demonstrate the concepts of real time control strategy based on FPGA, dSpace.

- b. Specific outcomes addressed by the course.

B3	Analyze and validate experimental results using appropriate techniques.
C1	Ability to formulate problems for electrical systems.
C2	Design process to satisfy project objective for electrical systems.
C3	Work with real time systems within realistic constraints.
C4	Build prototypes, wherever needed, that meet design specifications.

SO1.1	Ability to identify and formulate problems for electrical systems
SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO2.2	Work with real time systems within realistic constraints
SO6.2	Analyze and validate experimental results using appropriate techniques.

7. Brief list of topics to be covered

- Hardware-in-loop
- Real-time operating systems
- Design of control systems

ELECTIVE-II

1. **Course number and name:** UEC408; Probability and Information Theory
2. **Credits and contact hours:** Credits: 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electronics and Communication Engineering
4. **Text book, title, author, and year**
 - Athanasios Papoulis, *Probability Random Variables and Stochastic Processes*, McGraw-Hill (1984) 4th ed.
 - Daigle, John N., *Queueing theory with applications to packet telecommunication* Springer (2005) 2nd ed.
 - Sklar Bernard, *Digital Communications: Fundamentals and Applications*, Prentice Hall (2001)
 - Peebles, P.Z., *Probability, random variables, and random signal principles*, McGraw-Hill (1980) 2nd ed.
 - Bertsekas, Dimitri P., Gallager, Robert G., *Data networks*, Prentice-Hall (1987) 4th ed.
 - Larson, A. and Schubert, B. O., *Stochastic Processes*, vol. I and II, Holden-Day (1979) 5th ed.
 - Gardener, W., *Stochastic Processes*, McGraw Hill (1986) 2nd ed.
 - David J. C. Mackay, *"Information Theory, Inference and Learning Algorithms"*, Cambridge University Press, 2003
 - a. Other supplemental materials
 - *IEEE Transactions on Information Theory*

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Probability Theory: Review of Probability, Bernoulli Trials, Bernoulli's Theorem, Concepts of Random Variables, Distribution and Probability Density Functions, Conditional Distributions, Binomial distribution, Random variables Functions of One Random Variable, its Distribution, Mean and Variance, Moments, Characteristic Functions; Two Functions of Two Random Variables, Joint Moments, Joint Characteristic Functions, Conditional Distributions, Conditional Expected Values, Normality, Center Limit Theorem, and Bayes' Theorem.

Stochastic Processes: Systems with Stochastic Inputs, Power Spectral Analysis of I/O Signals, Poisson Points, Cyclostationary Processes, Poisson Sum Formula, Ergodicity, Mean Square Estimation, Markov Chains, and Random-Walk Model.

Estimation & Hypothesis Testing: Time and Ensemble Averages, Covariance Functions. Simple Binary Hypothesis Tests, Decision Criteria, Neyman Pearson Tests, Bayes' Criteria, z-Score, and p-Value Test

Statistical Modeling of Noise: Probability Density of a Jointly-Gaussian Random Vector, Wide-Sense-Stationary (WSS) Processes, Poisson Process Noise, Noise Statistics in Linear Time-Invariant Systems, Noise Power Spectral Densities, Signal-to-Noise-Ratio in Presence of AWGN and Interferences.

Introduction to Queueing Systems: Characteristics of Queueing Process, Poisson Points in Random Intervals, Renewal Processes, Arrivals and Departures, Little's Theorem, Immediate Service and M/G/1 system.

Information Theory: Unit of Information, Rate of Information, Entropy, Joint Entropy and Conditional Entropy, Mutual Information, Shannon-Hartley Theorem, Channel Capacity Calculations of Different Channels, Source Coding- Coding Efficiency, Shannon-Fano

Coding, Huffman Coding.

b. Prerequisites

- UMA004, UMA007.

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Selected Elective

6. Specific goals for the course

a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Apply the probabilistic concepts as well as properties of the random variables
- Perform the spectral analysis of stationary stochastic processes, for the modelling of real-time desired signals and spurious-signals/noise
- Incorporate the estimation and hypothesis testing principles to find remedial solutions
- Utilize the features/characteristics of queueing theory in communication systems
- Employ information theory and coding concepts, to improve information symbol transmission rate, and also use it for data compression

b. Specific outcomes addressed by the course

A1	An ability to identify, formulate and solve complex engineering problems by applying principles of engineering, science and mathematics.
E.1	An ability to identify, formulate and solve complex engineering problems by applying principles of engineering, science and mathematics.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO6.3	Able to analyse engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Probability theory
- Stochastic processes
- Estimation & hypothesis testing
- Statistical modeling of noise
- Introduction to queueing systems
- Information theory

1. **Course number and name:** UEE521; Electric Machine Design
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical and Instrumentation Engineering.

4. **Text book, title, author, and year**

Text Books / Reference Books

- Ramamoorthy, M., *Computer Aided Design of Electrical Equipment*, Eastern Press Private Limited (1989).
- A.K. Sawhney, *a Course in Electrical Machine Design*, Dhanpat Rai & CO. (2013).
- Say, M.G., *Design and Performance of Machines*, CBS Publications (1981).
- Hamdi, E.S., *Design of Small Electrical Machine*, John Wiley and Sons (1994).
- Smith, S.P. and Say, M.G., *Electrical Engineering Design Manual*, Chapman and Hall (1984).
- Walker, J.H., *Large AC Machines: Performance and Operation*, BHEL (1997).

- a. Other supplemental materials

- Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Introduction: Design of Machines, Factors, limitations, Modern trends. Materials: Conducting, magnetic and insulating materials.

Magnetic Circuits: Calculations of mmf for air gap and teeth, real and apparent flux densities, iron losses, field form, leakage flux, specific permeance.

Heating and Cooling: Modes of heat dissipation, Temperature gradients, types of enclosures, types of ventilation, conventional and direct cooling, amount of coolants used, Ratings.

Armature Windings: Windings for DC and AC machines and their layout.

Design of Transformers: Output equation, Types of transformer windings, design of core and windings and cooling tank, performance calculations.

Concepts and Constraints in Design of Rotating Machines: Specific loading, output equation and output co-efficient, effects of variation of linear dimension.

Skeleton Design of Rotating Machines: Calculation of D and L for DC, induction and synchronous machines, length of air gap, design of field coils for DC and synchronous machines, selection of rotor slots of squirrel cage induction motors, design of bars and ends, design of rotor for wound rotor for induction motors, design of commutator and inter poles for DC machines.

Computer Aided Design of Electrical Machines: Analysis and synthesis approaches, design algorithms, Introduction to optimization techniques, Implementing computer program for design of three phase induction motor, Familiarization with applicable IEEE/IS standards.

- b. Prerequisites

- UEE301, UEE401.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Selected Elective

6. **Specific goals for the course**

a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Design DC machines.
- Design transformers with reduced losses.
- Calculate the losses and efficiency in the machines.
- Analyze and synthesis of computer aided design of electrical machines.
- Design three phase induction motor.

b. Specific outcomes addressed by the course.

E2	Develop appropriate models to formulate solutions.
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SO1.1	Ability to identify and formulate problems for electrical systems
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7. Brief list of topics to be covered

- Magnetic Circuits
- Heating and Cooling
- Armature Windings
- Design of Transformers
- Concepts and Constraints in Design of Rotating Machines
- Skeleton Design of Rotating Machines
- Computer Aided Design of Electrical Machines

1. **Course number and name:** UEE524; Power Quality Monitoring and Conditioning
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical and Instrumentation Engineering.

4. **Text book, title, author, and year**

Text Books / Reference Books

- Kennedy, B., *Power Quality Primer*, McGrawHill (2000).
- Beaty, H. and Santoso, S., *Electrical Power System Quality*, McGrawHill (2002).
- Bollen, M.H.J., *Power Quality Problems: Voltage Sag and Interruptions*, IEEE Press (2007).
- a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Overview and definition of power quality (PQ): Sources of pollution and regulations, Power quality problems, Rapid voltage fluctuations voltage unbalance, Voltage dips and voltage swells, Short duration outages.

Definitions Voltage sag analysis and mitigation: Sag caused by motor starting, Sag caused by utility fault clearing, Sag mitigation, Sag magnitude and duration calculations, RMS voltage, Calculation in 1-phase systems, Equipment performance in presence of sag, Computers, AC and DC drives.

Harmonics: Effects-within the power system, Interference with communication harmonic measurements, Harmonic elimination.

Harmonic distortion: Power Overview system harmonics, Harmonic analysis, Harmonic sources-the static converters, Transformer magnetization and non-linearities, Rotating machines, Arc furnaces, Fluorescent lighting, Total harmonic distortion, rms and average value calculations, Effects of harmonic distortion.

Principles for controlling harmonics: Locating sources of harmonics, Passive and active filters, Harmonic filter design.

Monitoring power quality: Monitoring essentials, Power quality measuring equipment, Current industry trends.

Power Conditioning: Electric power conditioning, Active and passive filters, IEEE, IEC, ANSI standards, Power acceptability curves.

- b. Prerequisites

- UEE406, UEE504.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Selected Elective

6. **Specific goals for the course**

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Reliably identify the sources of various power quality problems.
- Explain about causes of harmonic and its distortion effect.
- Estimate the impact of various power quality problems on appliances.
- Educate the harmful effects of poor power quality and harmonics.

- Decide the compensators and filters to keep the power quality indices within the standards.
- b. Specific outcomes addressed by the course.

E1	Classify information to identify engineering problems.
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SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
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7. Brief list of topics to be covered

- Overview and definition of power quality
- Definitions Voltage sag analysis and mitigation
- Harmonics
- Harmonic distortion
- Principles for controlling harmonics
- Monitoring power quality
- Power Conditioning

1. **Course number and name:** UEI714; Digital Signal Processing Fundamentals
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Proakis, J.G. and Manolakis, D.G., *Digital Signal Processing*, Prentice Hall of India (1996).
- Rabiner, C.R. and Gold, B., *Theory and Applications of Digital Signal Processing*, Prentice Hall of India (2000)
- Antonion, A., *Digital Filters: Analysis Design and Application*, Prentice-Hall of India (1999).
- Oppenheim, A.V. and Schafer, R.W., *Digital Signal Processing*, Prentice-Hall of India (1998).
- Helmut, U. and Willibald, W., *Protection Techniques in Electrical Engg. Systems*, Marcel Dekker Inc. (2001)
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Introduction: Definition, conversion from analog signal to digital signal, advantages and disadvantages of digital signal processing, Basic Terminologies.

z-Transform: Region of Convergence (ROC), Properties of z-transform, Initial and Final Value theorems, Partial Sum, Parseval's Theorem, z-transform of standard sequences, Inverse z-transform, Pole-Zero plot, System function of LTI system, Causality and Stability in terms of z-transform.

DFT and FFT: Discrete Fourier Series, Discrete Fourier Transform and its Properties, Efficient Computation of DFT using FFT algorithms, Linear Filtering Approach to Computation of DFT.

Digital Filter Structure: Describing Equation, Structures for FIR Systems, Structures for IIR Systems, Representation of Structures using Signal Flow Graph.

Design of Digital Filters: Introduction, Difference between analog and digital filters, Implementation of digital filter, Types of filters, LTI systems as filters, Design of IIR filters from analog filters, IIR filter design using Butterworth Approximation, Frequency transformation, FIR filters design, Least square filter design, Designing digital filter from pole-zero placement, Butterworth filter design using Bilinear transformation, FIR filter design using windows, Design of filters using pole-zero combination, Analysis of coefficient quantization effects in FIR filters, Analysis of round-off errors, Dynamic range scaling, Low sensitivity digital filters, Limit cycles in IIR filters.

Hardware Architecture of DSP Processor: Desirable features of DSP processors, Types of architectures, Internal architecture, Features, System interface and Instruction set of ADSP-21xx, ADSP-21xx, Development tools, TMS DSP processor.

Applications: Dual-tone multi frequency signal detection, Spectral analysis using DFT, Short term DFT, oversampling, Protection.

- b. Prerequisites

- UMA004, UEI501.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Required

6. Specific goals for the course

- a. Specific outcomes of instructions.

After the completion of the course, the students will be able to:

- Explain the digital signal processing concepts and stability analysis of digital system.
- Demonstrate the hardware architecture of DSP Processor.
- Design digital filter and harmonic mitigation.
- Carryout spectrum analysis using DFT.
- Apply DSP concepts for power system purposes such as relaying, protection and metering.

- b. Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
C4	Build prototypes where needed, that meet design specifications.
E1	Classify information to identify engineering problems.
K2	Able to develop systems for engineering applications.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO2.1	Design process to satisfy project objective for electrical systems and build prototypes, wherever necessary, that meet design specifications.
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Z-Transform
- DFT and FFT
- Digital Filter Structure
- Design of Digital Filters
- Hardware Architecture of DSP Processor
- Applications

1. **Course number and name:** UEE803; Load Dispatch and Communication
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering
4. **Text book, title, author, and year**
 - Carlson, A.B., Crilly; P.B., Rutledge, J.C., *Communication Systems*, McGraw Hill ISE (2002).
 - Chakraborti, A., Soni, M.L., Gupta, P.V. and Bhatnagar, U.S., *a Text Book on Power System Engineering*, Dhanpat Rai and Co. (P) Ltd. (2008).
 - Nagrath, I.J. and Kothari, D.P., *Modern Power System Analysis*, Tata McGraw Hill (2006).
 - Pabla, A.S., *Electric Power Distribution*, Tata McGraw Hill (2008).
 - Wang, X., *Modern Power System Planning*, McGraw Hill ISE (1994).
 - Gupta, B.R., *Power System Analysis and Design*, S.Chand and Company Limited (2009).
 - Wadhwa, C.L., *Electrical Power Systems*, New Age International (P) Limited, Publishers (2008).
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - b. Brief description of the content of the course (catalog description)

Estimation of power system: Introduction, State estimation: Topology, Bad data handling, Observability analysis, Least square estimation, Steady state estimation of power systems, Tracking state estimation of power systems, Some computational considerations, External system equivalencing, Treatment of bad data, Network observability, Pseudo measurements.

States of power system: Introduction, Components of modern energy control center.

Load prediction: Estimation of average and trend terms, Periodic and stochastic components, Difference model forecasting.

Load dispatch centres: Introduction, functions, SLDC/RLDC, NLDC, Pre-dispatch, During-dispatch, post-dispatch functions. Supervisory control and data acquisition (SCADA) system RTU, Front end computers, Main computers, Visual display units, Mimic boards, Energy management System-functions.

System management: Introduction, Load management, Load shedding, Voltage/Frequency control, Reactive power management, Grid management, Operational co-operation, Import/Export of energy, Role of tariff in system operation, Maintenance, On-line maintenance, Grid disturbance/Case studies, Software tools.

Communication system: Introduction, Types-PLCC, Microwave, Leased lines, Fibreoptics, Satellite, V-SAT, Communication comparison, Characteristics, Modules, Planning criteria, Selection criteria, RTUs, Modems, Baudrate, Communication protocols, Data exchange, System noise and interference, Integrated communication system.
 - c. Prerequisites
 - UEE304.
 - d. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Selected Elective

6. Specific goals for the course

a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Carry out state estimation and computational methods related to power system.
- Analyze various components of modern energy control centre.
- Carry out the prediction and forecasting related to load.
- Conduct studies related to various component of load dispatch centre.
- Decide the impact of system management.
- Decide the different modules for communication system.

b. Specific outcomes addressed by the course

A1	An ability to identify, formulate and solve complex engineering problems by applying principles of engineering, science and mathematics.
C2	An ability to apply engineering design to produce solutions that meet specified need with consideration of public health, safety, and welfare as well as global, cultural, social, environmental and economic factors.
K1	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

SO1.1	Ability to identify and formulate problems for electrical systems.
SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.2	Work with real-time systems within realistic constraints.
SO2.3	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.
SO4.2	Evaluate engineering solutions considering environmental constraints.
SO6.1	Identify the constraints, assumptions and models for the experiments.
SO6.3	Able to analyse engineering problems and develop systems for engineering applications.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Estimation of power system
- States of power system
- Load prediction
- Load dispatch centres
- System management
- Communication system

1. **Course number and name:** UEE841; Industrial Electronics
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from Department of Electrical and Instrumentation Engineering.

4. **Text book, title, author, and year**

Text Books / Reference Books

- Dubey, G.K., *Power Semiconductor Controlled Drives*, Prentice Hall inc. (1989).
- Paul, B., *Industrial Electronic and Control*, Prentice Hall of India Private Limited (2004).
- J.M.D. Murphy, F.G. Turnbull, *Power Electronic Control of Ac Motors*, Pergamon (1990).
- Sen, P.C., *Thyristor DC Drives*, John Wiley and Sons (1981).
- a. Other supplemental materials
- Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Conventional DC and AC Traction: Electric traction services, Nature of traction load, Coefficient of adhesion, Load sharing between traction motors, Main line and suburban train configurations, Calculation of traction drive rating and energy consumption. Important features of traction drives, Conventional DC and AC traction drives, Diesel electric traction.

Static converters for Traction: Semi-conductor converter-controlled drive for AC traction, Semiconductor chopper-controlled DC traction.

Illumination: Nature of light, Basic laws of illumination, Light sources and their characteristics, Light production by excitation and ionization, Incandescence and fluorescence, Different types of lamps, Their construction, Operation and characteristics, Applications, Latest light sources, Design of illumination systems.

Electric Heating: Introduction to electric heating, Advantages of electric heating, Resistance heating, Temperature control of furnaces, Induction and dielectric heating.

Power Supplies: Performance parameters of power supplies, Comparison of rectifier circuits, Filters, Regulated power supplies, Switching regulators, Switch mode converter.

Power factor Control: Static reactive power compensation, Shunt reactive power compensator, Application of static SCR controlled shunt compensators for load compensation, Power factor improvement and harmonic control of converter fed systems, Methods employing natural and forced commutation schemes, Methods of implementation of forced commutation.

Motor Control: Voltage control at constant frequency, PWM control, Synchronous tap changer, Phase control of DC motor, Servomechanism, PLL control of a DC motor.

- b. Prerequisites
 - UEE504, UEE801.
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Selected Elective

6. **Specific goals for the course**

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Simulate and analyse the semiconductor-controlled ac and DC drive system.
 - Design and develop an illumination system for domestic, industry and commercial sites.
 - Design an electric heating system for industrial purposes.
 - Equip the skill to design and develop a regulated power supply.
 - Simulate and analyse the series and shunt compensators for power factor improvement in drive system.
- b. Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
E2	Develop appropriate models to formulate solutions.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.

7. Brief list of topics to be covered

- Conventional dc and ac Traction
- Static converters for Traction
- Illumination
- Electric Heating
- Power Supplies
- Power factor Control
- Motor Control

1. **Course number and name:** UEE850; Smart Grid
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- *INIEWSKI, Smart Grid Infrastructure and Networking, McGraw-Hill Education India Pvt.Ltd (2012), 1st Edition.*
- [James Momoh](#), *Smart Grid: Fundamentals of Design and Analysis, IEEE Computer Society Press (2012).*
- *Ekanayake J., Jenkins N., Liyanage K., Wu, J., Yokoyama A., Smart Grid: Technology and applications, Wiley Publications.*
- *Momoh J., Smart Grid: Fundamentals of design and analysis, John Wiley & Sons. Flick T., Morehouse J., Securing the smart grid: Next generation power grid security, paperback).*

- a. Other supplemental materials

- Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Communication Technologies for Power System: Fiber Optical Networks, WAN based on Fiber Optical Networks, IP based Real Time data Transmission, Substation communication network, Zigbee.

Information System for Control Centers (ICCS): ICCS Configuration, ICCS communication Network, ICCS Time Synchronization, E-Commerce of Electricity, GIS, GPS.

Integration, Control and Operation of Distributed Generation: Distributed Generation Technologies and its benefits, Distributed Generation Utilization Barriers, Distributed Generation integration to power grid.

Monitoring the smart grid: Load dispatch centers, wide-area monitoring system (WAMS), PMU; Smart sensors/telemetry, advanced metering infrastructure (AMI); smart metering; smart grid system monitoring; communication infrastructure and technologies; self-healing.

Micro grid: Integration of distributed energy sources; concept, operation, control and protection of Micro grid.

Hybrid Power Systems: Integration of conventional and non-conventional energy sources.

- b. Prerequisites

- UEE605.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Selected Elective

6. **Specific goals for the course**

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Explain various aspects of the smart grid, including, Technologies, Components, Architectures and Applications.
- Explain communication infrastructure of smart grid.
- Explain various integration aspects of conventional and non-conventional energy

sources.

- Explain distributed generation coordination including monitoring of smart grid using modern communication infrastructure.
 - Analyze Microgrid as a hybrid power system with advantages and challenges in future.
- b Specific outcomes addressed by the course.

E2	Develop appropriate models to formulate solutions.
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SO1.1	Ability to identify and formulate problems for electrical systems
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7. Brief list of topics to be covered

- Communication Technologies for Power System
- Information System for Control Centers (ICCS)
- Integration, Control and Operation of Distributed Generation
- Monitoring the smart grid
- Micro grid
- Hybrid Power Systems

1. **Course number and name:** UEI841; Advanced Control Systems
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from the Department of Electrical and Instrumentation Engineering.
4. **Text book, title, author, and year**

Text Books / Reference Books

- Slotine & Li, *Applied Non-Linear Control*, Englewood Cliffs, NJ: Prentice-Hall, (1991).
- Bandyopadhyay, M.N., *Control Engineering: Theory and Practice*, Prentice-Hall of India Private Limited (2003).
- Ogata, K., *Discrete-time Control Systems*, Pearson Education (2005).
- a. Other supplemental materials
 - Nil

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Nonlinear Control Systems: Introduction to Nonlinear systems and their properties, Common Non-linearities, Describing functions, Phase plane method, Lyapunov's method for stability study, concept of Limit Cycle.

Optimal Control Theory: Introduction, Optimal control problems, Mathematical procedures for optimal control design: Calculus of variations, Pontryagin's optimum policy, Bang-Bang Control, Hamilton-Jacobi Principle.

z-Plane Analysis of Discrete-Time Control Systems: Introduction, Impulse sampling and data hold, Reconstructing original signal from sampled signals, concept of pulse transfer function, Realization of digital controllers.

Design of Discrete-time Control Systems: Introduction, Stability analysis of closed-loop systems in the z-plane, Transient and steady state response analysis, Design based on the root-locus method, Design based on the frequency-response method.

State-Space Analysis: Introduction, State-space representations of discrete-time systems, Solving discrete-time state-space equations, Pulse transfer function matrix, Discretization of continuous time state space equations, Lyapunov stability analysis, Controllability and Observability, Design via pole placement, State observer design.

- b. Prerequisites

- UEI501.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Selected Elective

6. Specific goals for the course

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Demonstrate non-linear system behavior by phase plane and describing function methods.
- Perform the stability analysis nonlinear systems by Lyapunov method develop design skills in optimal control problems.
- Derive discrete-time mathematical models in both time domain (difference equations, state equations) and z-domain (transfer function using z-transform).
- Predict and analyze transient and steady-state responses and stability and sensitivity of both open-loop and closed-loop linear, time-invariant, discrete-time control systems.

- Acquire knowledge of state space and state feedback in modern control systems, pole placement, design of state observers and output feedback controllers.
- b Specific outcomes addressed by the course.

A1	Apply mathematics (partial differentiation, linear algebra, complex variables, Laplace transform probability statistic etc.) to obtain analytical numerical and statistical solutions
A3	Apply numerical and statistical methods in analyzing data
K2	Able to develop systems for engineering applications.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Nonlinear control system
- State space analysis
- Optimal control theory
- Discrete time control system

GENERIC ELECTIVES

1. **Course number and name:** UBT509; Biology for Engineers
2. **Credits and contact hours:** 3 and 3
3. **Instructor's or course coordinator's name:** Faculty from Department of Biotechnology
4. **Text book, title, author, and year**

- Nelson, D.L., Cox, M.M., *Lehninger: Principles of Biochemistry*, WH Freeman (2008) 5th ed.
- Dharmi, P.S., Srivastava, H.N. Chopra, G., *A Textbook of Biology*, Pradeep Publications (2008).
- Das, H.K., *Textbook of Biotechnology*, John Wiley & Sons (2004) 3rd Edition.
- Gardner, E.J., Simmons, M., Peter, S.D., *Principles of Genetics*, John Wiley & Sons (2008)
- Albert, B., *Essential Cell Biology*, Taylor & Francis, London (2009)

a. Other supplemental materials

- Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Characteristics of life: Living versus non-living organisms, origin of life, theory of evolution, diversity of life, classification of life into animals, plants, fungi, protists, archea and bacteria. Phylogenetics and its relationship with evolution.

Introduction to biological systems: basic unit of life, cellular organelles and their functions, important biomacromolecules (carbohydrates, lipids, proteins and nucleic acids) and their properties.

Cell membrane: Membrane structure, selective permeability, transport across cell membrane, active and passive transport, membrane proteins, type of transport proteins, channels and pumps, examples of membrane transport in cell physiology.

Classical and molecular genetics: Heredity and laws of genetics, genetic material and genetic information, Structure and properties of DNA, central dogma, replication of genetic information, universal codon system, encoding of genetic information via transcription and translation.

b. Prerequisites

- None

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Generic Elective

6. **Specific goals for the course**

- a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Describe living-systems and differentiate them from non-living systems
- Explain the theory of evolution and apply it non-living world
- Apply properties of nucleic acids in molecular recognition-based diagnostics
- Familiarized with various transport mechanisms across cell membranes
- Explain how genetic information is stored, replicated and encoded in living organisms.

b. Specific outcomes addressed by the course

A1	Apply mathematics (partial differentiation, vector calculus, linear algebra, complex variables, Laplace transform, probability, statistics, discrete mathematics etc.) to obtain analytical, numerical and statistical solutions.
A2	Demonstrate and apply knowledge of fundamentals, scientific and/or engineering principles towards solving engineering problems.
I1	Able to use resources to adopt new technologies not included in curriculum.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.

7. Brief list of topics to be covered

- Characteristics of life
- Biological Systems
- Cell membrane
- Classical and molecular genetics

1. **Course number and name:** UCS001; Introduction to Cyber Security
2. **Credits and contact hours:** 3 and 3
3. **Instructor's or course coordinator's name:** Faculty from Department of Computer Science and Engineering.
4. **Text book, title, author, and year**
 - Pfleeger, C.P., *Security in Computing*, Prentice Hall, 5th edition (2010)
 - Schneier, B., *Applied Cryptography*, Second Edition, John Wiley & Sons (1996)
 - Rhodes-Ousley, M., *Information Security: The Complete Reference*, Second Edition, *Information Security Management: Concepts and Practice*. New York, McGraw-Hill, (2013).
 - Whitman, M.E. and Herbert J. M., *Roadmap to Information Security for IT and Infosec Managers*, Course Technology, Boston, MA (2011).
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Introduction: Introduction to Computer Security, Threats, Harm, Vulnerabilities, Controls, Authentication, Access Control, and Cryptography, Authentication, Access Control, Cryptography

Programs and Programming: Unintentional (Non-malicious) Programming Oversights, Malicious Code—Malware, Countermeasures

Web Security: User Side, Browser Attacks, Web Attacks Targeting Users, Obtaining User or Website Data, Email Attacks

Operating Systems Security: Security in Operating Systems, Security in the Design of Operating Systems, Rootkit

Network Security: Network Concepts, Threats to Network Communications, Wireless Network Security, Denial of Service, Distributed Denial-of-Service Strategic Defenses: Security Countermeasures, Cryptography in Network Security, Firewalls, Intrusion Detection and Prevention Systems, Network Management

Cloud Computing and Security: Cloud Computing Concepts, moving to the Cloud, Cloud Security Tools and Techniques, Cloud Identity Management, Securing IaaS

Privacy: Privacy Concepts, Privacy Principles and Policies, Authentication and Privacy, Data Mining, Privacy on the Web, Email Security, Privacy Impacts of Emerging Technologies, Where the Field Is Headed

Management and Incidents: Security Planning, Business Continuity Planning, Handling Incidents, Risk Analysis, Dealing with Disaster

Legal Issues and Ethics: Protecting Programs and Data, Information and the Law, Rights of Employees and Employers, Redress for Software Failures, Computer Crime, Ethical Issues in Computer Security, Incident Analysis with Ethics

Emerging Topics: The Internet of Things, Economics, Computerized Elections, Cyber Warfare.

- b. Prerequisites

- UCS049.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Generic Elective

6. Specific goals for the course

a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Understand the broad set of technical, social & political aspects of Cyber Security and security management methods to maintain security protection
- Appreciate the vulnerabilities and threats posed by criminals, terrorist and nation states to national infrastructure
- Understand the nature of secure software development and operating systems
- Recognize the role security management plays in cyber security defense and legal and social issues at play in developing solutions

b. Specific outcomes addressed by the course.

F2	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and well-being of society.
H1	Aware of societal and global changes due to engineering innovations.
I1	Able to use resources to adopt new technologies not included in curriculum.
I2	Ability to identify directions for continuing education opportunities.

SO2.3	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.
SO4.4	Aware of societal and global changes due to engineering innovations.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.
SO7.2	Ability to identify directions for continuing education opportunities.

7. Brief list of topics to be covered

- Web Security
- Operating Systems Security
- Network Security
- Cloud Computing and Security
- Privacy
- Management and Incidents
- Legal Issues and ethics

1. **Course number and name:** UEN004; Technologies for Sustainable Development
2. **Credits and contact hours:** 3 and 3
3. **Instructor's or course coordinator's name:** Faculty from the School of Energy and Environment.
4. **Text book, title, author, and year**
 - Kirkwood, R,C, and Longley, A,J, (Eds.), *Clean Technology and the Environment*, Chapman & Hall, London (1995),
 - World Bank Group; *Pollution Prevention and Abatement Handbook – Towards Cleaner Production*, World Bank and UNEP; Washington DC (1998),
 - Modak, P,, Visvanathan, C, and Parasnis, M,, *Cleaner Production Audit, Course Material on Cleaner Production and Waste Minimization*; United Nations Industrial Development Organization (UNIDP) (1995),
 - Rao, S, and Parulekar, B,B,, *Energy Technology: Non-conventional; Renewable and Conventional*; Khanna Pub,(2005) 3rd Ed,
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Concept of Sustainability and Industrial Processes: Industrialization and sustainable development; Cleaner production (CP) in achieving sustainability; Source reduction techniques - Raw material substitution; Process modification and equipment optimization; Product design or modification; Reuse and recycling strategies; Resources and by-product recovery from wastes; Treatment and disposal; CDM and Pollution prevention programs; Good housekeeping; CP audits,

Green Design: Green buildings - benefits and challenges; public policies and market-driven initiatives; Effective green specifications; Energy efficient design; Passive solar design; Green power; Green materials and Leadership in Energy and Environmental Design (LEED)

Renewable and Emerging Energy Technologies: Introduction to renewable energy technologies- Solar; wind; tidal; biomass; hydropower; geothermal energy technologies; Emerging concepts; Biomolecules and energy; Fuel cells; Fourth generation energy systems,

- b. Prerequisites

- UEN002.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Generic Elective

6. **Specific goals for the course**

- a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- comprehend basic concepts in source reduction, waste treatment and management
- Identify and plan cleaner production flow charts/processes for specific industrial sectors
- Examine and evaluate present and future advancements in emerging and renewable energy technologies.

- b. Specific outcomes addressed by the course.

H3	Evaluate engineering solutions considering environmental constraints.
I1	Able to use resources to adopt new technologies not included in curriculum. I2. Ability to identify directions for continuing education opportunities.
I2	Able to use resources to adopt new technologies not included in curriculum. I2. Ability to identify directions for continuing education opportunities.
J2	Recognize the impact of engineering decisions on energy resources and environment.

SO4.1	Recognize the impact of engineering decisions on energy resources and environment.
SO4.2	Evaluate engineering solutions considering environmental constraints.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.
SO7.2	Ability to identify directions for continuing education opportunities.

7. Brief list of topics to be covered

- a. Concept of Sustainability and Industrial Processes
- b. Green Design
- c. Renewable and Emerging Energy Technologies

1. **Course number and name:** UHU006; Introductory Course in French
2. **Credits and contact hours:** 3 and 3.
3. **Instructor's or course coordinator's name:** Faculty from the School of Humanities and Social Sciences.
4. **Text book, title, author, and year**
 - *Alter ego-1 : Méthode de français* by Annie Berthet, Catherine Hugot, Véronique M. Kizirion, Beatrix Sampsonis, Monique Waendendries, Editions Hachette français langue étrangère.
 - *Connexions-1 : Méthode de français* by Régine Mérieux, Yves Loiseau, Editions Didier
 - *Version Originale-1: Méthode de français* by Monique Denyer, Agustin Garmendia.
 - *Marie-Laure Lions-Olivieri*, Editions Maison des Langues, Paris 2009
 - *Latitudes-1 : Méthode de français* by Régine Mérieux, Yves Loiseau, Editions Didier
 - *Campus-1 : Méthode de français* by Jacky Girardet, Jacques Pécheur, Editions CLE International.
 - *Echo-1 : Méthode de français* by J. Girardet, J. Pécheur, Editions CLE International.
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Communicative skills: Greetings and Its Usage, asking for and giving personal information, How to ask and answer questions, How to talk over the phone, Exchange simple information on preference, feelings etc. Invite, accept, or refuse invitation, fix an appointment, Describe the weather, Ask for/give explanations, Describe a person, an object, an event, a place.

Grammar: Pronouns: Pronom sujets (Je/ Tu/Il/Elle/Nous/Vous/Ils/Elles), Nouns: Genders, Articles: Definite article and Indefinite articles, Verbs: Regular verbs (-er, -ir ending) Irregular verbs (-re ending), Auxiliary verbs (avoir, être, aller). Adjective: Description, Adjective possessive, Simple Negation, Tense: Present, Future, Questions, Singular & plural.

Vocabulary: Countries and Nationalities, Professions, Numbers (ordinal, cardinal), Colours, Food and drinks, Days of the week, Months, Family, Places.

Phonetics: The course develops the ability, to pronounce words, say sentences, questions and give orders using the right accent and intonation. To express surprise, doubt, fear, and all positive or negative feelings using the right intonation. To distinguish voiced and unvoiced consonants. To distinguish between vowel sounds.

- b. Prerequisites
 - UHU003.
 - c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Generic Elective
6. **Specific goals for the course**
 - a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Communicate in simple everyday situations acquiring basic grammatical structure and vocabulary.
- To develop oral and reading comprehension skills as well as speaking and writing.
- Demonstrate understanding of simple information in a variety of authentic materials

such as posters, advertisement, signs etc.

- Discuss different professions, courses and areas of specialization.
- Write simple messages, letters, composition and dialogues. Complete simple forms and documents.
- Express feelings, preferences, wishes and opinions and display basic awareness of francophone studies.
- Units on pronunciation and spelling expose students to the different sounds in the French language and how they are transcribed.

b. Specific outcomes addressed by the course.

F1	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
G2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
I3	Recognize the need to embrace personal responsibility for lifelong learning.

SO3.2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
SO3.3	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
SO7.3	Recognize the need to embrace personal responsibility for lifelong learning

7. Brief list of topics to be covered

- Communicative skills
- Grammar
- Vocabulary
- Phonetics:

1. **Course number and name:** UHU007; Employability Development Skills
2. **Credits and contact hours:** 3 and 3
3. **Instructor's or course coordinator's name:** Faculty from the School of Humanities and Social Sciences.
4. **Text book, title, author, and year**
 - *Harvard Business Essentials; Creating Teams with an Edge; Harvard Business School Press (2004)*
 - *Edward de B., Six Thinking Hats; Penguin Life (2016)*
 - *Daniel, G., Working with Emotional Intelligence; Bantam Books (2000)*
 - *Aggarwal, R.S., Quantitative Aptitude for Competitive Examinations; S Chand (2017)*
 - *Agarwal, A., An expert guide to problem solving: with practical examples; CreateSpace Independent Publishing Platform (2016)*
 - *William, D., The Logical Thinking process; American Society for Quality (2007)*
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Emotional Intelligence: Understanding Emotional Intelligence (EI); Daniel Goleman's EI Model: Self Awareness, Self-Regulation, Internal Motivation, Empathy, Social Skills; Application of EI during Group Discussions & Personal Interview; Application of EI in personal life, student life and at the workplace

Team Dynamics & Leadership: Understanding the challenges of working within a team format in today's complex organizational environments; Stages of team formation; Appreciating forces that influence the direction of a team's behaviour and performance; Cross-functional teams; Conflict in Teams- leveraging differences to create opportunity Leadership in the team setting & energizing team efforts; Situational leadership; Application of team dynamics & collaboration in Group Discussions; Application of team dynamics at the workplace

Complex Problem Solving: Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions; Understanding a working model for complex problem solving - framing the problem, diagnosing the problem, identifying solutions & executing the solutions; Appreciation of complex problem solving at the workplace through case studies

Lateral Thinking: Understanding lateral thinking & appreciating the difference between vertical & lateral thinking, and between convergent & divergent thinking; Understanding brain storming & mind-maps; Solving of problems by an indirect and creative approach, typically through viewing the problem in a new and unusual light; Application of lateral thinking during Group Discussions & Personal Interviews; Application of lateral thinking at the workplace

Persuasion: Role of persuasion in communication; Application of ethos-pathos-logos; Using persuasive strategies to connect with individuals & teams to create competitive advantage **Quantitative Reasoning:** Thinking critically and applying basic mathematics skills to interpret data, draw conclusions, and solve problems; developing proficiency in numerical reasoning; Application of quantitative reasoning in aptitude tests

Verbal Reasoning: Understanding and reasoning using concepts framed in words; Critical verbal reasoning; Reading Comprehension; Application of verbal reasoning in aptitude tests

Group Discussion (GD): Illustrating the do's and don'ts in Group Discussions; Specific thrust on types of GD topics; GD evaluation parameters; Understanding the challenge in a case discussion; SPACER model

Personal Interview (PI): Interview do's and don'ts; PI evaluation parameters; The art of introduction; Managing bounce questions; Leading the panel in a PI

b. Prerequisites

- UHU003, UHU005.

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Generic Elective

6. Specific goals for the course

a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Understand the broad set of technical, social & political aspects of Cyber Security and security management methods to maintain security protection
- Appreciate the vulnerabilities and threats posed by criminals, terrorist and nation states to national infrastructure
- Understand the nature of secure software development and operating systems
- Recognize the role security management plays in cyber security defense and legal and social issues at play in developing solutions

b. Specific outcomes addressed by the course.

A2	Demonstrate and apply knowledge of fundamentals, scientific and/or engineering principles towards solving engineering problems.
D1	Share responsibility and information schedule with others in team.
D2	Participate in the development and selection of ideas.
F1	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
F2	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and well-being of society.
G2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
I1	Able to use resources to adopt new technologies not included in curriculum.
I2	Ability to identify directions for continuing education opportunities.
I3	Recognize the need to embrace personal responsibility for lifelong learning.

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO2.3	Able to evaluate ethical issues that may occur in professional practice using professional codes of ethics ensuring protection of organization, human safety and wellbeing of society.
SO3.2	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve engineering problems.
SO3.3	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
SO5.1	Share responsibility and information schedule with others in team.
SO5.2	Participate in the development and selection of ideas on a team whose members together provide leadership.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.
SO7.2	Ability to identify directions for continuing education opportunities.
SO7.3	Recognize the need to embrace personal responsibility for lifelong learning

7. Brief list of topics to be covered

- Emotional Intelligence
- Team dynamics & leadership
- Complex problem solving
- Lateral thinking
- Persuasion
- Quantitative and verbal reasoning
- Group discussion and personal interview

1. **Course number and name:** UHU008; Introduction to Corporate Finance
2. **Credits and contact hours:** 3 and 3
3. **Instructor's or course coordinator's name:** Faculty from the School of Humanities and Social Sciences
4. **Text book, title, author, and year**
 - Brealey, R. A., Myers, S.C., Allen, F., *Principles of Corporate Finance (9th edition)*, The McGraw-Hill, London, (2006).
 - Ehrhardt, M.C., Brigham, E.F., *Financial Management: Theory and Practice (10th edition)* South Western-Cengage, New York (2011)
 - Van Horne, J.C., Wachowicz, J.M., Kuhlemeyer, G.A., 2005, *Fundamentals of Financial Management*, Pearson, Vancouver (2010)
 - Pandey, I. M., *Financial management*, Vikas Publishing House Pvt. Ltd., Noida (2011)
 - Elton, E.J. and Gruber, M.J., *Modern Portfolio Theory and Investment Analysis, (7th Edition)*, John Wiley and Sons, New York (2007)
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Introduction to corporate finance: Finance and corporate finance. Forms of business organizations, basic types of financial management decisions, the goal of financial management, the agency problem; the role of the financial manager; basic types of financial management decisions.

Financial statements analysis: Balance sheet, income statement, cash flow, fund flow financial statement analysis Computing and interpreting financial ratios; conducting trend analysis and Du Pont analysis.

The time value of money: Time value of money, future value and compounding, present value and discounting, uneven cash flow and annuity, discounted cash flow valuation.

Risk and return: Introduction to systematic and unsystematic risks, computation of risk and return, security market line, capital asset pricing model.

Long-term financial planning & Financial Decisions: Various sources of long-term financing, the elements and role of financial planning, financial planning model, percentage of sales approach, external financing needed. Cost of capital, financial leverage, operating leverage. Capital structure, theories of capital structure net income, net operating income & M&M proposition I and II.

Short-term financial planning and management: Working capital, operating cycle, cash cycle, cash budget, short-term financial policy, cash management, inventory management, credit management.

Capital budgeting: Concepts and procedures of capital budgeting, investment criteria (net present value, payback, discounted payback, average accounting return, internal rate of return, profitability index), incremental cash flows, scenario analysis, sensitivity analysis, break-even analysis,

Dividend policy: Dividend, dividend policy, Various models of dividend policy (Residual approach, Walter model, Gordon Model, M&M, Determinants of dividend policy.

Security valuation: Bond features, bond valuation, bond yields, bond risks, stock features, common stock valuation, and dividend discount & dividend growth models. Common stock yields, preferred stock valuation.

- b. Prerequisites
 - None
- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
 - Generic Electives

6. Specific goals for the course

- a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Ability to evaluate critically corporate financial management practices with the aim of proposing and implementing improvements.
 - Apply the methods and procedures of financial management, with particular reference to investment evaluation corporate evaluation and risk management.
 - Applying the knowledge to estimate a company's cost of capital; determine whether a company is creating or destroying value.
 - Applying the various theories of corporate finance to design a company's optimal mix of debt and equity financing; and compensate shareholders in the most convenient way.
 - Apply the methods and procedures to value stocks and bonds; assess the risk and return of assets.
- b. Specific outcomes addressed by the course.

C3	Work with real time systems within realistic constraints.
H1	Aware of societal and global changes due to engineering innovations.
H2	Analyze economic tradeoffs in engineering systems.
I2	Ability to identify directions for continuing education opportunities.

SO2.2	Work with real time systems within realistic constraints.
SO4.1	Recognize the impact of engineering decisions on energy resources and environment.
SO4.3	Analyze economic trade-offs in engineering systems
SO7.2	Ability to identify directions for continuing education opportunities.

7. Brief list of topics to be covered

- Introduction to corporate finance
- Financial statements analysis
- The time value of money
- Risk and return
- Long-term financial planning & Financial Decisions
- Short-term financial planning and management
- Capital budgeting
- Dividend policy
- Security valuation

1. **Course number and name:** UHU009; Introduction to Cognitive Science
2. **Credits and contact hours:** 3 and 3
3. **Instructor's or course coordinator's name:** Faculty from the School of Humanities and Social Sciences
4. **Text book, title, author, and year**
 - Bermúdez, J.L., *Cognitive Science: An Introduction to the Science of the Mind* (2nd Ed.), Cambridge, UK: Cambridge (2014).
 - Friedenberg, J.D, and Silverman, G, *Cognitive Science: An Introduction To The Study Of Mind*, Sage Publications:, London (2014)
 - Thagard, P., *Mind: An introduction to Cognitive Science*, MIT Press, (2005)
 - Thagard, P., (1998) *Mind Readings: Introductory Selections on Cognitive Science*, MIT Press, Cambridge, Mass
 - a. Other supplemental materials
 - Nil
5. **Specific course information**
 - a. Brief description of the content of the course (catalog description)

Overview of Cognitive Science: Newell's big question, Constituent disciplines, Interdisciplinary approach, Unity and diversity of cognitive science,

Philosophy: Philosophy of Mind, Cartesian dualism Nativism vs, empiricism, Mind-body problem, Functionalism, Turing Test, Modularity of mind, Consciousness, Phineas Gage, Physicalism.

Psychology: Behaviorism vs, cognitive psychology, The cognitive revolution in psychology, Hardware/software distinction, Perception and psychophysics, Visual cognition, Temporal dynamics of visual perception, Pattern recognition, David Marr's computational theory of vision, Learning and memory, Theories of learning, Multiple memory systems, Working Memory and Executive Control, Memory span, Dissociations of short- and long-term memory, Baddeley's working memory model.

Linguistics: Components of a grammar, Chomsky, Phrases and constituents, Productivity, Generative grammars, Compositional syntax, Productivity by recursion, Surface- and deep structures, Referential theory of meaning, Compositional semantics, Semantics, Language acquisition, Language and thought.

Neuroscience: Brain anatomy, Hierarchical functional organization, Decorticate animals, Neuroimaging, Neurophysiology, Neuron doctrine, Ion channels, Action potentials, Synaptic transmission, Synaptic plasticity, Biological basis of learning, Brain damage, Amnesia, Aphasia, Agnosia, Parallel Distributed Processing(PDP), Computational cognitive neuroscience, The appeal of the PDP approach, Biological Basis of Learning, Cajal's synaptic plasticity hypothesis, Long- term potentiation (LTP) and depotentiation (LTD), NMDA receptors and their role in LTP, Synaptic consolidation, Vertical integration, The Problem of representation, Shannon's information theory.

Artificial Intelligence: Turing machines, Physical symbol systems, Symbols and Search Connectionism, Machine Learning, Weak versus strong AI, Subfields, applications, and recent trends in AI, Turing Test revisited, SHRDLU, Heuristic search, General Problem Solver (GPS), Means-ends analysis.

Cognitive architectures: Tripartite architecture, Integration, ACT-R Architecture Modularity

- b. Prerequisites

- UHU005.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program
- Generic Elective

6. Specific goals for the course

- a. . Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Identify cognitive science as an interdisciplinary paradigm of study of cross-cutting areas such as Philosophy, Psychology, Neuroscience, Linguistics, Anthropology, and Artificial Intelligence.
- Explain various processes of the mind such as memory and attention, as well as representational and modelling techniques that are used to build computational models of mental processes;
- Acquire basic knowledge of neural networks, linguistic formalism, computing theory, and the brain.
- Apply basic Artificial Intelligence techniques to solve simple problems.

- b. Specific outcomes addressed by the course.

D1	Share responsibility and information schedule with others in team.
D2	Participate in the development and selection of ideas.
F1	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
I1	Able to use resources to adopt new technologies not included in curriculum.
I3	Recognize the need to embrace personal responsibility for lifelong learning.

SO3.3	Able to interact with the people in organizations, industries and/or professional societies in a professional manner to achieve their goals.
SO5.1	Share responsibility and information schedule with others in team.
SO5.2	Participate in the development and selection of ideas on a team whose members together provide leadership.
SO7.1	Able to use resources to adopt new technologies not included in curriculum.
SO7.3	Recognize the need to embrace personal responsibility for lifelong learning

7. Brief list of topics to be covered

- Overview of Cognitive Science:
- Psychology
- Neuroscience
- Artificial Intelligence

1. **Course number and name:** UPMA061; Advanced Numerical Methods
2. **Credits and contact hours:** 3 and 3
3. **Instructor's or course coordinator's name:** Faculty from School of Mathematics
4. **Text book, title, author, and year**
 - *Gerald, C.F. and Wheatley, P.O., Applied Numerical Analysis, Pearson Education (2008) 7th ed.*
 - *Gupta, S.R., Elements of Numerical Analysis, MacMillan India (2009).*
 - *Atkinson, K.E., An introduction to Numerical Analysis, John Wiley (2004) 2nd ed.*
 - *S.D. Conte, S.D. and Carl D. Boor, Elementary Numerical Analysis: An Algorithmic Approach, Tata McGraw Hill (2005).*
 - *Jain M. K., Iyengar. S.R.K. and Jain, R.K. Numerical Methods for Scientific and Engineering Computation, New Age International (2008) 5th ed.*
 - a. Other supplemental materials
 - Nil

5. **Specific course information**

- a. Brief description of the content of the course (catalog description)

Non-Linear Equations: Methods for multiple roots, Muller's, Iteration and Newton-Raphson method for non-linear system of equations and Newton-Raphson method for complex roots.

Polynomial Equations: Descartes' rule of sign, Birge-vieta, Giraffe's methods.

System of Linear Equations: Cholesky and Partition methods, SOR method with optimal relaxation parameters.

Eigen-Values and Eigen-Vectors: Similarity transformations, Gerschgorin's bound(s) on eigenvalues, Given's and Rutishauser methods.

Interpolation and Approximation: Cubic and B – Spline and bivariate interpolation, least squares approximations, Gram-Schmidt orthogonalisation process and approximation by orthogonal polynomial, Legendre and Chebyshev polynomials and approximation.

Differentiation and Integration: Differentiation and integration using cubic splines, Romberg integration and multiple integrals.

Ordinary differential Equations: Milne's, Adams-Moulton and Adam's Bashforth methods with their convergence and stability, Shooting and finite difference methods for second order boundary value problems.

- b. Prerequisites

- UMA007.

- c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Generic Elective

6. **Specific goals for the course**

- a. Specific outcomes of instruction

After the completion of the course, the students will be able to:

- Find multiple roots of equation and apply Newton -Raphson's method to obtain complex roots as well solution of system of non - linear equations.
- Learn how to obtain numerical solution of polynomial equations using Birge - Vitae and Giraffe's methods.
- Apply Cholesky, Partition and SOR methods to solve system of linear equations.

- Understand how to approximate the functions using Spline, B-Spline, and least square. approximations
 - Learn how to solve definite integrals by using cubic spline, Romberg and initial value problems and boundary value problems numerically.
- b. Specific outcomes addressed by the course

A1	Apply mathematics (partial differentiation, vector calculus, linear algebra, complex variables, Laplace transform, probability, statistics, discrete mathematics etc.) to obtain analytical, numerical and statistical solutions.
B1	Identify the constraints, assumptions and models for the experiments.
C3	Work with real time systems within realistic constraints.
K2	Able to develop systems for engineering applications.
K3	Able to analyze engineering problems using software tools.

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO2.2	Work with real time systems within realistic constraints
SO6.1	Identify the constraints, assumptions and models for the experiments.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Nonlinear equations
- Polynomial equations
- Eigen-values and eigen vectors
- Interpolation and approximation
- Differentiation and Integration
- Ordinary differential equations

1. **Course number and name:** UMA066; Graph Theory and Applications
2. **Credits and contact hours:** 3.5 and 4
3. **Instructor's or course coordinator's name:** Faculty from School of Mathematics
4. **Text book, title, author, and year**
 - *Donald H Sanders, Computers Today, Tata Mcgraw Hill.*
 - *A Siloerschatz and Peter B Galvin, Operating System Concepts, Addison-Wesley*
 - *S Tanenbaum, Computer Networks, PHI.*
 - *Elrnasri&. Navathe, Fundamentals of Database Systems, Pearson Education Asia.*
 - *William Stallings, Network Security Essentials Applications and Standards, Pearson Education Asia.*

a. Other supplemental materials

 - Nil
5. **Specific course information**

a. Brief description of the content of the course (catalog description)

Role of operating system in efficiently managing systems resources: Process management Memory Management, File & Device management, Multiprogramming & Multiprocessing Time-sharing Systems

System Software: Introduction to compiler, assembler, debugger, linker, working on Linux/Unix/Solaris Utilities like gzip, tar, C Language, make Environment Xtern, Vi editor

Programming Paradigm: Object oriented VS Structured programming approach, Concepts of class, object, method, inheritance, polymorphism, abstraction, and encapsulation.

Networking and Internetworking concepts: Protocols TCPIIP suite, IP addressing scheme; DNS, UDP; TELN'ET, ITP, HTTP.

Devices: Repeaters, bridges, routers, gateways, switches, Networking Security

Web Page: Web browsers; development tools, web server, Email server

Web Technologies: Introduction to ASP, .Net, ActiveX, Java script, VBScript

Threats: Virus Threats, Hacking Prevention Mechanism: Anti-Viruses, Secured Network, Firewalls, Proxy servers

Concepts of Software Engineering: Life Cycle, Importance of analysis and design in software development process, SDLC models: Prototyping, Spiral Model, Combined Paradigm

Concepts of RDBMS: Exposure of Backend and Front-end Products, Introduction to E-R diagrams, SQL (Simple), SQL (Nested), Normalization.

Algorithms: Algorithm complexity, efficiency, BIG(O) notation, Searching and Sorting algorithms.

b. Prerequisites

 - UCS049.

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

 - Generic Elective

6. Specific goals for the course

a. After the completion of the course, the students will be able to:

- Explain the concept of operating system management.
- Explain the concept of system software.
- Differentiate between various programming Paradigm.
- Apply concepts of web page, software engineering and RDBMS in realapplications.

b. Specific outcomes addressed by the course

E2	Develop appropriate models to formulate solutions
K2	Able to develop systems for engineering applications

SO1.2	Apply sciences and mathematics and to obtain analytical, numerical and statistical solutions.
SO6.3	Able to analyze engineering problems and develop systems for engineering applications.

7. Brief list of topics to be covered

- Operating system
- System Software
- Networking
- Programming Paradigm
- Web page

1. **Course number and name:** UPH063; Nano Science and Nano Materials
2. **Credits and contact hours:** 3 and 4
3. **Instructor's or course coordinator's name:** Faculty from the School of Physics and Materials Science
4. **Text book, title, author, and year**

- Booker, R., Boysen, E., *Nanotechnology*, Wiley India Pvt. Ltd. (2008).
- Rogers, B., Pennathur, S., Adams, J., *Nanotechnology*, CRS Press (2007).
- Bandyopadhyay, A.K., *Nano Materials*, New Age Int., (2007).
- Niemeyer, C. N., and Mirkin, C.A., *Nanobiotechnology Concepts, Applications and Perspectives*, Wiley VCH, Weinheim, Germany (2007).

a. Other supplemental materials

- Nil

5. Specific course information

a. Brief description of the content of the course (catalog description)

Fundamental of Nanoscience: Features of Nano system, Free electron theory and its features, Idea of band structures, Density of states in bands, Variation of density of state and band gap with size of crystal.

Quantum Size Effect: Concepts of quantum effects, Schrodinger time independent and time dependent equation, Electron confinement in one-dimensional well and three-dimensional infinite square well, Idea of quantum well structure, Quantum dots and quantum wires.

Nano Materials: Classification of Nano Materials their properties, Basic concept relevant to application, Fullerenes, Nanotubes and nano-wires, Thin films chemical sensors, Gas sensors, Vapour sensors and Bio sensors.

Synthesis and processing: Sol-gel process, Cluster beam evaporation, Ion beam deposition, Chemical bath deposition with capping techniques and ball milling, Cluster assembly and mechanical attrition, Sputtering method, Thermal evaporation, Laser method.

Characterization: Determination of particle size, XRD technique, Photo luminescence, Electron microscopy, Raman spectroscopy, STEM, AFM.

Applications: Photonic crystals, Smart materials, Fuel and solar cells, Opto- electronic devices.

b. Prerequisites

- UES012.

c. Indicate whether a required elective or selected elective (as per Table 5.1) course in the program

- Selected Elective

6. Specific goals for the course

a. After the completion of the course, the students will be able to:

- Explain quantum size effect on the properties of materials at nanoscale.
- Design and characterize materials at nano-scale.

b. Specific outcomes addressed by the course

A2	Demonstrate and apply Knowledge of fundamental Scientific and/or Engineering principles towards solving Engineering problems.
E1	Classify information to identify engineering problems
E3	Use analytical, computational and/or experimental methods to obtain solutions

SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.
SO1.1	Ability to identify and formulate problems for electrical systems.
SO1.3	Apply knowledge of fundamentals, scientific and/or engineering principles towards solving complex engineering problems using analytical, computational and/or experimental methods.

7. Brief list of topics to be covered

- Fundamental of Nanoscience
- Quantum Size Effect
- Synthesis and processing
- Nano material